Hanzo — Technical Paper Compendium

Sovereign AI, Post-Quantum, FHE & Agentic Engineering

Owned, self-hostable AI and cryptography for regulated industries, critical infrastructure, and government. Click any title to jump to the paper; the PDF bookmark sidebar also navigates the suite. The full suite is published at papers.hanzo.ai.

Hanzo Industries — American-owned applied-AI lab, Techstars '17, founded 2014. Approved for public release; distribution unlimited.

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Hanzo Cloud for Reindustrialization

Sovereign AI, robotics, spatial 3D & post-quantum infrastructure for the American industrial base

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The sovereign AI operating layer for the new American industrial base — factories, robots, vehicles, sensors, drones, ships, and defense systems. We pair open-source AI, spatial 3D, robotics control, post-quantum cryptography, FHE, and low-latency on-chain coordination so operators can **build, automate, simulate, secure, and command** real-world systems — factory floor to contested edge — with no closed or foreign AI dependency.

Why Hanzo

- **Sovereign industrial cloud** — self-hostable agents, models, tools, memory, IAM/KMS, audit, and deploy; cloud, factory, edge, or air-gapped.
- **Spatial 3D + robotics** — digital twins, perception, simulation-to-field, fleet coordination, low-latency control, edge inference.
- **Low-latency machine coordination** — on-chain identity, provenance, command logs, encrypted state, settlement, multi-party trust.
- **Post-quantum + FHE-native security** — PQC for comms and identity; FHE to compute over ciphertext (coalition analytics, sealed telemetry).
- **Agentic engineering + cost collapse** — modernize a contractor's stack in place: fewer services, lower cost, sovereign and auditable.

The proof

Regulated capital-markets modernization — FINRA/SEC-approved, SOC 2 (client anonymized)

A regulated capital-markets modernization showed what the agentic stack can do. By the platform's own engineering audit: **334 repositories**, **31,058 commits**, **15,294** from Hanzo's agentic CTO workflow, and **114M raw git-log lines** — spanning framework-wide refactors, dependency upgrades, automation, and production code — in an **81-day** window, measured at **$0.0006 per line** by the audit's methodology. The same modernization cut cloud spend ∼**90%** (∼**$35K/mo to** ∼**$3.5K/mo,** ∼**$380K/yr**), consolidated **200+ microservices (57 core) into 3 binaries**, and closed **93 security findings** — into a SOC 2 / FINRA/SEC-regulated production environment on post-quantum crypto, FHE, and native-GPU on-chain systems.

By the numbers: 334 repos · 31,058 commits · 15,294 agentic commits · 114M raw git-log LoC · **$0.0006/LoC** · 57 services → 3 binaries · ∼$35K/mo → ∼$3.5K/mo (∼90%) · 93 findings closed · 50 Lean proofs · FIPS 203/204/205 · 11.5M tx/s ZAP.

Why this matters for defense contractors: Hanzo modernizes approved legacy stacks with a smaller team, faster iteration, lower cloud cost, fewer services, stronger auditability, and post-quantum / FHE-native security.

For combat systems: spatial awareness, low-latency agent control, post-quantum identity, encrypted telemetry, FHE-protected decision rules, and auditable on-chain command history — deployable at the edge when links are degraded or denied.

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22 technical papers — sovereign AI, FHE, post-quantum, security, compliance, robotics, and measured edge-AI inference.

Hanzo Industries — American-owned applied-AI lab, Techstars '17, founded 2014. Open-source · post-quantum · self-hostable.

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Sovereign vs. Closed: Hanzo AI versus Centralized Cloud AI

Zach Kelling — Hanzo Team

Version 1.0 2026

Abstract

We compare two structurally different ways to obtain frontier AI: a closed, centrally hosted model accessed as a metered cloud service (typified by Anthropic's Claude), and a sovereign, self-hostable stack with owned model weights (Hanzo). The distinction is not primarily one of model quality—closed frontier models are excellent—but of *deployment model and control*. For regulated industries, critical infrastructure, and government (including defense), the binding constraints are data residency, operation in disconnected and critical environments, supply-chain provenance, post-quantum security, auditability, and cost at scale. A closed cloud service cannot satisfy several of these by construction: it cannot run air-gapped, its weights cannot be inspected or owned, and sensitive data must leave the enclave to be processed. We argue that for these workloads the relevant choice is between foreign open-weight models and an American open-source alternative, and we position Hanzo as the latter—owning both the model (the Zen family) and the cryptography (a production post-quantum stack). We give a dimension-by-dimension comparison and a decision framework for when each model of consumption fits. These constraints are most acute in financial services, healthcare, critical infrastructure, and government systems—defense among them—but the framework applies to any organization that must control where its data and models live. We also make the productivity case concrete: on a regulated capital-markets build where every engineer had the same frontier chat model, the client's own engineering audit found that one engineer working through Hanzo's agentic stack out-produced entire contractor teams by the audit's own unit-cost metrics—the differentiator being the owned agentic *stack*, not the model.

Executive Summary. This is not a claim that our model is smarter than Claude. It is a claim about *where* and *how* you can run it. Claude is an excellent frontier model delivered as a managed cloud service: you call it over the internet, your data goes to the vendor, you cannot see or own the weights, and it cannot run on a disconnected device. That is the right product for many commercial uses and the wrong one for a workload that cannot leave a controlled environment. Hanzo is the opposite shape: open, owned-weight models you run on your own hardware—in a data center, at the edge, or air-gapped—with post-quantum security built in, your data never leaving, and cost driven down because you are not paying a per-token markup or a hyperscaler margin. America's frontier AI has narrowed to two closed labs; the only openly available frontier weights today are largely foreign. For regulated industries, critical infrastructure, and government buyers—defense included—who need open, sovereign, controllable AI, that makes Hanzo the American alternative to both the closed duopoly and foreign open weights.

Contents

1 The Real Axis of Comparison

Discussions of "which AI is better" usually compare benchmark scores. For commercial chat and coding that is a reasonable proxy. For regulated and sovereign deployment it is the wrong axis. A model that scores two points higher on a reasoning benchmark is of no use inside a controlled environment if it can only be reached by sending the data outside that environment.

The decisive axis is the *deployment and trust model*: who holds the weights, where inference physically runs, whether the data leaves the enclave, whether the system works without connectivity, what cryptography protects it, and what it costs at scale. On this axis a closed cloud service and a sovereign self-hosted stack are not competitors with different scores—they are different categories of thing. Claude is the leading example of the first category; Hanzo is built as the second.

We state plainly what closed cloud AI does well: it is operationally simple (no infrastructure to run), it tracks the frontier of raw capability, and its provider invests heavily in alignment and safety research. None of that is in dispute. The argument here is narrow and structural: several requirements that are mandatory for regulated, critical-infrastructure, and government workloads (defense among them) cannot be met by a model you do not hold and cannot run yourself.

2 One Engineer on the Hanzo Stack vs. a Team on Claude

The cleanest evidence for this paper's thesis is not a benchmark—it is a controlled natural experiment that already happened in production. On a regulated capital-markets platform, every engineer on the program had access to the same frontier chat model (Anthropic's Claude, via `claude.ai`). The variable that differed was the *tooling around* that model: one engineer worked through Hanzo's owned, agentic stack (`hanzo.ai`)—an orchestration layer that turns the model into an autonomous build-test-ship loop across hundreds of repositories—while the contractor teams used the model the conventional way, as an interactive assistant alongside hand-driven development.

Crucially, this is *not* a claim that one group had a smarter model. They did not. Everyone held the same frontier weights. What separated the outcomes was whether the model was wrapped in an agentic stack or merely chatted with. The platform's own engineering audit measured the result.

The measured comparison. The audit covered 334 repositories and 31,058 commits over an 81-day window, attributing every commit from raw version-control history and normalizing cost against each contributor's monthly spend. We report the three metrics the audit treats as defensible: total commits shipped, unit cost per line of code ($/LoC), and shipping rhythm per dollar of monthly spend (commits per $1,000). The contractor pods are anonymized as Team A, Team B, and Team C.

Group (same frontier model)	**Commits**	**$ / LoC**	**Commits / $1k**
One engineer on the Hanzo agentic stack	15,294	$0.0006	588
Team A (contractor pod, conventional tooling)	2,112	$0.179	105
Team B (contractor pod, conventional tooling)	498	$0.560	10
Team C (contractor pod, conventional tooling)	1,415	$0.153	32

Table 1: The client's own engineering audit, anonymized. All contributors used the same frontier chat model; only the single engineer's work was multiplied by Hanzo's agentic stack. By the audit's own unit-cost metric, that engineer's cost per line of code was roughly 250× to 900× lower than the contractor pods', and that engineer alone shipped more commits than the three contractor pods combined.

Reading the numbers honestly. The audit is explicit that the raw lines-of-code figure for the agentic-stack engineer is inflated by framework-wide refactors, bulk dependency upgrades, and automation identities operating under the same attribution. We therefore do *not* headline raw lines authored; the defensible signals are (i) the *commit count*, which reflects discrete units of shipped, reviewed work; (ii) the *cost per line*; and (iii) the *relative ratio between groups*, since every group's lines were counted by the identical method, so whatever inflation exists applies uniformly and the 250×–900× unit-cost gap survives it. Even read at its most conservative, one engineer plus an agentic stack delivered a team's worth of audited, shipped output at a fraction of the cost per line.

Why the stack wins (the mechanism). The audit is one measured data point; the reason to expect it to reproduce is structural, not anecdotal. Interactive chat is human-paced and serial: a prompt is written, an answer is read, code is pasted, the loop turns once, and the engineer is the bottleneck on every turn. An agentic stack inverts the bottleneck—the model is wrapped in an autonomous build–test–ship loop that reads the repository, makes the change, runs the tests, reads the failures, and iterates, across many repositories at once, while the engineer supervises rather than types. Throughput then scales with the orchestration and the available compute, not with one person's typing speed. Three structural factors compound:

- **The model is one component, not the product.** The stack supplies the memory, tool use, sandboxes, retrieval, identity/secrets, and CI integration that turn a good answer into a merged, tested commit. Chat supplies only the answer; a human supplies the rest, by hand.
- **Parallelism over serialism.** Many agent loops run concurrently across the codebase; a chat session is one conversation at a time. The productivity gap is the gap between a fleet and a single operator.
- **Owned and unmetered.** Self-hosted on owned hardware, the loop runs at full throttle without a per-token meter deciding how much iteration is affordable—and iteration is exactly what produces tested, reviewed work.

The same frontier weights, accessed two ways, land in two different productivity regimes. The model is a commodity input; the agentic stack is the durable advantage—which is why the gap is a property of the *method*, reproducible by any team that adopts the stack, not a property of one exceptional engineer. The measured 250×–900× is the proven floor; the theoretical case says it travels.

What switching gets you. For an engineering organization the practical read is direct: adopting Hanzo's stack force-multiplies the engineers already on payroll—same people, same frontier model, now wrapped in an owned orchestration layer—so legacy modernization, regulated rebuilds, and net-new work all ship at a fraction of the cost per line, on infrastructure the organization owns and can audit. The switch is low-friction (the stack wraps the model access a team already has) and the gain is structural (it is the method, not the model). That is the case for moving from renting a chat window to owning an agentic stack.

What this is, in one line. This is the `hanzo.ai`-versus-`claude.ai` comparison made concrete. It is not "our model is smarter"—everyone had Claude. It is that an *owned agentic stack* turns one engineer into a team's worth of shipped, audited output, at a fraction of the cost per line. The same ownership that buys sovereignty in the rest of this paper—self-hostable weights, post-quantum security, data that never leaves the enclave—is what buys this productivity edge. You own the stack; you do not rent the chat. A buyer adopting Hanzo gets both halves of that sentence at once.

3 Dimension-by-Dimension

Dimension	Closed cloud AI (e.g. Claude)	Hanzo (sovereign OSS)
Model weights	Not disclosed; not transferable	Owned (Zen family, 0.8B–1T+); inspectable
Where inference runs	Vendor cloud only	Your hardware: data center, edge, air-gapped
Your data	Leaves the enclave to the vendor	Never leaves; stays where it lives
Disconnected / offline	Not possible (requires connectivity)	First-class (offline on embedded ARM)
Provenance	Vendor-controlled	American-owned, full provenance
Post-quantum security	Not a product property	Native (ML-KEM-768 + ML-DSA-65)
Auditability	Black box	Open stack; cryptographically signed actions
Customization	Prompting / limited fine-tune via API	Full fine-tune on your data, in your enclave
Cost model	Per-token metered	Owned + self-hosted; no markup or cloud margin
Lock-in	High (API + data gravity)	Low (you hold the stack)

Table 2: Closed cloud AI versus a sovereign self-hostable stack. The comparison is of deployment model, not of raw benchmark quality.

3.1 Data residency and the enclave boundary

The single most consequential difference is the enclave boundary. To use a closed cloud model, the data being reasoned over must cross the boundary to the vendor. For protected health information, customer financial records, regulated controlled data (CUI included), engineering files, or operational data this is often simply prohibited. Hanzo runs the model inside the boundary, so the data never crosses it. This is not a policy preference that can be waived with a contract clause; it is a property of where computation physically happens.

3.2 Disconnected and offline operation

A cloud service is unavailable the moment the link drops. On a remote industrial site, aboard a vessel, in an air-gapped facility, or at any field location where connectivity is intermittent or denied—disconnected defense operations included—there is no link to call. Hanzo's models are quantized to run fully offline on hardware ranging from a laptop to an embedded processor, so capability persists in exactly the environments where a cloud service cannot operate at all.

3.3 Provenance and the supply chain

For an open, controllable alternative to a closed cloud model, today's practical options are foreign open-weight models or Hanzo. Regulated enterprises and government buyers (defense among them) care about who produced the weights and whether they can be trusted and inspected. Hanzo's position is specific: an American-owned, open-source frontier stack with owned weights, so an organization can have open, auditable AI without depending on either the closed domestic duopoly or foreign-origin weights.

3.4 Post-quantum security

Security against a future quantum adversary is a property of the whole stack, not a feature of a chat API. Hanzo's transport, identity, and key management are post-quantum by construction (all three NIST standards in production, formally verified). A closed chat service does not expose this as something the customer controls.

3.5 Cost at scale

A metered API charges per token on every inference, forever, with the vendor's margin baked in. Owning the model and self-hosting removes both the per-token markup and the hyperscaler margin; combined with an efficient transport layer (the ZAP protocol's 91% memory and 96% energy reductions) this is the mechanism behind roughly an order-of-magnitude lower infrastructure cost at scale. A companion case study documents a regulated client whose cloud spend fell approximately 97% after migrating to this stack.

3.6 Agentic throughput, restated as cost

The productivity comparison in Section 2 is also a cost argument. Hanzo's founding CEO/CTO, Zach Kelling, was—by public usage tracking (`claudecount.com`)—the **#1 user of Anthropic by token count worldwide in 2025**; the audited output behind that throughput, one engineer outproducing entire contractor pods on the same frontier model, is documented above. The point here is what changed afterward: that same agentic throughput now runs on Hanzo's *owned*, self-hostable stack rather than a metered cloud API. The capability that topped a closed provider's global usage is exactly the capability Hanzo packages as a sovereign, owned alternative, so a buyer gets the productivity without the per-token meter, the data egress, or the dependency—frontier-grade agentic output, owned instead of rented.

4 When Each Model Fits

This is a decision framework, not a verdict. Closed cloud AI is the right choice when the workload is non-sensitive, connectivity is reliable, operational simplicity outweighs control, and there is no requirement to own or inspect the model. A sovereign self-hosted stack is the right—often the only—choice when any of the following hold: the data cannot leave a controlled environment; the system must work disconnected or in degraded conditions; provenance and auditability are required; post-quantum protection is mandated; or per-token cost at scale is untenable. Regulated industries, critical infrastructure, and government workloads—defense included—concentrate in the second column. That is the gap Hanzo is built to fill.

5 Conclusion

The honest comparison is not "Hanzo's model beats Claude." It is that a closed cloud model and a sovereign self-hostable stack are different categories, and the requirements of regulated, critical-infrastructure, and government workloads (defense among them)—data residency, disconnected operation, provenance, post-quantum security, auditability, and cost—select the second category by construction. Within that category the choice narrows to foreign open weights or an American open-source alternative. Hanzo is built to be that alternative: own the model, own the cryptography, run it where the data lives.

Agentic Engineering at Regulated Scale: Building a Post-Quantum, FHE-Native Securities Platform

Zach Kelling — Hanzo Team

Version 1.0 2026

Abstract

We document the engineering capability behind Hanzo's stack by way of its hardest production proof: an agent-driven team built a fully regulated securities venue—an SEC-registered alternative trading system (ATS) with broker-dealer and transfer-agent functions—on a next-generation foundation of production post-quantum cryptography, a high-performance native-GPU blockchain, and fully-homomorphic-encryption (FHE) privacy. The build *consolidated* rather than sprawled—200+ microservices collapsed into 3 compiled binaries, with 3.07M lines of code reduced to 1.06M active (∼3.5M lines of dead code deleted)—reaching production in record time with a small team, and passed FINRA/SEC approval and SOC 2 compliance. This is not template web software; it is among the most compliance-heavy and cryptographically advanced software domains that exists. We describe the throughput of the agentic engineering method that produced it, and we expand on the broad applicability of its FHE layer—computing on data one is not permitted to see—across regulated industries and government, defense included.

Executive Summary. The fastest way to judge an engineering organization is what it has shipped under real constraints. Hanzo's founding CTO, Zach Kelling, was ranked the **#1 user of Anthropic by token count worldwide in 2025** (per public usage tracking at `claudecount.com`)—a marker of an engineer operating agentic AI at the frontier of throughput. That throughput is not spent on brochure sites: the flagship production system built with Hanzo's agentic stack is a **fully regulated securities exchange**—an SEC-registered ATS with broker-dealer and transfer-agent functions—running on **post-quantum cryptography**, a **native-GPU high-performance blockchain**, and **fully homomorphic encryption** for privacy. The build *consolidated* rather than sprawled—200+ microservices collapsed into 3 compiled binaries, 3.07M lines of code cut to 1.06M active—shipped fast with a small team, and cleared FINRA/SEC approval and SOC 2. Regulated financial infrastructure is one of the hardest, most audited software domains there is; clearing it on a cryptographically advanced foundation is the proof that agent-driven engineering is production-grade, not a demo—and that it is exactly the capability needed to modernize the software stacks of regulated enterprises and government alike, defense among them.

Contents

1 The Engineer and the Throughput

Agentic engineering multiplies a strong engineer; it does not replace one. Hanzo's founding CTO works at the frontier of both AI/ML and cryptography—the two disciplines this stack fuses—and operates agentic AI at a volume that placed him, by public token-count tracking (`claudecount.com`), as the top Anthropic user in the world in 2025. The relevant point is not the ranking for its own sake; it is what that volume was spent producing. Raw AI usage that yields disposable prototypes proves nothing. Raw AI usage that yields a regulated, audited, cryptographically advanced production system proves the method works where it is hardest.

2 What Was Built

The flagship system is a regulated securities venue, not a content site. In U.S. securities terms it spans three regulated functions that ordinarily take separate firms years to stand up:

- **Alternative Trading System (ATS)** — an SEC-registered trading venue (Regulation ATS), matching and executing orders.
- **Broker-Dealer (BD)** — FINRA-regulated, with the supervisory, recordkeeping, and capital obligations that entails.
- **Transfer Agent (TA)** — SEC-registered, maintaining the authoritative record of securities ownership and transfers.

These functions are governed by some of the most exacting compliance regimes in software. The notable engineering fact is that the build *consolidated* rather than sprawled: a legacy estate of **200+ microservices** on a public cloud was collapsed into **3 compiled binaries**, and the codebase went from **3.07M lines to 1.06M active** (∼70% smaller, with ∼3.5M lines of dead code deleted; the trading/auth core itself shrank from ∼264,000 to ∼44,000 lines, with 1.3 GB of vendored dependencies eliminated and a single 48 MB binary subsuming 17+ formerly separate services). It reached production **in record time**, with a **small team**, and was carried through **FINRA/SEC approval** and **SOC 2** compliance—leaner than the system it replaced, not larger.

3 One Engineer, a Team's Output

The result that matters most is the ratio of output to headcount. The agentic method let a *single* AI-augmented engineer carry the bulk of a regulated production platform that a conventional organization staffs as a full team—and the platform's own engineering audit documents it. Across an **81-day** window spanning **334 repositories** and **31,058 commits**, the audit attributed **15,294 commits** and **114M raw git-log lines**—framework-wide refactors, dependency upgrades, automation, and production code—to the agentic CTO workflow, measured at **$0.0006 per line** by the audit's methodology. Within that, a focused **10-day remediation sprint** produced **3,694 commits**, removed **∼3.5 million lines of dead code** across 181 repositories, and authored **100% of the critical- and high-severity security fixes**—single-handedly—while consolidating 200+ microservices into three binaries. The security and compliance gaps were closed on the path to approval (a source audit of **93 findings**, 22 critical, plus **780+ dependency vulnerabilities** remediated to zero actionable), through FINRA/SEC approval and SOC 2.

The economic consequence is direct and measured. When one engineer, multiplied by the agentic stack, produces what a team produces, the **cost per line of shipped, audited, production code**

collapses—here, **$0.0006 per line** by the audit's own methodology, orders of magnitude below conventional team rates. For any regulated enterprise or government program (defense included) the implication is the whole argument: the same regulated, post-quantum, FHE-native system, delivered by a fraction of the headcount at a fraction of the cost—and owned, not rented. This is what "agentic engineering" means in practice—not a chatbot writing snippets, but one expert operating at the throughput of a department.

4 The Foundation: Post-Quantum, GPU Blockchain, FHE

What makes the build a genuine engineering flex is the foundation it runs on— each element of which is independently hard:

- **Production post-quantum cryptography.** All three NIST PQC standards (ML-KEM/203, ML-DSA/204, SLH-DSA/205) in a live system with hybrid classical/quantum security—so the chain of custody for securities records is protected against a future quantum adversary.
- **Native-GPU high-performance blockchain.** A chain serving as the authoritative, immutable, post-quantum-anchored record of transfers, with GPU-accelerated cryptographic operations for throughput.
- **Fully homomorphic encryption (FHE).** GPU-accelerated FHE (BFV/BGV/CKKS/TFHE) with threshold decryption, so privacy-sensitive computation runs *on ciphertext*—no single component ever holds both the plaintext and the result.

A team that can compose post-quantum cryptography, a GPU blockchain, and FHE into a system that clears securities regulators is a team that can modernize a high-assurance software stack—in healthcare, finance, critical infrastructure, or government, defense included—to the same foundation.

5 FHE for High-Stakes Computation: Computing on Data You Cannot See

Fully homomorphic encryption is the capability with the broadest reach across high-stakes domains, so it is worth drawing out. FHE allows arbitrary computation directly on encrypted data: the operator performs the computation and returns an encrypted result without ever decrypting the inputs, and only the data owner (or a threshold of keyholders) can decrypt the output. The system doing the work never sees the plaintext. In the securities platform this protects order and ownership data; the same primitive answers a range of problems—in regulated industry, multi-party commerce, and government (defense included)—that have no clean classical solution:

- **Private healthcare and population analytics.** Hospitals and researchers can compute statistics, risk scores, or model inferences over patient records while the records themselves stay encrypted—enabling collaboration on protected health information without exposing it.
- **Multi-party financial computation.** Banks, exchanges, and counterparties can compute a joint result—netting, exposure, fraud signals—over their combined data while each party's books stay encrypted and private, so no participant has to surrender its raw positions to get the answer.

- **Confidential supply-chain analytics.** Suppliers and buyers can reconcile inventory, demand, or provenance across organizations without any party revealing its underlying commercial data.
- **Computation on untrusted infrastructure.** Sensitive workloads can run on shared or third-party cloud infrastructure without ever exposing the data to the host—widening where a regulated workload may legally run.
- **Encrypted ML inference.** A model can classify encrypted images, documents, or signals: the data owner never exposes the input, and the model operator never sees it—useful when the data is sensitive, the model is sensitive, or both.
- **Cross-organization data fusion without disclosure.** Multiple parties can correlate multi-source data while each source stays encrypted, so a breach of the fusion system does not expose the underlying inputs. In a defense setting this is the cross-domain and multi-party (coalition) case—a guard evaluates a release policy against encrypted content, or allies compute a joint result, without ever seeing the data (developed in the cross-domain paper).

FHE's historical objection is speed; the practical answer in this stack is GPU acceleration of the underlying number-theoretic transforms, plus threshold decryption so no single party holds the key. "Compute on data you are not allowed to see" is a capability high-stakes domains have wanted for decades—defense among them; here it is a production component, not a research slide.

6 Why Open Source Was the Accelerant

The obvious question is how a small team shipped a regulated, cryptographically advanced system of this scale so quickly. The answer is open source—specifically, an *owned* open-source stack composed by an agentic engineering method.

Hanzo did not assemble the platform from a patchwork of third-party SaaS. It composed it from coherent open-source building blocks that Hanzo itself authors and controls—identity and access management, key management, the data layer, the gateway and ingress, the ZAP transport, the post-quantum and FHE cryptography, the operator, and the Zen models—each reusable, inspectable, and owned. The agentic stack is the multiplier; the owned OSS foundation is the leverage. You do not rebuild identity, key management, or a gateway for each new system; you compose owned, hardened components and point the agents at the application on top. That is how a regulated platform of this scope reaches production fast—and lean.

For any high-assurance buyer, open source inverts the usual security argument. A closed system asks you to trust a vendor you cannot inspect; an open one lets you read every line, verify it—here, with machine-checked proofs—confirm there are no hidden backdoors, and run it on hardware you control. Auditable beats opaque whenever the stakes are high. Open source is therefore not a cost-cutting compromise: it is simultaneously the source of **velocity** (compose, do not rebuild), of **security** (inspectable and formally verifiable), and of **sovereignty** (own it, modify it, run it anywhere, including air-gapped). The three properties a regulated or government program most wants—defense included—are the same three that made the build fast.

Stated plainly as a hook: *Hanzo builds massive, secure systems quickly because it builds on an open-source stack it owns—so the speed, the security, and the sovereignty all come from the same source.*

7 Why It Matters for Regulated and Government Modernization

The throughput and the difficulty compound. The same agentic engineering method that shipped a regulated, post-quantum, FHE-native securities exchange fast and small is the method Hanzo brings to any regulated-enterprise or government modernization—healthcare, finance, critical infrastructure, and defense alike: take an existing, audited (and, for government systems, already-accredited) stack and modernize it in place—to a sovereign, post-quantum, self-hostable footprint with FHE-grade privacy—faster and cheaper than a traditional rebuild, and owned by the operator. The securities venue is the existence proof that the method holds up under the most demanding regulatory and cryptographic constraints production software offers; every other high-assurance modernization—defense among them—is the same engine pointed at a new mission.

Cloud Economics of a Sovereign OSS Stack: A Regulated Capital-Markets Migration Case Study

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Abstract

We document the infrastructure economics of migrating a regulated capital-markets platform from a sprawling multi-project Google Cloud (GCP) footprint to a consolidated, self-hostable open-source stack built with Hanzo's agentic engineering system. The client—a FINRA/SEC-regulated securities platform, anonymized here—was spending, by *actual billing* once a startup-credit program expired, approximately $35,158/month (≈ $421,900/year) across eleven billing accounts, the great majority consumed by legacy GKE clusters and GPU virtual machines left over from an earlier architecture. Consolidation onto the new stack and owned hardware reduced spend to approximately $3,500/month (≈ $42,000/year)—a ∼90% reduction, on the order of $380,000/year saved—and removed the dependency on a single cloud vendor. The same agentic stack also *built* the platform: a full SEC-registered ATS with broker-dealer and transfer-agent functions, consolidated from 200+ microservices and 3,071,930 active lines of a prior implementation down to 1,056,566 active lines (≈70% less code; ∼3.5 M lines of dead code deleted) running as 3 compiled Go binaries plus a small set of OSS platform services, carried through FINRA/SEC approval and SOC 2. We decompose where the savings come from, report the figures with their provenance, and argue the same mechanism applies directly to any regulated enterprise or government modernization—defense among them— carrying a metered-API-plus-hyperscaler-sprawl cost structure.

Executive Summary. A regulated financial client appeared to be spending only a modest amount on Google Cloud because a startup-credit program was absorbing the bill. When those credits expired, the *actual* cost surfaced: about $35,158 a month—roughly $926 a day of newly cash-billed spend—almost all of it on old GKE clusters and GPU machines the new architecture had made unnecessary. Migrating to a consolidated, self-hostable open-source stack on owned hardware cut that bill to about $3,500 a month (≈90%, on the order of $380,000 a year), and removed the dependency on a single cloud vendor; ongoing consolidation targets a further floor near $1,200/month (nearly 97 percent off the original bill) as the last legacy projects are retired. Just as important is *how little code* now runs that platform: the agentic engineering stack collapsed 200+ legacy microservices into 3 compiled Go binaries plus a small set of OSS platform services, cutting the active codebase from ∼3.07 M to ∼1.06 M lines (deleting roughly 3.5 M lines of dead code and eliminating 1.3 GB of vendored dependencies)—and still cleared the bar of a heavily regulated, audited domain (FINRA/SEC approval, SOC 2). For any organization carrying a similar cost structure—a healthcare provider, a financial firm, a critical-infrastructure operator, or a government contractor—the lesson is concrete: the same approach modernizes an existing stack to a sovereign, self-hosted footprint that is far cheaper to run, smaller to maintain, and faster to build.

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1 Before: A Sprawling, Credit-Masked Footprint

The starting state is typical of an organization that grew fast on Google Cloud under promotional credits. Headline spend looked small because a startup-credit program was absorbing it; an audit of the *actual* GCP billing console—after the credits expired—put true spend at approximately **$35,158/month** (≈$421,900/year), concentrated in legacy infrastructure rather than the live product. When the credits lapsed, roughly **$27,783/month** of previously credit-masked spend—about $926/day—flipped from a hidden line to cash. The footprint spanned **185 GCP projects** (27 billing-enabled, only ∼4 actually needed) across **11 billing accounts**:

Billing account	**$/month**	**$/year**
Legacy / startup-credit account (the waste)	27,783	333,396
Production account A	4,040	48,480
Secondary product group	3,087	37,044
New-stack non-prod (already efficient)	248	2,976
Total (actual billing)	**∼35,158**	**∼421,900**

Table 1: Actual monthly GCP spend by billing account, taken from the billing console after startup credits expired (11 billing accounts in total; the four material ones shown). A single legacy account accounted for ∼79% of the total; the current-generation stack was already a rounding error.

The legacy spend bought capacity the new design no longer needed: more than **125 GKE nodes across 20 clusters** (single clusters of 19, 17, and 14 nodes; 64 of those nodes were 18 GB machines), two standing **GPU virtual machines** (an A100-class instance at ∼$2,900/month and an L4-class instance at ∼$800/month, ∼$3,700/month together), **5 Cloud SQL PostgreSQL instances**, and **∼18.5 TB** of persistent disk—several terminated VMs were even still paying for idle attached storage. Two problems compound here: **cost** (most of the spend bought unused capacity) and **sovereignty** (the entire footprint lived inside one hyperscaler—GKE, Cloud SQL, BigQuery, GCS, Artifact Registry, Cloud Build—with the data gravity and lock-in that implies).

2 The Migration: Two Documented Steps

The migration consolidated the workload onto Hanzo's open-source stack—the same self-hostable components (identity, key management, data, gateway, agent runtime) that run sovereign and disconnected—and onto owned hardware for the compute-heavy services. The savings landed in two measured steps:

Decommissioning was clean precisely because the legacy capacity was unused: shutting down the two GPU VMs removed ∼$3,700/month on its own; the largest legacy GKE clusters and dozens of dead projects accounted for most of the rest. The net effect is a reduction from ∼$35,158 to ∼$3,500 per month (≈90%) and, equally important, the elimination of single-vendor lock-in: the resulting stack runs on owned hardware, in another cloud, or air-gapped.

3 Where the Savings Come From

The reduction is not a one-time clean-up; it is structural, and it reproduces:

- **No metered model markup.** Owned models (the Zen family) replace per-token API calls, removing a recurring per-inference charge.

State	$/month	vs. start
Actual billing (credits expired)	∼35,158	—
Step 1 — retire legacy: stop 2 GPU VMs (−$3,700), delete legacy GKE clusters, close dead projects	∼6,489	−82%
Step 2 — consolidate & right-size onto the owned stack and owned hardware	∼3,500	−90%
Target — finish consolidation (retire last legacy projects)	∼1,200	≈−97 pts

Table 2: Reduction to date. Step 1 (retiring unused legacy) alone removed ∼82% of spend with no impact on the live product; Step 2 consolidated the remaining workload onto the self-hostable stack, landing at ∼$3,500/month (≈$380,000/year saved). The ∼$1,200/month line is a forward *target* as the last legacy projects are retired—not yet realized.

- **No hyperscaler margin.** Self-hosting on owned hardware removes the cloud provider's margin on compute, storage, and egress.
- **Efficient transport.** The ZAP protocol's 91% memory and 96% energy reductions and 30× throughput mean the same work needs far less hardware (see the ZAP paper).
- **Consolidation.** One coherent stack replaces a sprawl of projects, clusters, and services—the single largest line item above was pure redundant overhead.

Together these are the mechanism behind the roughly order-of-magnitude infrastructure savings, and they apply to any organization carrying a metered-API plus hyperscaler-sprawl cost structure.

4 The Build: Consolidation, Not Volume

The economics of *running* the platform are only half the story; the economics of *building* it are the other half—and here the measured story is *doing more with dramatically less code*. The legacy estate was a sprawl of **200+ microservices** (a single legacy auth monolith alone fronted 391 route handlers; ∼80+ distinct deployable services, 49 of them active Node.js processes at the time of migration). The same agentic engineering stack that now runs the platform also built it, and the artifacts are small: **3 compiled Go binaries** (ATS / broker-dealer / transfer-agent) alongside ∼10 OSS platform services and 5 validator nodes. Measured git-over-git, the active codebase fell from **3,071,930** to **1,056,566** lines (≈70% less; roughly 3.5 M lines of dead code deleted):

This inverts the usual "big system = big codebase" intuition: agent-driven engineering produced a *leaner* system than the one it replaced, which is why it is cheaper to run (fewer moving parts), cheaper to maintain (less code), and faster to deploy (<2 min rolling deploys, versus ∼1 hr manual or 15–30 min canary before). The memory story is the same shape: the legacy fleet held ∼1 TB+ of *provisioned* cluster RAM (64 nodes × 18 GB) against an application working set of roughly 10 GB; the consolidated Go binary holds its working set in ∼50 MB (≈99.5% less), and cold start dropped from ∼10 s per Node.js service to <1 s. On security, an internal source audit found **93 findings** across 127+ repositories (22 critical, 30 high, 32 medium, 9 low) plus ∼780+ npm dependency vulnerabilities; the consolidated rebuild remediated that entire class to *zero actionable* at the architecture level (header hardening, strict CORS allowlisting, JWKS-verified JWTs, secrets moved out of plaintext environment, and the 1.3 GB vendored dependency tree—the source of most of the npm vulns—eliminated outright).

Measure	Legacy	New stack	Change
Deployable services	200+	3 Go binaries	∼60× fewer
Active lines of code (whole estate)	3,071,930	1,056,566	≈70% less
Lines of code (trading/auth core)	∼263,803	∼44,308	∼6× less
Vendored dependencies (flagship svc)	1.3 GB	0	eliminated
Provisioned cluster RAM	∼1 TB+	∼50 MB working set	∼99.5% less
Deploy artifact (flagship)	—	48 MB binary	replaces 17+ services
Deploy time	∼1 hr manual	<2 min rolling	—
Cold start (per service)	∼10 s	<1 s	∼10× faster
Automated tests (new stack)	n/a tracked	700+ Go tests	—

Table 3: Measured engineering footprint, new stack vs. the legacy implementation it replaced, git-verified. The whole-estate line count fell from ∼3.07 M to ∼1.06 M active lines; the trading/auth core specifically went ∼6× smaller. A single 48 MB Go binary subsumes 17+ formerly separate services.

5 The Compliance Bar

The load-bearing proof point is that this lean, cheap-to-run system is not a prototype: it is a regulated securities venue—an SEC-registered alternative trading system with broker-dealer and transfer-agent functions—carried through **FINRA/SEC approval** and **SOC 2** compliance, in record time, with a **small team**. Agent-driven engineering shipped and passed regulatory scrutiny in financial services, among the most compliance-heavy software environments that exists.

6 Implications for Regulated and Government Modernization

The same motion transfers directly to any regulated-enterprise or government modernization—a healthcare system, a bank, a utility, or a government contractor (defense included). An organization with an existing, audited (and, for government systems, already-accredited) software stack does not need to rebuild from zero; the agentic stack can modernize that stack in place—to a post-quantum, sovereign, self-hostable footprint—while the consolidation cuts recurring infrastructure cost by an order of magnitude, shrinks the codebase that must be maintained (and, where applicable, accredited), and removes single-vendor dependency. The wedge is low-friction (modernize what you already have, with proofs), the savings are immediate and structural, and the result is software the operator owns and can run where the data lives. A regulated financial platform is the existence proof; the same pattern applies anywhere the cost structure and the compliance burden repeat.

A note on figures. Cost figures are drawn from the client's actual GCP billing-console data after promotional credits expired (a more defensible basis than list-price estimates); infrastructure, code, and security figures are drawn from internal infrastructure, code, and source-security audits. Line counts, service counts, and dependency sizes are measured (git-verified) where stated. Where a number is a forward target rather than a realized result (e.g. the ∼$1,200/ month consolidation floor) it is labeled as such; where a source labels a number projected or self-reported, we have not relied on it here. All identifiers are removed; the client is not named.

Confidential Computing by Default: An FHE-Native Chain and Agent Runtime

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Abstract

Every system that processes sensitive data has a moment where it holds the data in the clear—and that moment is the insider risk, the cross-domain leak, and the blast radius of every breach. Fully homomorphic encryption (FHE) removes that moment: computation runs directly on ciphertext, and only a threshold of keyholders can decrypt the result. This paper describes Hanzo's design for making FHE the *default* mode of computation across the two layers that normally demand plaintext—a blockchain's state and rules, and an autonomous agent's actions—so that state stays encrypted on-chain, policies are evaluated over ciphertext, and agents act on data they are never able to read. We give the cryptographic basis (BFV/BGV/CKKS/TFHE with threshold decryption), the GPU acceleration that makes it practical, the post-quantum anchoring that protects it for the long term, the applications it unlocks across regulated industry, multi-party commerce, and government (defense included), and an explicit, honest assessment of which components are production today versus an active research frontier.

Executive Summary. Normally, to compute on data you must first decrypt it—so somewhere in every system there is a component holding the secret in the clear, and that component is what an adversary, an insider, or a subpoena goes after. FHE lets the computer do the work *while the data stays encrypted*: it takes in ciphertext, produces a ciphertext answer, and never sees the plaintext. Hanzo pushes this idea through the whole stack. The blockchain stores encrypted state and can enforce rules on encrypted data; agents can act on encrypted data and return encrypted answers; and only a quorum of keyholders can ever decrypt—no single party holds the key. This is directly useful wherever the stakes are high: hospitals can compute over patient records without exposing them, financial counterparties can compute a joint result without sharing their books, and sensitive workloads can run on infrastructure you do not fully control without exposing the data to the host. The same shape serves government and defense—allies can compute a joint answer without sharing raw data, and a cross-domain guard can decide whether something is releasable without seeing its content. FHE's historical objection is speed; the practical answer here is GPU acceleration plus running only the sensitive part of a computation under encryption. The result is a stack where confidentiality is the default state of data in use, not just data at rest or in transit.

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1 The Plaintext Gap

We protect data in two of its three states well and the third state badly. Data *at rest* is encrypted on disk; data *in transit* is encrypted on the wire. But data *in use*—the moment a program actually computes on it—is almost always plaintext in memory. Every database query that filters records, every policy engine that decides access, every model that runs inference, every smart contract that checks a balance, does so on cleartext. That cleartext moment is the structural weakness behind a large share of real incidents: the compromised process, the malicious insider, the memory-scraping implant, the cross-domain guard that has to read the data to rule on it, the cloud host that can see its tenants' workloads.

Confidential computing has tried to shrink this gap with hardware enclaves (trusted execution environments). Enclaves help, but they relocate trust rather than remove it: you must trust the silicon vendor, the attestation chain, and the absence of side channels, and the data is still plaintext *inside* the enclave. Fully homomorphic encryption attacks the gap directly. With FHE the data is never decrypted to be computed on at all—not in the enclave, not in memory, nowhere. The computation is performed on ciphertext and yields a ciphertext result. The plaintext gap closes.

2 FHE in One Page

Fully homomorphic encryption is an encryption scheme with an extra property: there exist operations on ciphertexts that correspond to operations on the underlying plaintexts. Concretely, given encryptions of a and b, anyone—without the key—can compute an encryption of $a + b$ and of $a \times b$. Addition and multiplication are a complete basis, so in principle any computable function can be evaluated homomorphically by expressing it as an arithmetic (or boolean) circuit. The party doing the computation learns nothing; only a holder of the secret key can decrypt the output.

The scheme families and what each is for.

- **BFV / BGV** — exact integer arithmetic. Right for counts, sums, comparisons, exact matching, and policy predicates over integers.
- **CKKS** — approximate arithmetic over real/complex numbers, with SIMD "batching" that packs thousands of values into one ciphertext. Right for statistics, signal processing, and machine-learning inference.
- **TFHE** — fast boolean gates and programmable bootstrapping. Right for branching, comparisons, and arbitrary logic, including data-dependent control flow.

Bootstrapping. Each homomorphic operation adds noise; after enough operations the noise would corrupt the result. *Bootstrapping* refreshes a ciphertext to a low-noise state and is what makes encryption "fully" homomorphic—it permits unbounded computation depth. Bootstrapping is the expensive step, and it is the principal target of the acceleration discussed in §7.

Threshold decryption. The secret key need not exist in one place. With *threshold* FHE the key is split across n parties so that any t of them can jointly decrypt but no coalition smaller than t can. Combined with FHE this gives a system in which *no single party ever holds both the data and the ability to reveal it*—the computation runs blind, and revealing the answer requires a quorum. This property is what makes the design defensible for multi-party and adversarial settings.

3 Confidential State: An FHE-Native Chain

A blockchain is normally the *opposite* of confidential: every validator re-executes every transaction against shared plaintext state, which is precisely why public chains leak everything. An FHE-native chain inverts this. State is stored as ciphertext; transactions carry encrypted inputs; and the state-transition rules are evaluated homomorphically, so validators advance the ledger *without ever seeing the values they are agreeing on.*

- **Encrypted state.** Balances, ownership records, order books, classifications—whatever the application holds—live on-chain as ciphertext. The authoritative record exists, is immutable and auditable, and is unreadable to anyone without a decryption quorum.
- **Rules on ciphertext.** A transfer rule such as "permit only if the sender's balance covers the amount" is a comparison; under BFV/TFHE that comparison is evaluated on the encrypted balance and encrypted amount, yielding an encrypted yes/no that gates the state update. The rule is enforced without the rule-evaluator learning the balance.
- **Threshold-gated disclosure.** When a value must be revealed—a settlement, an audit, a court order, a cross-organization release decision—a threshold of keyholders cooperates to decrypt exactly that value, and the decryption itself is recorded. Disclosure becomes an explicit, quorum-authorized, logged event rather than a default.
- **Post-quantum anchoring.** The chain's signatures and transport are post-quantum (ML-KEM/ML-DSA), so the encrypted record and its provenance survive a future quantum adversary—essential because ciphertext recorded immutably today is exactly the "harvest now, decrypt later" target (see the full-spectrum cyber and ZAP papers).

The chain thus provides a tamper-evident, post-quantum, immutable audit trail *of encrypted facts*: you can prove what happened and in what order without ever exposing what the facts were, until a quorum decides otherwise.

4 Confidential Action: FHE-Native Agents

The second layer that normally demands plaintext is automation. An autonomous agent that triages intelligence, screens transactions, routes logistics, or enforces a policy has to "see" the data to act on it—and so the agent runtime becomes another plaintext concentration point, now with the additional risk that the model itself might memorize or exfiltrate. FHE breaks this too.

An FHE-native agent operates on encrypted inputs and emits encrypted outputs. It can evaluate the encrypted state of the chain, apply encrypted rules, and produce an encrypted decision or an encrypted action—all without the agent operator (or the model host) ever seeing the plaintext. Where the agent must run a learned model rather than a fixed rule, CKKS-based encrypted inference lets a (suitably structured) model classify or score encrypted inputs and return an encrypted result. The agent is then *operator-blind*: it does useful work on data it cannot read, on infrastructure that cannot read it either.

This composes with Hanzo's broader agentic runtime (the agentic-AI and edge papers): the same orchestration of specialized agents and tool calls runs, but the data plane underneath it is ciphertext. Autonomy and confidentiality stop being a trade-off.

5 Programmable Confidentiality: Rules and Policies on Ciphertext

The unifying idea is that a *rule* is just a function, and any function can in principle be evaluated under FHE. So the policies that govern a system—not only its data—can run on ciphertext. This is the capability the rest of the paper builds on, because most high-consequence decisions are policy decisions over sensitive inputs:

- **Releasability** — "may this datum cross to that domain / that external partner?" decided on the encrypted datum (a cross-domain or coalition guard is the defense instance; see cross-domain paper).
- **Eligibility / authorization** — "does this party satisfy the conditions?" decided without revealing the party's attributes.
- **Thresholds and limits** — "is the aggregate over/under the line?" decided without revealing the individual contributions.
- **Matching** — "do these two sealed values correspond?" (sealed-bid auctions, private set intersection, deconfliction) decided without opening either side.

In each case the *decision* is computed and enforced while the *inputs* remain encrypted, and only the (small, controlled) decision output is ever a candidate for threshold decryption. Confidentiality becomes programmable: you write the rule once and it runs blind.

6 High-Stakes Applications

The combination of an encrypted ledger and operator-blind agents answers a set of problems—across regulated industry, multi-party commerce, and government (defense included)—that have no clean classical solution:

- **Private healthcare and population analytics.** Hospitals and researchers compute statistics, risk scores, or model inferences over patient records while the records stay encrypted, so collaboration on protected health information never requires exposing it.
- **Multi-party computation without disclosure.** Independent parties compute a joint result—netting and exposure between financial counterparties, overlapping indicators between organizations, combined logistics between partners—over their combined data while each party's raw data stays encrypted. Nobody surrenders sources or proprietary data to get the joint answer, and the chain records that the computation occurred. (In a defense setting this is coalition analytics among allies.)
- **Releasability on ciphertext.** A guard evaluates a release policy against encrypted content and decides whether it may move between domains without ever seeing the content, removing the guard itself as a data tap—the pattern behind both commercial data-sharing controls and a defense cross-domain guard.
- **Private data fusion.** Multi-source correlation runs while the sources stay encrypted, so a breach of the fusion system does not expose the underlying inputs—whether those are commercial datasets or intelligence collection.

- **Encrypted ML inference.** A model classifies encrypted images, documents, or signals: the data owner never exposes the input and the model operator never sees it—useful when the data is sensitive, the model is sensitive, or both.
- **Encrypted operational state.** Records, work lists, and movement tables—an order book, a supply chain, or, in a defense setting, command, targeting, and logistics state—live as encrypted on-chain state with an immutable, post-quantum audit trail; agents act on them under encryption; reveal is quorum-gated and logged.
- **Computation on untrusted infrastructure.** Sensitive workloads run on third-party, shared, or commercial-cloud hardware without exposing the data to the host—widening where a regulated workload can legally compute.

7 Why the Speed Objection No Longer Holds

FHE's reputation for impracticality is a decade out of date, and three things close the gap in this stack:

1. **GPU acceleration.** The dominant cost in FHE is the number-theoretic transform (NTT) at the heart of polynomial multiplication and bootstrapping. These are massively parallel and map naturally onto GPUs; the same native-GPU foundation that accelerates the chain's cryptography accelerates the FHE layer, turning bootstrapping from seconds toward the practical range.
2. **Batching.** CKKS and BGV pack thousands of plaintext slots into one ciphertext, so a single homomorphic operation does thousands of useful operations at once—ideal for the vector and tensor workloads in analytics and inference.
3. **Selective FHE.** Most computations have a small sensitive core inside a large non-sensitive shell. You do not run the whole program under FHE—only the predicate or the field that must stay private. The expensive primitive is spent precisely where it buys confidentiality and nowhere else.

The honest framing is not "FHE is now as fast as plaintext"—it is not—but "the workloads that matter for confidential rules, matching, gating, and bounded inference are now practical, and GPU acceleration keeps widening the set."

8 Key Custody Without a Single Point of Trust

A confidentiality system is only as strong as its key handling. The design holds the decryption key nowhere: threshold decryption splits it t-of-n across independent parties (validators, regulators, partner organizations, or coalition members) so that decrypting any value requires an explicit quorum and no minority—and no single compromised host, insider, or seized server—can reveal anything. This is the structural answer to "who can see the data?": by construction, *nobody*, until enough authorized parties agree (validators, agencies, partner organizations, or coalition members), and that agreement is itself recorded on the chain.

9 Composition With the Rest of the Stack

FHE here is not a bolt-on library; it is one primitive in a stack Hanzo owns end-to-end, which is what makes the "native" claim real:

- **Native-GPU chain** provides the encrypted, immutable, post-quantum ledger and the GPU substrate the FHE layer rides on.
- **Post-quantum transport and identity** (ZAP, ML-KEM/ML-DSA) protect the ciphertext in motion and bind it to verifiable identities.
- **Agentic runtime** supplies the operator-blind automation that acts on the encrypted state.
- **Owned models** (the Zen family) supply the inference that, under CKKS, can run on encrypted inputs.

Because all four are owned and open, the confidentiality guarantee is auditable rather than asserted: any high-assurance customer—a regulated enterprise, a government agency, or a defense program—can inspect the whole path the data takes and confirm there is no point where it is silently in the clear.

10 Honest Maturity Assessment

This is a capability paper, and the credible version of it is explicit about what runs today versus what is an active frontier. No overstatement:

- **Production-grade today:** the GPU-accelerated FHE primitives (BFV/BGV/CKKS/TFHE), threshold decryption, the native-GPU post-quantum chain, and the agentic runtime each exist and run. Bounded-depth confidential workloads —rule and policy predicates, comparisons, matching/deconfliction, aggregate thresholds, and inference on suitably structured models—are practical now.
- **Active frontier:** fully general, arbitrary-depth FHE-native programmability at plaintext-class throughput is not solved—by anyone—and we do not claim it. The performance envelope is widening (faster bootstrapping, better GPU kernels, hybrid MPC/FHE designs), and the architecture is built so that workloads move from "frontier" to "practical" as that envelope grows, without redesign.

The strategic point stands regardless of where the line currently sits: Hanzo owns every layer the line runs through, so as FHE performance improves the gains land in a stack a customer already controls—rather than waiting on a closed vendor to expose the capability.

11 Conclusion

Confidentiality of data *in use* is the unsolved third of the problem, and it is the one that matters most for multi-party computation, private analytics, cross-organization data sharing, and computing on infrastructure you do not own—and, in government and defense, for coalition operations, cross-domain movement, and intelligence fusion. FHE closes the plaintext gap by computing on ciphertext; threshold decryption removes the single point of trust; GPU acceleration and selective application

make the important workloads practical; and post-quantum anchoring protects the encrypted record for the long term. Hanzo's contribution is to make this the default across a chain and an agent runtime it owns end-to-end, so that "private ciphertext for every operation" is an architectural property of the stack rather than a feature of one application. Compute on data you are not allowed to see—and let the rules, the ledger, and the agents all run blind.

Hanzo AI for Security: Automated, Agentic Security Auditing at Estate Scale

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Abstract

Security debt does not live in the system you were asked to look at; it lives in the systems you did not know existed. Modern software estates are sprawls of hundreds of microservices, dozens of repositories, and dependency trees no human has ever read end to end—and the vulnerabilities hide precisely in that sprawl: the forgotten service with a wildcard CORS policy, the credential committed to a `.env` four years ago, the auth bypass list that a database write can edit, the package with a published CVE that no audit ever ran against. This paper describes how Hanzo applies an agentic AI stack to the problem: a fleet of specialized review agents that fan out across an entire estate, discover its real extent, find vulnerabilities at a rate no human team can match, scan every dependency tree, and re-audit continuously as code changes—always with a human in the loop for triage and sign-off. We ground the description in a real engagement (anonymized): a FINRA/SEC-regulated securities platform that engaged for a narrow remediation scope and was found to be running more than 200 microservices across 127+ repositories. Across eleven audit rounds the stack surfaced 143 findings (22 Critical), the estate's ∼780 dependency vulnerabilities were driven to zero actionable, and the platform was consolidated from 200+ services into three Go binaries—deleting roughly 3.5 million lines of dead code and, with it, most of the attack surface. The outcome was FINRA/SEC approval and SOC 2. We are explicit throughout about what the machine does versus what a human must still decide.

Executive Summary. You cannot secure what you have not enumerated, and the hard part of real-world security is that nobody has enumerated the estate. A team asks for help hardening a handful of services and the actual footprint turns out to be hundreds of services across scores of repositories, most of them undocumented, several of them forgotten, all of them in scope for the regulator and the attacker even if they were out of scope for the conversation. Hanzo's answer is to point an agentic AI stack at the whole thing. Specialized agents—one reasoning about authentication, one about secrets, one about injection, one about access control, one about dependencies—run in parallel across every repository, build the map nobody had, and report findings far faster than a human auditor can read code. The machine is good at breadth, recall, and tirelessness; it is not the final authority. A human engineer triages every finding, discards the false positives, decides severity and exploitability, and signs off remediation. In the engagement this paper anonymizes, that pairing took a securities platform from 93 source-code findings (22 Critical, 30 High) across 127+ repositories and ∼780 dependency vulnerabilities (18 critical, 350+ high) to zero actionable dependency vulnerabilities and zero open critical or high findings over eleven rounds—while consolidating 200+ microservices into three compiled binaries and deleting ∼3.5 million lines of dead code, a reduction in attack surface that no amount of patching can equal. The same properties that made this possible—owned models, an open and auditable stack, secrets in a KMS, post-quantum identity, and a cryptographically signed, immutable audit trail—are exactly the properties a SOC 2 audit, a FINRA/SEC examination, an ISO 27001 certification, and (for government systems) a continuous-authorization posture all require.

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1 The Problem: Security Debt Hides in the Sprawl

A security review has an implicit assumption baked into it: that someone knows what is being reviewed. In a small, well-documented system that assumption holds. In the systems that actually run regulated businesses and critical infrastructure, it does not. These are estates—accreted over years by rotating teams, spread across many repositories, deployed as dozens or hundreds of microservices, wired together by webhooks and shared databases, and standing on dependency trees that pull in thousands of transitive packages. No single person holds the map. The org chart that built it has turned over. The documentation, if it exists, describes the system someone intended to build, not the one that is running.

This is where vulnerabilities live. Not usually in the service everyone is watching—in the one nobody remembers:

- **The forgotten service.** A proof-of-concept that quietly went to production, an internal tool exposed to the public internet, a duplicate of a service that was "replaced" but never decommissioned. It still has credentials, still talks to the database, and is still in scope for an attacker.
- **The credential in history.** An AWS key, a payment-processor secret, an OAuth client secret, a personal access token—committed to a `.env` or a build layer years ago, never rotated, still valid, sitting in a git history that an attacker can clone.
- **The structural auth flaw.** A bypass-route list loaded from a mutable database table, so that any database write can create an authentication bypass; a JWT verified through an asynchronous callback that can race the next middleware; a tenant identity derived from a client-controllable header.
- **The dependency CVE.** A published, well-known vulnerability in a package like a common HTTP client, a JWT library, an ORM, or a web framework —trivially fixed by a version bump, but invisible because no one ever ran a dependency audit across the estate in years of operation.
- **The cipher that is not encryption.** Sensitive regulated data "protected" by a reversible substitution scheme that a developer mistook for encryption, propagated across many services, and that survived years of operation because no review ever questioned it.

None of these are exotic. Every one of them is a finding a competent auditor would flag on sight—*if* the auditor's eyes ever reached that file. The binding constraint is not auditor skill; it is auditor bandwidth against estate size. A human team can deeply review a handful of services in the time a quarter allows. An estate has hundreds. The math does not close, so coverage is sampled, sampling misses the forgotten service, and the breach—or the regulator—finds what the sample did not. The problem is fundamentally one of *scale*: breadth of coverage, recall across an enormous surface, and the stamina to re-check everything every time the code moves. Those are exactly the properties a human team lacks and an agentic system has.

2 Agentic Security Auditing

Hanzo treats an audit not as a single review but as a swarm of them. The unit of work is one specialized agent assigned to one repository (or one service) with one analytical lens. Many such

agents run concurrently, so the wall-clock time to sweep an estate scales with how many workers you are willing to run, not with how many repositories the estate contains. The model's job is breadth and recall; the human's job is judgment. The two are deliberately kept in separate lanes.

2.1 Specialized agents, run in parallel

A single model prompted to "find security bugs" is a generalist and behaves like one—shallow, distractible, prone to missing a whole class of issue while chasing another. The stack instead runs a set of agents each holding one threat model in working memory, so each pass is deep in one dimension rather than shallow in all of them:

- **Authentication & session.** Token issuance and verification, expiry and rotation, algorithm pinning, bypass routes, rate limiting on auth endpoints, OTP handling, fail-open versus fail-closed middleware.
- **Secrets & credentials.** Plaintext keys in environment files, source, build layers, and manifests; hardcoded tokens and passphrases; secrets reachable in client bundles; secrets that should live in a KMS.
- **Injection & output handling.** SQL/NoSQL built by string concatenation, command injection, and cross-site scripting through unsanitized rendering of user-controlled data.
- **Access control & tenancy.** Missing ownership checks (IDOR), object lookups that resolve across tenant boundaries, authorization enforced in the UI but not the API, header-driven tenant switching.
- **Transport & configuration.** CORS allowlisting (especially wildcard-with-credentials), security headers (CSP, `X-Content-Type-Options`, HSTS), TLS posture, and webhook signature/replay protection.
- **Dependencies & supply chain.** The full transitive tree of every package manifest, matched against published advisories, with a remediation path (override, upgrade, replace, or remove) for each.

Because the agents share a finding schema—identifier, severity, location, category, evidence, suggested fix—their outputs merge into a single estate-wide ledger that deduplicates the same flaw recurring across services (the wildcard CORS that appears in a dozen places, the same vulnerable package in every tree) and makes the systemic pattern legible rather than buried in a hundred separate reports.

2.2 Automated discovery: building the map first

The first thing the stack produces is not a finding but an *inventory*. Before reviewing anything, agents crawl the version-control organizations, enumerate every repository, classify each by language and stack, read whatever deployment configuration exists, and reconstruct the actual service graph—which services are live, which are duplicates, which are abandoned, what talks to what. This is the step a human team usually skips because it is tedious, and it is exactly the step that surfaces the forgotten service. In the engagement this paper describes, this discovery pass is what turned a "narrow remediation" into a true accounting of the estate: it found a footprint of more than 200 microservices across 127+ repositories where the engagement had been scoped to a handful. You cannot audit, certify, or—for a government system—accredit a system whose extent you have not established; automated discovery establishes it.

2.3 Dependency scanning across the whole tree

Dependency risk is the highest-leverage target for automation because it is pure breadth: every repository, every manifest, every transitive edge, matched against a moving database of advisories. A human will not do this across hundreds of repositories; a machine does it exhaustively and repeatably. The stack resolves each tree, identifies every package with a known advisory, and—critically—does not stop at a list. For each vulnerable package it determines the minimal action: a transitive *override* to a patched version, a direct *upgrade*, a *replacement* of a deprecated or unmaintained package with a maintained equivalent, or *removal* of a dependency that was dead weight pulling in a vulnerable transitive package. The residual—advisories with no upstream fix—is documented with an explicit reachability argument (why the affected code path is not exploitable in context) rather than silently ignored. "Zero actionable" means every advisory has been either fixed or formally adjudicated, which is the state an auditor and a regulator will accept.

2.4 Continuous re-audit

An audit is a photograph; security is a film. A finding fixed today regresses tomorrow when someone reintroduces a wildcard CORS in a service the original sweep did not cover, or a dependency bump pulls a fresh CVE, or a refactor reopens a closed injection path. Because each agent run is automated and cheap relative to human review, the same sweep runs continuously—in CI on every change and on a schedule across the whole estate—so regressions are caught as events rather than discovered in the next annual assessment. Every closed finding becomes a standing regression check: the specific bypass that was fixed is asserted to stay fixed, the specific CORS origin is asserted to stay allowlisted, the specific package is asserted to stay above its patched version. This converts "we audited it once" into "it is audited on every commit," which is the substantive content of *continuous authorization*.

2.5 Honest about the human in the loop

The machine is not the auditor of record, and claiming otherwise would be both false and dangerous. Automated static analysis over-reports: it flags string-built queries that are in fact parameterized, "secrets" that are test fixtures, and IDORs guarded by a check it could not see. It under-reports too: business-logic flaws, broken authorization semantics that are individually valid but collectively wrong, and abuse cases that require understanding intent are not reliably found by pattern-matching. So the division of labor is explicit. The agents deliver breadth: they read everything, miss little of the mechanical classes, and never get bored on repository two hundred. A human security engineer delivers judgment: triaging each finding, discarding false positives, confirming exploitability, setting severity, threat-modeling the critical paths by hand, and signing off remediation. The honest claim is not "AI replaces the auditor." It is "AI gives one auditor the reach of fifty, and the auditor still decides." The final sign-off in the case study was human, and the staged penetration re-tests that production money movement demanded were human; the machine made that human able to cover an estate that would otherwise have been impossible to cover.

3 Case Study: From a Narrow Scope to a 200-Service Estate

The following is drawn from a real engagement with a FINRA/SEC-regulated digital securities platform. All identifying details—company, product, people, services, repositories, domains—are removed; the numbers and the shape of the work are as they happened.

3.1 The scope that wasn't

The engagement began as a targeted vulnerability assessment of a small set of services. The automated discovery pass (§2.2) immediately changed the picture: an organization-wide crawl enumerated an estate of **more than 200 microservices across 127+ repositories**, spanning an API gateway, trading and matching engines, a custody/payments tier, KYC/identity services, a sprawl of frontends, and a long tail of proof-of-concept and duplicate repositories that were still live. The "narrow scope" was a keyhole view of an estate an order of magnitude larger—and, as is typical, the keyhole did not happen to be pointed at the worst of it.

3.2 What the swarm found

A parallel static-analysis pass—one worker per repository, with the specialized lenses of §2.1—produced an internal source-code audit of **93 findings: 22 Critical, 30 High, 32 Medium, 9 Low**, concentrated in the gateway, trading, and custody services. The findings were not subtle once seen; they were invisible only because no one had looked across the whole surface. By category:

- **Authentication.** A bypass-route list loaded from a mutable database table (any DB write could mint an auth bypass); JWT verification via an async callback that could race the next middleware; tenant identity taken from a client-controllable header, allowing tenant switching; tokens without expiry; no rate limiting on OTP/credential endpoints; fail-open middleware that admitted requests when the auth service was unreachable.
- **Secrets.** Plaintext cloud keys, payment-processor secrets, and database passwords committed to environment files across the custody and payment tiers; an OAuth client secret shipped inside a frontend JavaScript bundle; a hardcoded webhook secret in source; build-time tokens leaked into container layers.
- **Injection.** SQL built by string concatenation in a payments-rail query path; cross-site scripting via unsanitized rendering of user-controlled data in multiple frontends.
- **Access control.** Missing ownership checks on order modification (IDOR); object lookups that crossed tenant boundaries; withdrawal flows lacking an approval gate; missing KYC and velocity gates on crypto withdrawals.
- **Transport & webhooks.** Wildcard CORS *with credentials* enabled across several services; missing or unverified webhook signatures on inbound custody and payment events; missing security headers.
- **Financial correctness.** Floating-point arithmetic on monetary values—an integrity and compliance defect as much as a precision one.

A later forensic pass added the kind of finding that only a full-history, full-estate sweep surfaces—a class of cryptographic misuse, in which data was "protected" by a scheme that was not in fact secure, that had propagated across multiple services and survived years of review because no prior audit ever reached every service. The specifics are immaterial here; the lesson is that estate-scale, full-history coverage finds what sampling structurally cannot.

3.3 The dependency debt

Running the dependency scanner (§2.3) across the estate exposed the technical debt of a team that had shipped volume for years without ever running an audit. The full ecosystem carried **∼780 known dependency vulnerabilities**, including **18 critical and 350+ high-severity**. Narrowing to the eight core services, **141** vulnerabilities sat in their trees—published CVEs in exactly the load-bearing packages you would least want them in: the HTTP client, the JWT library, the ORM, the web framework. Every one was fixable; none had been fixed, because no one had looked.

3.4 Remediation to zero actionable

Remediation followed the discipline of §2.3. The 141 vulnerabilities in the core services were driven to **0 actionable**, and the full ∼780-vulnerability ecosystem was driven to **0 actionable** as well—every advisory either fixed or formally adjudicated as unreachable. The mechanical work was exactly the catalog automation is built for: transitive overrides to patched versions, deprecated-package migrations (a v2 cloud SDK to its modular v3, an unmaintained HTML minifier to a maintained fork, an old crypto library swap that dropped a vulnerable transitive dependency), and removal of dead dependencies that existed only to pull in vulnerable transitive packages. The handful of residual advisories with no upstream fix were documented with reachability arguments, not buried.

On the source-code side, the findings were closed across **eleven audit rounds**. Round one surfaced the initial 93; subsequent rounds drove remediation, caught regressions (including a reintroduced wildcard CORS found during a verification pass), hardened the multi-tenant authentication path, purged the legacy service framework, and re-verified each closed finding. Across all eleven rounds the series recorded **143 findings, 115 fixed**, with the remainder accepted or mitigated as medium/low items carrying no exploitable path. The remediation classes were the standard hardening moves, applied estate-wide rather than service-by-service:

- Secrets moved out of plaintext environment files and into a KMS, with machine-identity access rather than shared static credentials.
- Wildcard CORS replaced by strict origin allowlists everywhere it appeared.
- Security headers added; webhook ingress hardened with HMAC signatures plus timestamp and nonce replay protection.
- JWTs verified synchronously with a pinned algorithm, expiry, and rotation; for the next-generation build, JWKS-verified tokens against a real identity provider.
- Access control corrected with explicit ownership checks and tenant-scoped queries at the middleware and datastore layers.

3.5 The real fix: deleting the attack surface

Patching findings hardens the code you keep. The larger security win was deciding how much code to keep at all. In parallel with remediation, the estate was consolidated: **200+ microservices collapsed into three Go binaries**, and the active codebase fell from **3.07 million lines to 1.06 million** —roughly **3.5 million lines of dead code deleted**, git-verified, with the superseded repositories archived. This is the most underrated security action available, because the cleanest defense against a vulnerability in a service is for that service not to exist. Each deleted service is one fewer credential to leak, one fewer dependency tree to patch, one fewer CORS policy to misconfigure,

one fewer forgotten endpoint on the public internet. Less code is less attack surface, and a 70% reduction in active code is a 70% reduction in the places a flaw can hide—a result no patch campaign can match because it removes the substrate the flaws live in.

3.6 Outcome

The pairing of automated breadth and human judgment took the platform from 93 source findings and ∼780 dependency vulnerabilities to zero open critical or high findings and zero actionable dependency vulnerabilities, on an estate it had not even fully enumerated at the start, while shrinking that estate by an order of magnitude. The platform achieved **FINRA/SEC approval and SOC 2**—the regulatory and audit outcomes that are the real test of whether security work was not just performed but *evidenced*.

4 Security by Construction in the OSS Stack

Auditing finds the holes; architecture decides how many there are to find. The same Hanzo open-source stack that runs the agents is built so that whole classes of finding cannot occur, and so that the evidence an auditor needs is produced as a byproduct of normal operation rather than reconstructed after the fact. The case study's remediation was, in effect, a migration toward these properties.

- **KMS for secrets.** Secrets live in a key-management service and are delivered to workloads by machine identity, not committed to environment files, source, or manifests. The single most common critical finding in the case study —plaintext credentials—is structurally prevented when there is no plaintext credential to commit. Access is mnemonic-derived and consensus-authorized, with no open fallback mode.

- **IAM for identity.** Authentication and authorization run through one identity provider over standard OIDC, with tokens verified against published JWKS. Bespoke per-service auth—the source of the bypass-list, the header-driven tenancy, and the race-prone verification in the case study—is replaced by a single audited path, so there is one place to get right instead of hundreds to get wrong.

- **Post-quantum identity and transport.** Service identities and inter-service transport use post-quantum primitives (ML-KEM for key establishment, ML-DSA for signatures, NIST FIPS 203/204). Every service-to-service call is mutually authenticated and forward-secret against a future quantum adversary, closing the "harvest now, decrypt later" exposure on traffic and on the audit record itself.

- **Cryptographically signed actions.** Consequential actions—a secret access, an administrative change, an agent's action—ride a signed envelope binding the actor's identity, the operation, a timestamp, and a nonce, signed with a post-quantum key. Authority is verifiable and non-repudiable: not "a log says this happened" but "this actor provably authorized this, and the signature proves it."

- **Immutable on-chain audit trail.** Security-relevant events are recorded to an append-only, tamper-evident ledger. The audit trail cannot be edited or deleted after the fact—the precise property that books-and-records and custody regulations demand, and the precise property an after-the-fact log store cannot guarantee.

The throughline is that confidentiality, identity, authority, and auditability are *architectural defaults*, not features bolted on per service. An estate built this way starts where the case-study estate finished, and the agentic auditor then has dramatically less to find—and what it does find, it finds against a system that is already producing the evidence to prove the fix.

5 Why It Matters for Audit, Certification, and Accreditation

Regulated enterprises, critical-infrastructure operators, and government systems—defense among them—face the estate-scale security problem in its most acute form: larger sprawls, longer lifetimes, higher consequences, and a formal audit or accreditation regime that demands evidence, not assertion. The capability described here maps directly onto those requirements, from a commercial SOC 2 or ISO 27001 audit through to a government Authorization to Operate.

- **Boundary and control evidence.** Every audit and accreditation regime—SOC 2, a FINRA/SEC examination, and (for government systems) an Authorization to Operate—requires an enumerated system boundary, an assessed control set, and a documented residual-risk posture. Automated discovery produces the boundary (including the forgotten services that would otherwise sit outside it), the agent swarm produces the assessment at a coverage no sampling approach reaches, and the signed, immutable audit trail produces the evidence—turning the evidence package from a periodic manual artifact into a continuously regenerated one.

- **Supply-chain assurance.** The dependency scanner enumerates the complete transitive software bill of materials, matches it against advisories, and drives it to zero actionable with a documented adjudication for every residual. This is the concrete substance behind software-supply-chain mandates (SBOM, known-vulnerability remediation), which now run from commercial procurement through to federal acquisition: not a one-time scan but a standing, re-run guarantee.

- **Continuous authorization.** The shift from periodic to continuous re-audit is exactly the shift modern frameworks are pushing toward—continuous SOC 2 monitoring on the commercial side and continuous ATO (cATO, ongoing authorization) on the government side. Every closed finding becomes a standing regression check; every commit is re-assessed; drift is an event, not a year-end surprise. The posture reflects the system as it is, not as it was at the last assessment.

- **Sovereign and auditable.** The entire stack—the review agents, the models behind them, the KMS, the IAM, the post-quantum cryptography, the ledger—is owned and open. It runs offline, at the edge, in an air-gapped or classified enclave, with no dependency on a foreign or black-box service and no telemetry leaving the boundary. The customer can inspect the auditor itself, which is the only honest basis for trusting an auditor's output.

The deeper point the case study makes for any high-assurance buyer is about *consolidation as defense*. A 200-service estate reduced to three auditable binaries is not only cheaper to run; it is a fundamentally smaller thing to defend, certify, and reason about. In a domain where every additional service is additional attack surface, additional audit or accreditation scope, and additional supply-chain exposure, the ability to find what you have, prove it secure, and then make it radically smaller is itself a security capability.

6 American Open Source for a Sovereign Security Posture

The stack described here is built and owned end-to-end by Hanzo AI, an American-owned applied-AI lab and venture studio (Techstars '17, founded 2014) responsible for more than 100 venture-funded startups and several unicorns, founded by an expert in both AI/ML and cryptography. Two facts make this relevant to any security-conscious buyer—a regulated enterprise, a critical-infrastructure operator, or a government and national-security program. First, sovereignty of the security tooling itself: an audit is only as trustworthy as the party that controls the auditor, and this stack—agents, models, key management, and cryptography—is American-owned, auditable, and free of dependence on foreign or black-box components, implemented to the same standards (FIPS 203/204/205, CNSA 2.0) the U.S. government mandates and inspectable by the party that has to rely on it. Second, sovereignty of the AI doing the work: U.S. frontier AI has narrowed to a closed commercial duopoly, while the only broadly open frontier weights available today are largely foreign, leaving regulated and defense buyers to either rent from the duopoly or take a dependency on foreign models. Hanzo is the American open-source frontier alternative—it owns both the Zen model family and this cryptographic stack, so the agents that read your code, the keys that protect your secrets, and the ledger that records your evidence can be deployed offline, at the edge, and under full audit, with nothing leaving your boundary. That this is engineering rather than research is evidenced in production by the case study itself: an American-owned agentic stack took a FINRA/SEC-approved, SOC 2 securities platform—an estate of 200+ microservices—to zero actionable dependency vulnerabilities and zero open critical findings, and consolidated it into three compiled binaries, with a small team.

7 Conclusion

Security debt hides in the part of the estate nobody is looking at, and the reason nobody is looking is that human review does not scale to hundreds of repositories and millions of lines. Hanzo's contribution is to make the looking scale: an agentic stack that first discovers the real extent of an estate, then sweeps it in parallel with specialized review agents, scans every dependency tree to zero actionable, and re-audits continuously as the code moves—always with a human engineer holding triage and sign-off. The anonymized case study is the proof: a narrow engagement that uncovered a 200+-service, 127+-repository estate; 143 findings (22 Critical) surfaced over eleven rounds; ∼780 dependency vulnerabilities driven to zero actionable; the estate consolidated into three Go binaries with ∼3.5 million lines of dead code deleted; and FINRA/SEC approval plus SOC 2 as the result. The same stack that audits is built so that whole classes of finding cannot occur in the first place—secrets in a KMS, identity in one IAM, post-quantum transport, signed actions, an immutable on-chain trail—and all of it is American-owned, open, and sovereign. For any audit, certification, or accreditation regime—SOC 2 and FINRA/SEC, and for government systems the ATO—that is the combination being asked for: enumerate what you have, prove it secure with evidence rather than assertion, keep proving it on every change, and make it smaller. Find everything; fix what matters; delete the rest; and let the auditor be a machine you are allowed to inspect.

Hanzo AI for Compliance: An OSS Cloud Built to Pass the Audit

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Abstract

Compliance is, in operational terms, an evidence-production problem. An auditor for SOC 2 or ISO 27001, an examiner for FINRA/SEC, a HIPAA assessor, or—for a government system—an assessor for an Authority to Operate (ATO) does not primarily ask whether a system *could* be secure; they ask the operator to *produce the artifacts* that demonstrate a control is designed and operating—access reviews, key-rotation records, immutable logs, change tickets, configuration baselines. Most stacks bolt auditing on after the fact, so producing those artifacts becomes months of manual screenshot-gathering, log-stitching, and spreadsheet reconciliation repeated at every cycle. This paper argues that Hanzo's sovereign open-source stack inverts that cost structure by emitting the evidence auditors ask for *by construction*: a centralized identity and access layer (IAM) produces access-control evidence natively; a key management service (KMS) eliminates plaintext secrets and produces key-custody evidence; a cryptographically signed, on-chain command/audit log produces tamper-evident, non-repudiable audit records; reproducible builds and policy-as-code produce change-management evidence; and 50 machine-checked Lean 4 proofs plus NIST Known-Answer-Test validation of FIPS 203/204/205 produce machine-checked cryptographic assurance rather than "we tested it." We give the control-framework mapping from these native capabilities to the SOC 2 Trust Services Criteria, to FINRA/SEC recordkeeping, and to the NIST SP 800-53 control families (AC, AU, IA, SC, CM) that underpin RMF and the ATO process. We then report an anonymized case study in which this same stack carried a regulated securities platform through FINRA/SEC approval and SOC 2 with a small team—while a 200+-microservice to 3-binary consolidation shrank the very surface that had to be audited. Throughout, claims are kept careful: the stack *provides evidence for* and *maps to* controls; certification is an assessor's judgment, not a vendor's.

Executive Summary. Passing an audit is mostly about producing proof. The painful part of SOC 2, FINRA/SEC, or a defense ATO is not deciding to be secure—it is assembling the mountain of evidence that shows, control by control, that the system is secure and stays that way: who can access what, where the secrets live and how often they rotate, an unalterable record of every privileged action, proof that the code in production is the code that was reviewed. When auditing is an afterthought, that evidence has to be reconstructed by hand every cycle, which is why compliance is expensive and recurring. Hanzo's stack is built so the evidence is a byproduct of running the system. Identity is centralized, so access reviews and least-privilege are queries, not archaeology. Secrets live in a key vault and are never in plaintext, so key-custody evidence exists by default. Every privileged action is cryptographically signed and written to an immutable, tamper-evident log, so the audit trail cannot be quietly edited—and the same property satisfies financial-recordkeeping rules that demand write-once records and supports non-repudiation. Builds are reproducible and policy is code, so change-control evidence is generated automatically. And the cryptography is not merely tested but *proved*—machine-checked theorems re-verified on every commit. The net effect is that SOC 2, ISO 27001, FINRA/SEC, HIPAA, and—for government systems—the RMF/ATO process all become tractable: less code to assess, a smaller audit boundary, and an evidence pipeline that is collected continuously rather than re-earned from scratch each year. For any regulated enterprise or government program—defense included—modernizing onto this stack means inheriting the audit pipeline, not building it.

Contents

1 Compliance Is an Evidence-Production Problem

A control framework—SOC 2, FINRA/SEC recordkeeping, NIST SP 800-53 under the Risk Management Framework (RMF)—is a catalog of assertions a system is required to satisfy. An audit or assessment is the exercise of *substantiating those assertions with artifacts*. The decisive distinction in practice is not between secure and insecure systems; it is between systems that can *produce the evidence* of their controls on demand and systems that cannot. An assessor's day is spent asking "show me": show me the list of who has administrative access, show me that it was reviewed last quarter, show me that this secret was rotated, show me an unalterable record that this privileged action happened and who performed it, show me that the binary running in production is the artifact your pipeline built from reviewed source.

When a stack is assembled from many independently-operated services with logging and identity bolted on afterward, answering "show me" is manual archaeology. Access lists are scattered across dozens of consoles; secrets are pasted into environment variables and config files; logs are mutable application logs spread across services with no integrity guarantee; the relationship between source, review, and the deployed artifact is reconstructed from memory. The result is the familiar compliance cost: a SOC 2 Type II observation window plus weeks of evidence collection; a FINRA/SEC examination that demands recordkeeping the architecture was never designed to produce; an ATO package whose body of evidence (the security controls assessment) is assembled by hand. And because nothing about the system *emits* this evidence, the cost recurs in full at every audit cycle and every re-authorization.

The thesis of this paper is structural: if the stack is built so that the evidence auditors ask for is a *native output* of normal operation, the cost collapses. The audit stops being a discovery project and becomes a query against artifacts the system already produces. Hanzo's stack is built this way deliberately, and the rest of this paper walks the specific capabilities, maps them to the frameworks, and shows—with an anonymized production case—what it does to the audit.

2 Evidence by Construction

The stack is consolidated around a small number of owned, open-source components. Each one, as a side effect of doing its job, emits a class of evidence that an auditor asks for. We take them in turn and name the control objective each supports.

2.1 Identity and Access (IAM): access-control evidence

All authentication and authorization route through one identity layer (Hanzo IAM) rather than per-service credentials. Authentication is centralized (OIDC/OAuth 2.1 with PKCE; one canonical set of endpoints); authorization is role- and scope-based; and because identity is a single source of truth rather than a federation of ad-hoc accounts, least-privilege is enforceable and *auditable*. The evidence an assessor wants for access control—the complete roster of principals and their entitlements, the record of access grants and revocations, the periodic access review—are queries against one system rather than a reconciliation across many.

This supports the SOC 2 Security (Common Criteria) logical-access criteria and maps directly to the NIST SP 800-53 **AC** (Access Control) and **IA** (Identification and Authentication) families: centralized account management, enforced least privilege, separation of duties expressed as roles, and unique attributable identities for every actor. Because there is one identity plane, an access review is *produced*, not assembled.

2.2 Secrets and Keys (KMS): confidentiality and key-management evidence

Secrets never live in plaintext. A key management service (Hanzo KMS) holds credentials and cryptographic keys under centralized custody with rotation, and services obtain secrets at runtime rather than embedding them. There are no long-lived secrets in source, in container images, or in configuration. The evidence auditors request for confidentiality and key management—where keys live, who can reach them, how custody is separated, when each was rotated—is the KMS's normal bookkeeping.

This supports the SOC 2 Confidentiality criteria and maps to the SP 800-53 **SC** (System and Communications Protection) family, specifically the key-establishment and key-management controls (SC-12/SC-13) and protection of secrets at rest. The structural property—*no plaintext secret anywhere in the pipeline*—is itself the control, and the KMS's rotation log is the evidence that it operates. A verification gate enforces the property mechanically: the build fails if a plaintext-secret pattern reappears, so "no secrets in code" is checked rather than asserted.

2.3 Immutable audit trail: audit-logging, integrity, and non-repudiation

This is the keystone. Every privileged action is cryptographically signed by the identity that performed it and written to an *append-only, on-chain, tamper-evident* command/audit log. Two properties follow that ordinary application logging cannot provide:

- **Tamper-evidence.** The log is hash-chained and consensus-anchored, so any retroactive edit, deletion, or reordering is detectable. An auditor does not have to trust that the log was not altered; the log's own structure demonstrates it.
- **Non-repudiation.** Because each entry is signed by the acting identity (post-quantum signatures; §2.4), the actor cannot later deny the action, and a reviewer cannot forge one. The signature binds *who* to *what* and *when*.

This maps to the SP 800-53 **AU** (Audit and Accountability) family in full— audit event generation (AU-2), content of records including actor and outcome (AU-3), protection of audit information from modification (AU-9), and non-repudiation (AU-10)— and to the SOC 2 Security monitoring criteria and Processing Integrity. Crucially, the same immutability is exactly what financial recordkeeping rules require: the write-once, non-rewriteable, non-erasable ("WORM-like") retention demanded for broker-dealer books and records is an emergent property of an append-only, hash-chained ledger, not a bolt-on archival appliance. One mechanism satisfies the defense integrity control *and* the securities recordkeeping rule.

2.4 Post-quantum identity and transport: forward-looking confidentiality

Identities, signatures on the audit log, and transport between services are post-quantum (ML-KEM for key establishment, ML-DSA for signatures; FIPS 203/204). The compliance-relevant point is durability of evidence: an audit record, a signed action, or an encrypted channel captured today must remain confidential and verifiable for the full retention horizon—which, for financial records and for classified material, is measured in years to decades. "Harvest now, decrypt later" makes classical-only confidentiality a liability for any record with a long required life. Post-quantum identity and transport make the audit trail's signatures and the data's confidentiality resistant to a future quantum adversary, which is the forward-looking posture that CNSA 2.0 mandates for national-security systems and that a prudent long-retention financial archive should adopt. This

supports the SC family's transmission-confidentiality and cryptographic-protection controls with algorithms chosen for the record's lifetime, not just today's threat.

2.5 Formal verification: machine-checked evidence, not "we tested it"

Most software is trusted because it was tested and nothing broke. The cryptographic core here goes further: its security properties are proved as theorems and re-checked by a machine on every code change. The suite comprises **50 machine-checked Lean 4 proofs**, including **ML-DSA EU-CMA** unforgeability (FIPS 204) and **ML-KEM IND-CCA2** security (FIPS 203), alongside consensus safety/liveness, threshold-signature, and protocol-correctness proofs; complementary **NIST Known-Answer-Test (KAT)** validation covers all three post-quantum standards (FIPS 203/204/205). For an assessor this is a different *kind* of evidence: not a test report asserting that some inputs produced expected outputs, but a machine-checked proof with an explicit, inspectable set of assumptions. A proof regression blocks the build, so the assurance is continuous rather than a point-in-time snapshot. This is the strongest available evidence for the SC family's cryptographic controls and for the "correctly implemented" burden in any cryptographic accreditation, and it is auditable end-to-end precisely because the stack is owned and open—there is no third-party black box whose internals are out of the prover's reach.

2.6 Reproducible builds, policy-as-code, and change control

The path from source to running artifact is itself evidence. Builds are reproducible and produced by a shared CI/CD pipeline; images are pinned to immutable, content-addressed tags (a semver tag or a commit-pinned `sha-` digest), never to a floating `:latest`; and deployment policy and infrastructure are expressed as code under version control. The consequence is that the relationship auditors probe for change management—reviewed source → built artifact → deployed instance—is a chain of immutable, attributable records rather than an after-the-fact reconstruction. Who changed what, who reviewed it, and exactly which artifact reached production are all recorded by the normal pipeline.

This maps to the SP 800-53 **CM** (Configuration Management) family—baseline configuration (CM-2), change control (CM-3), and configuration of the deployed system—and to the SOC 2 change-management criteria. Because policy is code, the configuration baseline is the repository state, and any drift is a diff.

3 Control-Framework Mapping

Table 1 maps each native capability to the SOC 2 Trust Services Criteria it provides evidence for, the NIST SP 800-53 control families it supports under RMF/ATO, and the corresponding FINRA/SEC recordkeeping relevance. The verbs are deliberate: a capability *provides evidence for* or *maps to* a control. Whether a control is satisfied is an assessor's determination on the specific deployment; the stack's contribution is to make the evidence native and the boundary small.

Native capability	**SOC 2 TSC**	**800-53 family**	**FINRA/SEC & ATO relevance**
Centralized IAM (OIDC; RBAC; least privilege; access reviews)	Security (CC6 logical access)	AC, IA	Supervisory access controls; attributable identities for the ATO access-control assessment
KMS (no plaintext secrets; central custody; rotation)	Confidentiality	SC (SC-12/13)	Protection of customer/non-public data; key-management evidence in the SCA
Immutable, signed, on-chain audit log	Security (monitoring); Processing Integrity	AU (AU-2/3/9/10)	WORM-like books-and-records retention; non-repudiation of privileged actions
Post-quantum identity & transport (ML-KEM / ML-DSA)	Confidentiality	SC (SC-8/12/13)	Long-retention record durability; CNSA 2.0 forward posture
Formal verification (50 Lean 4 proofs; NIST KAT)	Security; Processing Integrity	SC, SA	Machine-checked "correctly implemented" evidence for the SCA
Reproducible builds; policy-as-code; pinned artifacts	Security; Availability (change mgmt)	CM (CM-2/3)	Change-control records; configuration baseline as code
Consolidation (fewer services / smaller boundary)	All five (reduced scope)	All families (smaller boundary)	Smaller examination surface; smaller authorization boundary

Table 1: Native stack capabilities mapped to SOC 2 Trust Services Criteria, NIST SP 800-53 control families (RMF/ATO), and FINRA/SEC recordkeeping relevance. "Maps to / provides evidence for"—not a certification claim.

Five criteria, one stack. The SOC 2 Trust Services Criteria are Security, Availability, Confidentiality, Processing Integrity, and Privacy. The capabilities above touch all five: *Security* via IAM, the audit trail, and reproducible builds; *Confidentiality* via KMS and post-quantum transport; *Processing Integrity* via the tamper-evident log and machine-checked proofs of the logic that moves records; *Availability* via policy-as-code change management and a consolidated, self-hostable footprint; and *Privacy* via centralized access control and confidentiality of non-public data. Because one set of mechanisms underlies all five criteria, the evidence collected once is reused across the report rather than re-gathered per criterion.

The same evidence translates. The reason one stack serves SOC 2, ISO 27001, FINRA/SEC, and HIPAA—and, for government systems, an ATO— simultaneously is that the underlying control objectives overlap heavily—access control, confidentiality, audit integrity, change control, cryptographic correctness—and differ mainly in the framework's vocabulary and the assessor's authority. An append-only signed ledger is "AU-9 protection of audit information" to an RMF

assessor and "non-rewriteable record retention" to a securities examiner; it is the same ledger. Centralized least-privilege identity is "AC-2/AC-6" to one and "supervisory and access controls" to another; it is the same IAM. Producing the evidence once and presenting it through each framework's lens is the efficiency the consolidated stack creates.

4 Case Study: One Stack Through FINRA/SEC and SOC 2

The argument is not theoretical. The same stack carried a regulated securities platform—an SEC-registered alternative trading system (ATS) with broker-dealer and transfer-agent functions—through FINRA/SEC approval and SOC 2, with a small team, in record time. Two facts from that engagement matter for the thesis of this paper.

The evidence was native. The regulated entity ran on this stack's centralized IAM, its KMS (no plaintext secrets), and its immutable signed audit log. The recordkeeping obligations that fall on a broker-dealer and transfer agent— non-rewriteable books and records, an attributable trail of who did what to which record, controlled and reviewed access—were satisfied by capabilities the platform already had, not by a separate compliance system layered on top. The audit and the regulatory approval drew on artifacts the running system produced as a matter of course.

Consolidation shrank what had to be audited. In parallel, the platform's architecture was consolidated from 200+ **microservices into 3 compiled Go binaries** (plus a small set of OSS platform services), collapsing roughly **3.07 million lines of code to about 1.06 million**—*∼3.5 million lines of dead code deleted*, around a 65–70% reduction. The consolidated rebuild also remediated the security-finding backlog of the legacy estate (on the order of 90+ findings) at the architectural level. This is the second, quieter half of why the audit was tractable: *every line of code and every service is attack surface that an auditor or accreditor must reason about.* Cutting the codebase by two-thirds and the runtime from a sprawl of services to three binaries shrinks the examination surface, the SOC 2 system description, and—for a government system—the authorization boundary. Less code plus centralized IAM/KMS/audit is a smaller, cheaper, faster audit. The consolidation and the evidence-by-construction property are the same lever viewed from two sides: one reduces *what* must be proven, the other makes the proof *native*.

Anonymization. The regulated entity is intentionally unnamed; the point is the stack's properties, not the client. What transfers to any future program is the mechanism: a small team reached a regulated, audited production posture quickly because the stack produced the evidence and the consolidation shrank the surface.

5 Continuous Compliance

A SOC 2 Type II report covers an observation window; an ATO is not a one-time event but a posture that must be *maintained* under continuous monitoring; a broker-dealer's recordkeeping is examined repeatedly. The expensive failure mode is treating each cycle as a fresh discovery project. Because this stack emits evidence continuously, the posture is maintained rather than re-earned:

- **Always-on collection.** The audit log is generated by operation, not by an end-of-period scramble; access reviews are queries against live IAM state; key-rotation records accrue from the KMS; build and deployment records accrue from the pipeline. The body of evidence is a continuous stream.

- **Continuous verification.** The Lean proofs and KAT vectors run on every commit, so cryptographic assurance is re-established automatically and a regression blocks deployment. Policy-as-code means configuration drift surfaces as a diff, not as an audit finding months later.
- **Agentic evidence assembly.** The same agentic engineering pipeline that built the consolidated system can collect, organize, and map evidence to controls continuously—turning the ongoing-authorization (cATO) ideal into routine output instead of a periodic mobilization.

The effect on the ATO process specifically is that re-authorization draws on an evidence pipeline that never stopped running. The authorization boundary is smaller (consolidation), the controls produce their own evidence (by construction), and the evidence is current (continuous), which is precisely the posture continuous authorization is meant to reward.

6 Why It Matters Across Regulated Industries and Government

Any organization modernizing onto this stack—a healthcare provider, a financial firm, a critical-infrastructure operator, or a government contractor or agency (defense included)—inherits the evidence pipeline rather than building one. The same properties that make a commercial SOC 2, ISO 27001, FINRA/SEC, or HIPAA audit tractable are the properties a government RMF assessment and an ATO package require: centralized least-privilege identity (AC/IA), no plaintext secrets and managed keys (SC), a tamper-evident non-repudiable audit trail (AU), reproducible builds and configuration-as-code (CM), and machine-checked cryptographic correctness (SC/SA). Because the stack is sovereign, open-source, and self-hostable, it can be deployed in the environments these regimes demand—on-premises, air-gapped, at the edge—without renting from a closed vendor, and an assessor can inspect the entire path the evidence takes rather than trusting a black box. The cryptography is post-quantum, so the records and identities survive the retention horizon that financial, healthcare, and government archives all require.

This compliance posture is one facet of a broader sovereign stack built by Hanzo AI—an American-owned applied-AI lab and venture studio (Techstars '17, founded 2014). Hanzo owns both halves of the modern trust problem: the frontier AI models (the *Zen* family) and the production post-quantum cryptography that secures and signs the systems they run in. This matters for who a regulated or accredited operator can build on. U.S. frontier AI has consolidated into a closed commercial duopoly, while the broadly *open* frontier model weights available today are largely of foreign origin—an untenable basis for a system that must be inspected, accredited, and run on sovereign infrastructure. Hanzo is the American open-source alternative that owns the model and the cryptography together, deployable without renting from the closed duopoly or depending on foreign open weights. The evidence-by-construction property documented here is the auditable backing for that claim: not a vendor assurance an accreditor must take on faith, but artifacts—signed logs, key-custody records, reproducible builds, machine-checked proofs—an accreditor can read.

7 Honest Scope

This is a capability paper, and the credible version is explicit about what the stack does versus what an assessor does. No overstatement:

- **The stack provides evidence; it does not self-certify.** SOC 2 attestation is issued by a CPA firm, a FINRA/SEC approval is a regulator's decision, and an ATO is granted by an Authorizing Official. The stack's contribution is to make the evidence native, current,

and inspectable, and to shrink the boundary being judged—which is what makes those determinations faster and cheaper, not automatic.

- **Mapping is not equivalence.** A capability *provides evidence for* or *supports* a control family; closing a specific control still requires the operator to configure, document, and operate it for the assessed deployment. The tables here are a starting correspondence, not a completed controls matrix.
- **Formal verification has a stated boundary.** The 50 Lean proofs rest on explicit cryptographic hardness axioms (standard practice), and model checking and protocol analysis are on a published roadmap rather than complete. The assurance boundary is the true one, not a marketing one.

The strategic point holds regardless of where any one line sits: because Hanzo owns and open-sources every layer the evidence runs through, the artifacts an auditor or accreditor needs are produced by the system itself and can be inspected end-to-end— which is the difference between an audit that is a discovery project and one that is a query.

8 Conclusion

Compliance is an evidence-production problem, and most stacks make it expensive by producing no evidence until an auditor asks. Hanzo's sovereign open-source stack inverts that: centralized IAM emits access-control evidence, KMS emits key-custody evidence, an immutable signed on-chain log emits tamper-evident and non-repudiable audit evidence, reproducible builds and policy-as-code emit change-control evidence, and 50 machine-checked proofs plus NIST KAT validation emit cryptographic assurance that is proved rather than asserted. The same evidence maps across SOC 2's five Trust Services Criteria, ISO 27001, FINRA/SEC recordkeeping, HIPAA, and the NIST 800-53 families that underpin RMF and the ATO process—so one stack serves the commercial, financial, healthcare, and government frameworks (defense included) at once. A production case shows it carrying a regulated securities platform through FINRA/SEC and SOC 2 with a small team, while a 200+-microservice to 3-binary consolidation shrank the surface that had to be audited (and, for a government system, accredited). For any regulated enterprise or government program—defense included—the consequence is direct: modernize on this stack and inherit the audit pipeline—sovereign, auditable, post-quantum, and continuous—rather than rebuild it from scratch each cycle.

ZAP: Zero-Copy Application Protocol for Contested, Edge, and Bandwidth-Limited Military Environments

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Abstract

We present ZAP (Zero-copy Application Protocol), a high-performance binary serialization and RPC framework achieving 2.2 ns parse latency with zero memory allocations, 11.5 million transactions per second end-to-end, and 91% memory reduction compared to JSON-RPC. ZAP provides two production implementations—Rust (AI agent communication, MCP gateway bridging) and Go (blockchain consensus wire protocol, VM-to-VM messaging)—sharing a common wire format with 16-byte headers and 8-byte aligned structs for direct pointer-arithmetic reads from network buffers. The protocol includes native post-quantum transport security (ML-KEM-768 + ML-DSA-65 hybrid handshake), decentralized identity via W3C DIDs derived from lattice public keys, distributed agent consensus with threshold voting, and mDNS-based ad-hoc network discovery. We analyze ZAP's suitability for contested electromagnetic environments, bandwidth-limited tactical links, GPU cluster coordination, and edge AI agent orchestration, demonstrating 96% energy reduction and 30× throughput improvement over standard protocols under identical network conditions.

Executive Summary. ZAP is the "plumbing" that lets AI agents, sensors, and computers talk to each other using far less bandwidth, memory, and power than today's standard formats. It matters most at the tactical edge: it moves the same information using roughly a tenth of the data, which both cuts the radio footprint an adversary can detect and lets coordination work over links as slow as an HF radio that ordinary protocols cannot use at all. Its security is post-quantum by design—traffic encrypted with ZAP stays protected even against a future quantum computer—and that protection is built into the protocol rather than bolted on. Concretely, it parses a message in 2.2 nanoseconds, handles 11.5 million transactions per second, and uses 91% less memory and 96% less energy than the common JSON-RPC approach, which on a battery-powered device translates directly into longer mission endurance. For program leaders, ZAP is the efficiency and security layer that makes real-time, machine-speed AI coordination feasible in contested, bandwidth-starved, and disconnected environments.

Contents

1 Introduction

Military networking at the tactical edge operates under constraints that expose fundamental weaknesses in standard data interchange protocols. JSON-RPC, gRPC/Protocol Buffers, and REST APIs assume abundant bandwidth, low latency, and the availability of garbage-collected runtimes. These assumptions fail in contested electromagnetic environments where every byte transmitted increases detection probability, every microsecond of processing delays decision cycles, and every memory allocation risks unpredictable latency spikes.

ZAP addresses these constraints through three design principles:

1. **Zero-copy reads**: Data is accessed directly from network buffers via pointer arithmetic. No deserialization pass, no object construction, no memory allocation.
2. **Minimal wire overhead**: 16-byte headers with 8-byte aligned struct fields. No field tags, no length-delimited strings for known schemas.
3. **Native PQ security**: Post-quantum key exchange and authentication built into the protocol handshake, not layered on top.

1.1 Why Existing Protocols Fail at the Edge

Protocol	Parse	Allocs	Bytes/op	Edge Failure Mode
JSON-RPC	5,579 ns	58	2,826	GC pauses, verbose encoding
Protobuf	500 ns	11–121	1,200	Varint overhead, copy-on-parse
gRPC	800 ns	30+	1,500	HTTP/2 framing, TLS overhead
Cap'n Proto	∼100 ns	0	800	No PQ crypto, complex schema
ZAP	**2.2 ns**	**0**	**256**	**None**

Table 1: Protocol comparison for tactical edge deployment.

1.2 Contributions

1. **Zero-copy wire format**: 16-byte header, 8-byte aligned structs, direct buffer reads at 2.2 ns.
2. **Dual implementation**: Rust (AI agents, MCP gateway) and Go (consensus, VMs) sharing identical wire format.
3. **Native PQ transport**: ML-KEM-768 + ML-DSA-65 hybrid handshake integrated at protocol level.
4. **Agent consensus**: Distributed voting with threshold agreement and DID-based identity.
5. **Ad-hoc discovery**: mDNS for tactical network formation without infrastructure.
6. **Environmental analysis**: 96% energy reduction enabling sustainable edge operation.

2 Wire Format Specification

2.1 Header

Every ZAP message begins with a 16-byte little-endian header:

Field	Offset	Size	Description
Magic	0	4 B	`ZAP\x00` (0x5A415000)
Version	4	2 B	Protocol version (currently 1)
Flags	6	2 B	Bitfield (see below)
Root Offset	8	4 B	Byte offset to root struct
Size	12	4 B	Total message size in bytes

Table 2: ZAP 16-byte wire header.

2.1.1 Flag Bits

Flag	Value	Meaning
`FlagNone`	0x0000	No flags set
`FlagCompressed`	0x0001	Payload is compressed
`FlagEncrypted`	0x0002	Payload is encrypted
`FlagSigned`	0x0004	Payload carries PQ signature

Table 3: ZAP header flag definitions.

2.2 Type System

Type	Wire Size	Access	Zero-Copy
Bool	1 B	Direct read	Yes
Int8–Int64	1–8 B	Direct read	Yes
Float32/64	4/8 B	Direct read	Yes
Text	8 B (off+len)	Slice into buffer	Yes
Bytes	8 B (off+len)	Slice into buffer	Yes
List(T)	8 B (off+count)	Stride iteration	Yes
Struct	4+ B	Nested offset	Yes

Table 4: ZAP type system with zero-copy access patterns.

All struct fields are 8-byte aligned, enabling direct pointer-arithmetic reads on any architecture without alignment fixups.

2.3 Zero-Copy Access Pattern

Listing 1: Zero-copy message access (Go implementation).

```
// No allocation, no copying, no deserialization
```

```
data := networkBuffer[:msgSize]
msg, _ := zap.Parse(data)  // 2.2 ns
root := msg.Root()

// Direct reads from buffer via offset calculation
nodeID := root.Uint64(0)      // 8 bytes at offset 0
timestamp := root.Int64(8)    // 8 bytes at offset 8
payload := root.Bytes(16)     // slice into buffer
```

The `Parse` function validates the header (magic, version, size bounds) and returns a handle to the root struct. No data is copied; all subsequent reads use offset arithmetic into the original buffer.

3 Schema Language

3.1 Whitespace-Significant Format

ZAP defines a clean, whitespace-significant schema language (preferred over Cap'n Proto `.capnp` format):

Listing 2: ZAP schema definition.

```
struct ConsensusVote
  validator_id  UInt64
  block_hash    Bytes
  round         UInt32
  signature     Bytes
  pq_signature  Bytes    # ML-DSA-65 (optional)

enum VoteType
  prevote
  precommit
  commit

interface ConsensusService
  vote (v ConsensusVote) -> (accepted Bool)
  get_block (hash Bytes) -> (block Bytes)
```

3.2 Compiler

The `zapc` compiler generates code for multiple target languages:

- **Rust**: Primary target, zero-copy readers and builders.
- **Go**: Used in Hanzo consensus and VM communication.
- **Python, TypeScript**: SDK generation for agent integration.
- **C, C++, Java**: Available for embedded and enterprise integration.

The compiler also supports legacy `.capnp` files with automatic migration via `capnp_to_zap()`.

4 Post-Quantum Transport Security

4.1 Cryptographic Primitives

Algorithm	Purpose	Key/CT Size
ML-KEM-768 (FIPS 203)	Key encapsulation	PK: 1,184 B, CT: 1,088 B
ML-DSA-65 (FIPS 204)	Digital signatures	PK: 1,952 B, Sig: 3,309 B
X25519	Classical ECDH	PK: 32 B
HKDF-SHA256	Key derivation	—
AES-256-GCM	Session encryption	Key: 32 B

Table 5: ZAP cryptographic primitives.

4.2 Hybrid Handshake Protocol

Algorithm 1 ZAP Post-Quantum Hybrid Handshake

Initiator generates ephemeral keys:
  $(\text{ek}^X, \text{epk}^X) \leftarrow$ X25519.Generate()
  $(\text{ek}^M, \text{epk}^M) \leftarrow$ ML-KEM-768.Generate()
Send: $\text{epk}^X \| \text{epk}^M \| \text{pk}^{\text{DSA}} \| \sigma$
  $\sigma =$ ML-DSA-65.Sign(sk^{DSA}, transcript)

Responder verifies σ, then:
  $\text{ss}^X \leftarrow$ X25519.DH($\text{sk}^X, \text{epk}^X$)
  $(\text{ct}^M, \text{ss}^M) \leftarrow$ ML-KEM-768.Encaps(epk^M)
  $K \leftarrow$ HKDF-SHA256($\text{ss}^X \| \text{ss}^M$)
Send: $\text{epk}_B^X \| \text{ct}^M \| \sigma_B$

Both: Derive session keys from K
Both: Destroy ephemeral private keys (forward secrecy)
Session: AES-256-GCM with counter nonces

4.3 Crypto Backend Chain

The Rust implementation supports two backends with automatic fallback:

1. **Post-quantum crypto FFI library** (preferred): FFI binding to the Hanzo cryptographic library (CIRCL-based, CGO-optimized). Shared with the consensus layer.

2. **pqcrypto** (fallback): Pure Rust PQ implementations for environments without the native crypto library.

3. **x25519-dalek**: Classical ECDH (always present for hybrid mode).

4.4 Handshake Overhead Analysis

Component	Size	Classical	PQ Hybrid
Ephemeral X25519 PK	32 B	✓	✓
Ephemeral ML-KEM PK	1,184 B	—	✓
ML-KEM ciphertext	1,088 B	—	✓
ML-DSA-65 PK	1,952 B	—	✓
ML-DSA-65 signature	3,309 B	—	✓
Per-direction		**32 B**	**∼7.5 KB**
Amortized/message		**0 B**	**0 B**

Table 6: Handshake overhead: one-time cost, zero per-message overhead.

The 7.5 KB handshake is a *one-time* cost per connection. All subsequent messages use AES-256-GCM with 28-byte overhead (12 B nonce + 16 B auth tag), identical to classical TLS.

5 Performance Analysis

5.1 Microbenchmarks

5.1.1 Go Implementation (Consensus Wire Protocol)

Operation	Time	Ops/sec	Allocs
Parse	2.2 ns	446M	0
Build	38.9 ns	25.7M	0
Consensus round	4.6 ns	218M	0

Table 7: ZAP Go implementation benchmarks (10-core, Go 1.26.1).

5.1.2 Rust Implementation (AI Agent Communication)

Operation	Time	Ops/sec	Allocs
Serialize tool call	322 ns	3.1M	2
Deserialize response	21 ns	47.6M	0
MCP gateway route	1.2 μs	833K	4
Agent consensus vote	890 ns	1.12M	2

Table 8: ZAP Rust implementation benchmarks.

Note: The Go and Rust implementations benchmark different operations. The Go "Parse" (2.2 ns) reads a pre-built message from buffer. The Rust "Serialize" (322 ns) constructs a new message. The Rust "Deserialize" (21 ns) performs schema-validated parsing, more expensive than raw buffer validation.

5.2 End-to-End Throughput

Scenario	ZAP	JSON-RPC
Single-thread serial	3.1M ops/s	179K ops/s
20-server MCP gateway	4.2M calls/s	140K calls/s
Consensus (5 validators)	11.5M TPS	246K TPS

Table 9: End-to-end throughput comparison.

5.3 Memory Efficiency

Metric (100K ops)	JSON-RPC	ZAP
Total allocated	304.01 MB	26.70 MB
Malloc count	6,001,513	200,003
Bytes/op	3,187	280
Mallocs/op	60	2
GC pauses (50K ops)	41	3
Memory saved	**277.3 MB (91.2%)**	
Allocs saved	**5,801,510 (96.7%)**	

Table 10: Memory profile: ZAP vs. JSON-RPC over 100K operations.

5.4 Environmental Impact

At scale (1 billion tool calls per day):

Metric	JSON-RPC	ZAP	Savings
Memory throughput	2.9 TB/day	0.3 TB/day	2.6 TB
Energy consumption	1.8 kWh/day	0.1 kWh/day	96%
CO_2 emissions	∼0.7 kg/day	∼0.03 kg/day	∼245 kg/year

Table 11: Environmental impact at 1B calls/day.

For battery-powered tactical edge devices, the 96% energy reduction translates directly to extended operational endurance.

6 Transport Layer

6.1 Supported Transports

Scheme	Transport	Status	Use Case
`zap://`	TCP	Complete	Default RPC
`tcp://`	TCP explicit	Complete	Direct TCP
`unix://`	Unix socket	Complete	Local IPC
`ws://`	WebSocket	Complete	Browser/cloud
`wss://`	WebSocket+TLS	Complete	Secure browser
`stdio://`	Stdio	Complete	MCP subprocess
`http://`	HTTP/SSE	Complete	Remote MCP
`udp://`	UDP	Complete	Low-latency

Table 12: ZAP transport schemes.

Frame format: 4-byte big-endian length prefix + payload (max 16 MB).

6.2 mDNS Discovery for Ad-Hoc Networks

ZAP nodes advertise services via mDNS with domain-specific service types:

Service Type	Purpose
`_hzd._tcp`	Blockchain consensus nodes
`_hz-gpu._tcp`	GPU compute nodes
`_hz-vm._tcp`	Virtual machine endpoints
`_zap-agent._tcp`	AI agent endpoints

Table 13: mDNS service types for ad-hoc discovery.

This enables zero-configuration tactical network formation: devices joining a mesh automatically discover available services without DNS infrastructure or pre-configured endpoints.

7 Decentralized Identity

7.1 W3C DID Support

ZAP implements W3C Decentralized Identifiers with three methods:

- **did:key**: Self-certifying from ML-DSA-65 public keys. No external registry required.
- **did:hanzo**: Anchored to the Hanzo chain. Verifiable via consensus.
- **did:web**: DNS-based for interoperability with existing PKI.

Listing 3: DID generation from ML-DSA public key.

```
// Post-quantum self-certifying identity
let identity = Identity::generate_pq()?;
let did = identity.to_did();
// did:key:z6Mk... (multibase-encoded ML-DSA-65 pubkey)
```

7.2 Stake-Weighted Trust

For consensus operations, DIDs can be associated with stake weights:

- Validators register stake via blockchain transaction.
- Agent consensus votes are weighted by stake.
- Sybil resistance: creating identity is free, but gaining voting weight requires stake.

8 Agent Consensus Protocol

8.1 Distributed Agent Voting

When multiple AI agents must agree on a decision (threat classification, course of action), ZAP provides distributed consensus:

Algorithm 2 ZAP Agent Consensus

Initiator broadcasts query Q to n agents
for each agent A_i **do**
    A_i computes response R_i independently
    A_i signs: $\sigma_i = \text{ML-DSA-65.Sign}(\text{sk}_i, Q \| R_i)$
    A_i returns $(R_i, \sigma_i, \text{DID}_i)$
end for
Collect responses until threshold t reached (default $t = \lceil 2n/3 \rceil$)
Verify all signatures
Determine consensus: majority response among verified votes
if consensus reached ($\geq t$ agree) **then**
    return (ConsensusResult, aggregated signatures)
else
    return (NoConsensus, individual responses)
end if

8.2 Defense Application

Agent consensus is critical for high-consequence decisions:

- **Threat classification**: Multiple AI models independently assess a sensor contact; consensus prevents single-model hallucination from triggering false alarms.
- **Targeting**: Distributed analysis with threshold agreement before engagement recommendation.
- **Intelligence fusion**: Multiple agents process different data sources; consensus identifies corroborated intelligence.

9 MCP Gateway

9.1 Multi-Server Aggregation

The ZAP MCP gateway bridges multiple Model Context Protocol servers into a unified tool surface:

- **Tool namespacing**: Each server's tools prefixed (e.g., `filesystem:read_file`).
- **Health checking**: Automatic reconnection on server failure.
- **Resource aggregation**: Resources and prompts from all servers accessible through single endpoint.
- **Protocol bridging**: Stdio, HTTP/SSE, WebSocket, and native ZAP servers unified.

9.2 Performance Advantage

Metric	MCP JSON-RPC	ZAP Gateway
Tool calls/sec (20 servers)	140,000	4,200,000
Serialization overhead	5,579 ns	322 ns
Memory per call	2,826 B	256 B

Table 14: MCP gateway performance: native vs. ZAP-bridged.

10 GPU Cluster Coordination

10.1 Zero-Copy Ciphertext Handling

ZAP's zero-copy design enables direct GPU ciphertext operations:

Listing 4: GPU FHE operation via ZAP.

```
gpuNode := zap.NewNode(zap.NodeConfig{
    NodeID:      "gpu-node-1",
    ServiceType: "_hz-gpu._tcp",
    Port:        7000,
})

gpuNode.Handle(MsgTypeFHEOp,
  func(ctx context.Context, from string,
       msg *zap.Message) (*zap.Message, error) {
    root := msg.Root()
    opType := root.Uint32(0)     // FHE operation
    ct := root.Bytes(4)          // Zero-copy ciphertext

    // GPU kernel dispatch - no memcpy needed
    result := gpuFHE(opType, ct)
    return buildResult(result), nil
})
```

Ciphertext data flows from network buffer to GPU kernel without intermediate copies, critical for FHE workloads where ciphertexts are 100×–1000× larger than plaintext.

11 Contested Environment Analysis

11.1 Low-Probability-of-Intercept (LPI)

Protocol	Bytes/call	RF Exposure	LPI Factor
JSON-RPC	2,826	100% (baseline)	1.0×
gRPC	1,500	53%	1.9×
Protobuf	1,200	42%	2.4×
ZAP	**256**	**9.1%**	**11×**

Table 15: RF exposure reduction per operation.

ZAP reduces RF exposure by 11× compared to JSON-RPC, directly improving low-probability-of-intercept and low-probability-of-detection (LPI/LPD) performance for tactical radio links.

11.2 Bandwidth-Limited Links

For a 9.6 kbps HF radio link:

Protocol	Calls/sec at 9.6 kbps	Latency/call
JSON-RPC	0.42	2.35 s
Protobuf	1.00	1.00 s
ZAP	**4.69**	**0.21 s**

Table 16: Throughput on 9.6 kbps HF radio link.

ZAP enables near-real-time AI agent coordination over HF radio links that would be unusable with JSON-RPC.

11.3 Jamming Resilience

- **Minimal transmission time**: 91% less data reduces jamming window.
- **Burst-compatible**: Small message size enables burst transmission between jamming pulses.
- **mDNS re-discovery**: Network reforms around jammed nodes via multicast DNS.
- **Multi-path**: RNS mesh provides alternative routes when primary links are degraded.

11.4 Electronic Attack Resistance

- **PQ encryption**: Intercepted traffic immune to quantum cryptanalysis.
- **Forward secrecy**: Each session uses ephemeral keys; compromising one session reveals nothing about others.
- **Zero metadata leakage**: Binary format reveals no field names, no schema structure, no type information to passive observers.
- **Traffic analysis resistance**: Fixed-size messages possible via padding flag.

12 Comparison with Military Standards

Feature	**Link 16 (TADIL-J)**	**VMF**	**ZAP**
Encoding	Fixed-word binary	Variable-length binary	Zero-copy binary
Bandwidth	238 kbps	Variable	Wire rate
Latency	12.8 ms slot	Variable	2.2 ns parse
Crypto	Type 1	HAIPE	PQ hybrid (FIPS)
AI Support	None	None	Native (agents, MCP)
Discovery	JTIDS	IP-based	mDNS ad-hoc
PQ-Safe	No	No	Yes (FIPS 203/204)

Table 17: ZAP vs. existing military data link protocols.

ZAP is not a replacement for Link 16 or VMF—it operates at a different layer (application vs. tactical data link). However, ZAP can serve as the application protocol carried *over* these links, providing AI agent coordination, consensus messaging, and PQ security that existing message standards lack.

13 Corona Consensus Integration

13.1 Post-Quantum Threshold Signing

ZAP integrates with the Corona protocol for distributed PQ threshold signatures:

- **Basis**: Practical two-round threshold signature from LWE (IACR ePrint 2024/1113).
- **Parameters**: $M = 8, N = 7, \bar{D} = 48, \kappa = 23, Q =$ `0x1000000004A01`.
- **Security**: NIST Level 3 (192-bit equivalent) under Ring-LWE hardness.
- **Performance**: $\sim$89 ms for threshold signature (within 3-second quantum bundle window).

ZAP consensus messages carry Corona shares as native struct fields, enabling post-quantum finality for distributed agent decisions.

14 Implementation Details

14.1 Rust Crate

Module	Size	Purpose
`client.rs`	28.5 KB	ZAP RPC client
`server.rs`	32.2 KB	ZAP RPC server
`gateway.rs`	45.2 KB	MCP gateway aggregator
`transport.rs`	34.6 KB	TCP, WS, stdio, UDP
`crypto.rs`	28.3 KB	ML-KEM, ML-DSA, X25519
`consensus.rs`	35.5 KB	Corona threshold signing
`identity.rs`	26.5 KB	W3C DIDs
`schema.rs`	79.9 KB	Schema compiler
`agent_consensus.rs`	14.2 KB	Agent voting protocol
Total	**∼9,915 lines**	

Table 18: ZAP Rust implementation structure.

14.2 Go Implementation

Minimal-dependency implementation for consensus and VM communication:

- `zap.go`: Zero-copy builder and parser (8.9 KB).
- `node.go`: mDNS discovery and async RPC (21.2 KB).
- `mcp/`: MCP bridge for 20+ tool servers.
- Single external dependency: `github.com/hanzoai/mdns`.

15 Strategic Context

ZAP is the transport layer of an integrated, sovereign AI stack built by Hanzo, an American applied-AI lab founded in 2014 (Techstars '17) whose founder works at the frontier of both AI/ML and cryptography. American frontier AI has consolidated into a closed monoculture—effectively two labs reachable only as metered cloud services—while the only openly available frontier model weights today are largely foreign; Hanzo is the American open-source frontier alternative, owning both the model (the Zen family, 0.8B–1T+ parameters, with owned weights) and the cryptography (the production post-quantum stack ZAP is part of). That ownership is what lets the whole stack run sovereign and disconnected, where the data lives. ZAP also supplies a measurable piece of the economics: its 91% memory and 96% energy reductions and 30× throughput, combined with owned models (no per-token API markup) and self-hosting (no hyperscaler margin), are the mechanism behind roughly an order-of-magnitude lower infrastructure cost in aggregate. The efficiency is not academic—the same agent-driven stack this protocol carries was used to deliver a FINRA/SEC-approved, SOC2-compliant securities platform to production—a full ATS/BD/TA stack consolidated from 200+ microservices into 3 compiled binaries—in record time with a small team, showing the approach holds up under the demands of a regulated, audited domain.

16 Conclusion

ZAP provides the fastest possible binary protocol for military edge environments, achieving 2.2 ns parse latency, zero memory allocations, and 11× RF exposure reduction compared to standard protocols. Its native post-quantum transport security (FIPS 203/FIPS 204), ad-hoc mDNS discovery, distributed agent consensus, and dual Rust/Go implementations make it uniquely suited for contested, bandwidth-limited, and disconnected tactical operations.

The protocol's 96% energy reduction extends battery endurance on tactical devices, while its zero-copy design eliminates the GC pauses and memory pressure that plague JSON-RPC and Protobuf in resource-constrained environments. ZAP enables real-time AI agent coordination over links as slow as 9.6 kbps HF radio—a capability no existing protocol provides.

References

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Reproducibility

Source code. `github.com/hanzoai/zap` (commit `TBD`).
Build. cargo build –release (Rust impl); go build ./... (Go impl).
Hardware tested. Apple M-series, x86_64 Linux + NVIDIA A100, ARM Cortex-A (edge).
Standards referenced. Cap'n Proto wire spec, ML-KEM-768, ML-DSA-65 (FIPS 203/204), W3C DID.
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FIPS 203/204/205 Compliant Cryptographic Architecture for Naval Information Dominance

Zach Kelling — Hanzo Team

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Abstract

We present a comprehensive cryptographic architecture implementing all three NIST post-quantum standards—ML-DSA (FIPS 204), ML-KEM (FIPS 203), and SLH-DSA (FIPS 205)—integrated into a production blockchain consensus system with hybrid classical/quantum security. The Quasar consensus protocol achieves sub-second finality through dual-certificate blocks combining BLS12-381 aggregate signatures with Corona lattice-based threshold signatures, providing an attack window of $\leq$50 ms. We detail six threshold multi-party computation schemes (CGGMP21, FROST, LSS, BLS, SR25519, TFHE) operating over a zero-copy wire protocol, GPU-accelerated cryptographic operations on Metal and CUDA, fully homomorphic encryption via a dedicated virtual machine, and a comprehensive key management system with Shamir secret sharing, HSM integration, and HIPAA/SOX/SEC compliance controls. The architecture provides defense-in-depth against both current and quantum adversaries, with 17 formal cryptographic proofs supporting security claims.

Executive Summary. A capable adversary is already recording encrypted traffic today to decrypt it later, once a quantum computer can break the classical cryptography that protects nearly all systems now in the field. This work is a complete, working cryptographic stack that closes that window: it implements all three of the U.S. government's new post-quantum standards (the NIST FIPS 203/204/205 algorithms that CNSA 2.0 mandates by 2030) and runs them in a live system rather than a slideware roadmap. It reaches a decision in under a second, refreshes its keys every few minutes so any single compromise exposes only minutes of data, and never reconstructs a full secret key in one place. For defense program leads and contractors, the takeaway is that the migration to quantum-resistant cryptography—usually framed as a multi-year future project—is here as deployed, tested infrastructure, with formal proofs behind the security claims and acceleration on commodity Apple and NVIDIA hardware.

Contents

1 Introduction

The transition to post-quantum cryptography represents the most significant cryptographic migration in the history of information security. The National Security Agency's CNSA 2.0 guidance mandates migration to quantum-resistant algorithms by 2030 for national security systems, while adversaries are presumed to be conducting harvest-now-decrypt-later (HNDL) attacks today.

This paper presents a cryptographic architecture that has moved beyond theoretical compliance to production deployment. Every component described herein is implemented, tested, and operational—not a research proposal but an engineering reality.

1.1 Scope of NIST Compliance

Standard	Algorithm	Purpose	Integration Points
FIPS 203	ML-KEM-768	Key encapsulation	RNS, ZAP, HPKE, KMS
FIPS 204	ML-DSA-65	Digital signatures	Consensus, RNS, ZAP, EVM
FIPS 205	SLH-DSA	Hash-based signatures	Long-term anchors, audit

Table 1: NIST PQ standard implementation coverage.

2 Post-Quantum Algorithm Implementations

2.1 ML-DSA: Module-Lattice Digital Signatures (FIPS 204)

ML-DSA provides EU-CMA (Existential Unforgeability under Chosen Message Attack) security under the hardness of Module-LWE and Module-SIS problems.

Mode	Public Key	Secret Key	Signature	Security
ML-DSA-44	1,312 B	2,560 B	2,420 B	NIST Level 2
ML-DSA-65	1,952 B	4,032 B	3,309 B	NIST Level 3
ML-DSA-87	2,592 B	4,896 B	4,627 B	NIST Level 5

Table 2: ML-DSA parameter sets.

Definition 2.1 (EU-CMA Security of ML-DSA)**.** For any probabilistic polynomial-time adversary $\mathcal{A}$ making at most q_s signing queries, the probability of producing a valid forgery is:

$$\Pr[\text{EU-CMA}_{\mathcal{A}}^{\text{ML-DSA}} = 1] \leq \text{negl}(\lambda)$$

under the Module-LWE assumption with parameters $(k, \ell, n = 256, q = 8380417)$.

Implementation: Cloudflare CIRCL library (v1.6.1) with CGO optimization and pure-Go fallback. Available as EVM precompile at address `0x0601` with gas cost 75,000–150,000 depending on mode.

2.2 ML-KEM: Module-Lattice Key Encapsulation (FIPS 203)

ML-KEM provides IND-CCA2 security via the Fujisaki-Okamoto transform applied to Module-LWE encryption.

Mode	PK	SK	CT	SS	Security
ML-KEM-512	800 B	1,632 B	768 B	32 B	Level 1
ML-KEM-768	1,184 B	2,400 B	1,088 B	32 B	Level 3
ML-KEM-1024	1,568 B	3,168 B	1,568 B	32 B	Level 5

Table 3: ML-KEM parameter sets (PK: public key, SK: secret key, CT: ciphertext, SS: shared secret).

2.3 SLH-DSA: Stateless Hash-Based Signatures (FIPS 205)

SLH-DSA provides quantum-safe signatures with no algebraic structure assumptions—security relies solely on the second-preimage resistance of the underlying hash function.

- 12 parameter sets: SHA2/SHAKE × 128/192/256-bit × small/fast variants.
- Signature sizes: 7,856–49,856 bytes (tradeoff: signature size vs. verification speed).
- EVM precompile at `0x0602`, gas 50,000–250,000.
- Primary use: long-term audit anchors where algebraic assumptions may not hold for decades.

3 Hybrid Post-Quantum Consensus: Quasar

3.1 Dual-Certificate Architecture

Quasar is a hybrid consensus protocol that produces blocks certified by both classical (BLS12-381) and post-quantum (Corona ML-DSA-65) signatures:

Algorithm 1 Quasar Dual-Certificate Block Finalization

Phase I — Classical Finality (target: 500 ms)
Validators sample k peers (mainnet $k = 21$)
Accumulate confidence: β_1 for preference, β_2 for decision
Aggregate BLS12-381 signatures into 96-byte certificate
Block finalized with classical certificate

Phase II — Quantum Finality (target: 3 s bundle)
Bundle 6 BLS-certified blocks into Merkle tree
Each validator computes Corona share (Round 1)
Exchange shares, finalize threshold signature (Round 2)
Quantum bundle certified with ML-DSA-65 lattice certificate (∼3 KB)

Block Header: $\text{BLSAgg}_{96\text{B}} \| \text{PQCert}_{3\text{KB}} \| \text{HMAC-SHA256(both)}$

3.2 Attack Window Analysis

Theorem 3.1 (Quasar Quantum Safety)**.** *An adversary with a quantum computer must forge a BLS12-381 aggregate signature within the 50 ms window between classical finality and quantum bundle creation. Even with Shor's algorithm, real-time BLS forgery on current quantum hardware requires wall-clock time exceeding $O(2^{128})$ quantum gates on current architectures, making real-time exploitation infeasible.*

3.3 Epoch-Based Key Rotation

Corona threshold keys rotate on epoch boundaries:

- **MinEpochDuration**: 10 minutes (rate-limited to prevent churn).
- **MaxEpochDuration**: 1 hour (maximum key age).
- **HistoryLimit**: 6 epochs (≈1 hour) for cross-epoch verification.
- **QuantumCheckpointInterval**: 3 seconds (frequent PQ anchors).

3.4 Consensus Protocol Family

Quasar orchestrates 10 sub-protocols:

Protocol	Files	Purpose
Quasar	30	PQ consensus orchestrator
Wave	10	Threshold voting, FPC dynamic selection
Photon	5	K-of-N peer selection with luminance tracking
Focus	5	Confidence tracking, windowed confidence
Prism	9	DAG vertex ordering and cutting
Nova	4	Linear blockchain consensus
Nebula	4	DAG consensus mode
Ray	6	Chain consensus protocol
Horizon	5	Event horizon finality detection
Flare	5	Certificate generation, DAG frontier

Table 4: Quasar consensus protocol family.

4 Threshold Multi-Party Computation

The MPC system provides distributed key generation and signing without ever reconstructing full private keys.

4.1 Supported Schemes

Scheme	Curve/Basis	Use Case	Chains
CGGMP21	secp256k1	Threshold ECDSA	20+
FROST	Ed25519, Ristretto255	EdDSA, SR25519	Polkadot, Solana
LSS	secp256k1	Dynamic resharing	All ECDSA
BLS	BLS12-381	Consensus signing	Hanzo
SR25519	Ristretto255	Substrate compat	Polkadot
TFHE	Lattice	Threshold FHE	T-Chain

Table 5: MPC threshold signature schemes.

4.2 Distributed Key Generation

Algorithm 2 Verifiable Secret Sharing DKG

Phase 1: Commitment
for each party P_i, $i = 1, \ldots, n$ **do**
    Sample random polynomial $f_i(x)$ of degree $t - 1$
    Broadcast commitment $C_i = g^{f_i(0)}$
    Send share $s_{ij} = f_i(j)$ to party P_j (encrypted P2P)
end for

Phase 2: Verification
for each party P_j **do**
    Verify received shares against commitments
    if share invalid **then**
        File complaint against sender
    end if
end for

Phase 3: Finalization
Public key: $\text{PK} = \prod_{i=1}^{n} C_i$
Each P_j holds share: $\text{sk}_j = \sum_{i=1}^{n} s_{ij}$
Shares encrypted with AES-256 and stored locally

Performance: DKG completes in ∼30 seconds for 3 nodes. Signing requires <1 second for threshold signatures.

Byzantine resilience: $t - 1$ compromised parties reveal no information about the secret key, where $t \geq \lfloor n/2 \rfloor + 1$.

4.3 Consensus-Embedded Transport

MPC messages are transported directly over the ZAP consensus wire protocol:

Listing 1: MPC message types in consensus transport.

```
MsgMPCBroadcast  = 60  // Broadcast to all parties
MsgMPCDirect     = 61  // Point-to-point encrypted
MsgMPCMembership = 62  // Ed25519 PoA validators
MsgMPCResult     = 67  // Signing result
```

This eliminates external dependencies (NATS, Consul, Redis) and provides the same availability guarantees as the consensus layer itself.

5 Fully Homomorphic Encryption

5.1 ThresholdVM Architecture

ThresholdVM provides FHE operations as a native virtual machine on the T-Chain, enabling computation on encrypted data without decryption:

- **Schemes**: BFV (integer arithmetic), BGV (modular arithmetic), CKKS (approximate real), TFHE (Boolean circuits).
- **Threshold decryption**: t-of-n parties must cooperate; no single entity sees plaintext.
- **GPU acceleration**: NTT/iNTT, polynomial multiplication, and TFHE bootstrap operations accelerated via the Hanzo GPU kernel library's Metal/CUDA kernels.
- **EVM integration**: Precompiles at `0x0700`–`0x0703` (Fheos, ACL, InputVerifier, Gateway).

Definition 5.1 (Semantic Security under LWE)**.** For any PPT adversary $\mathcal{A}$, the advantage in distinguishing encryptions of two chosen messages is:

$$\mathrm{Adv}_{\mathcal{A}}^{\text{IND-CPA}}(\lambda) \leq \mathrm{negl}(\lambda)$$

under the Learning with Errors assumption with parameters (n, q, χ).

5.2 TFHE-KMS Bridge

The TFHE-KMS bridge enables encrypted policy evaluation:

1. Client encrypts sensitive query under FHE public key.
2. ThresholdVM evaluates policy predicate homomorphically.
3. Result (encrypted Boolean) is threshold-decrypted by MPC cluster.
4. Client receives only the policy decision, not the underlying data.

6 GPU-Accelerated Cryptography

The Hanzo GPU kernel library provides GPU kernels for cryptographic operations across Metal (Apple Silicon), CUDA (NVIDIA), and WebGPU:

Category	**Operations**
Elliptic Curves	BLS12-381, BN254, secp256k1, Ed25519: point add/double/scalar-mul, pairing, MSM
Post-Quantum	ML-DSA sign/verify, ML-KEM encaps/decaps, NTT/iNTT
ZK Proofs	KZG commitments, polynomial multiplication, Poseidon2 hashing
Hash Functions	SHA3-256/512, BLAKE3, Goldilocks field arithmetic
FHE	TFHE bootstrap, BFV/CKKS modular polynomial ops

Table 6: GPU-accelerated cryptographic operations.

The stable C ABI (`hz-accel/c_api.h`) enables FFI bindings from Go, Rust, and Python.

7 Zero-Knowledge Proof Systems

The ZK precompile suite (`0x0900`–`0x0932`) provides:

- **KZG4844**: Polynomial commitments for data availability (EIP-4844 compatible).
- **Groth16**: Succinct proofs with constant-size verification (∼200K gas).
- **PLONK/fflonk**: Universal setup proofs (∼250K gas).
- **Halo2**: Recursive proofs without trusted setup (∼300K gas).
- **Privacy pools**: Confidential UTXO model with nullifiers and range proofs.
- **ZK rollup**: Batch verification for off-chain computation (∼500K gas).

8 Key Management Architecture

8.1 Hanzo KMS

Enterprise key management with 8 core subsystems:

1. **Shamir Secret Sharing**: GF(2^8) arithmetic for root key distribution (t-of-n threshold).
2. **Transit Engine**: Encryption-as-a-service (AES-256-GCM, ChaCha20-Poly1305, Ed25519, ECDSA, RSA) with key versioning and auto-rotation.
3. **HPKE Key Wrapping**: RFC 9180 hybrid public key encryption for CEK distribution, including ML-KEM-768 + X25519 hybrid mode.
4. **Seal/Unseal Barrier**: AES-256-GCM encryption barrier with auto-unseal via AWS/GCP KMS.
5. **ACL Policy Engine**: Path-based capabilities (deny, CRUD, list, sudo) with template variables and LRU cache.
6. **Token System**: Service, batch, and recovery tokens with HMAC-SHA256 integrity and parent-child revocation cascade.
7. **Dynamic Secrets**: Auto-created temporary credentials for PostgreSQL, MySQL, Redis, MongoDB.
8. **TFHE-KMS Bridge**: FHE integration for encrypted policy evaluation.

8.2 Compliance Controls

- **HIPAA**: Break-glass emergency decryption tokens (time-limited), WORM hash-chained audit logs.
- **SEC 17a-4 / SOX**: 6–7 year retention enforcement, regulator escrow shards, immutable audit.
- **GDPR**: Per-org encryption keys, graceful secret revocation.
- **Cloud logging**: AWS CloudWatch + S3 Object Lock, GCP Cloud Logging + GCS WORM, Azure Monitor.

9 Ring Signatures and Anonymity

9.1 LatticeLSAG: Post-Quantum Ring Signatures

LatticeLSAG extends Linkable Spontaneous Anonymous Group signatures to ML-DSA-65:

- **Anonymity**: Verifier cannot determine which ring member signed.
- **Linkability**: Key images enable detection of double-signing (same signer produces same image).
- **Spontaneous**: Any set of public keys can form a ring without coordination.
- **Post-quantum**: Based on Module-LWE hardness, resistant to Shor's algorithm.
- **Application**: Anonymous reporting, covert intelligence sharing, whistleblower protection.

10 Software-Defined Networking Architecture

The combination of zero-trust overlay networking and hot-reloadable gateway configuration constitutes a software-defined networking (SDN) architecture with clear control plane / data plane separation.

10.1 Control Plane

- **Policy engine**: The Hanzo Gateway applies declarative routing, authentication, and rate-limiting policies defined in KMS-backed configuration. Policy updates propagate to all gateway instances within seconds without connection interruption.
- **Zero-trust controller**: The Hanzo ZT controller maintains the identity fabric—enrolling endpoints, issuing short-lived certificates, and distributing routing tables to edge routers. All control traffic is encrypted with ML-KEM-768 + X25519 hybrid key exchange.
- **Service mesh**: Mutual TLS between all services is enforced by the overlay, with certificate rotation on 15-minute intervals. Services are addressed by identity, not IP, eliminating lateral movement via network scanning.

10.2 Data Plane

- **Overlay transport**: Traffic between endpoints traverses encrypted tunnels (WireGuard or zero-trust fabric), with no reliance on network-layer security. The overlay is transparent to application protocols.
- **Policy-based routing**: Ingress and egress rules are evaluated per-packet based on caller identity, destination service, and request metadata. Routing decisions are made at the edge, not at a central choke point.
- **Hot reconfiguration**: Gateway routing tables, TLS certificates, and access control lists reload without process restart or connection drop, enabling continuous deployment of network policy changes under operational tempo.

This SDN architecture enables network segmentation, microsegmentation, and policy enforcement that adapts in real time to threat conditions—critical for contested electromagnetic environments where network topology may change rapidly.

11 Formal Verification

The architecture is supported by 17 formal cryptographic proofs:

Proof	Claim
proof-crypto-bls	BLS12-381 aggregate signature security
proof-crypto-cggmp21	CGGMP21 threshold ECDSA correctness
proof-crypto-mldsa	ML-DSA EU-CMA reduction to Module-LWE
proof-crypto-tfhe	TFHE semantic security under LWE
proof-crypto-verkle	Verkle tree binding and hiding
proof-protocol-wave	Wave consensus liveness and safety
proof-protocol-nova	Nova linear consensus termination
proof-bridge-teleport	Cross-chain message integrity
proof-warp-delivery	Warp messaging delivery guarantee

Table 7: Selected formal proofs (9 of 17 shown).

12 E2E Post-Quantum Integration Map

Layer	Classical	PQ Hybrid	Status
Consensus finality	BLS12-381	Corona ML-DSA-65	Live
RNS mesh transport	X25519	ML-KEM-768 + ML-DSA-65	Live
TLS 1.3 key exchange	X25519	X25519+ML-KEM-768	Live
SessionVM messaging	—	ML-KEM-768 + ML-DSA-65	Live
ZAP protocol	X25519	ML-KEM-768 + ML-DSA-65	Live
KMS encryption	HPKE-X25519	Hybrid HPKE	Implemented
EVM precompiles	ECDSA	ML-DSA, ML-KEM, SLH-DSA	Implemented

Table 8: End-to-end post-quantum integration status.

13 Naval Cyber Operations Scenarios

13.1 Defensive Cyber Operations (DCO)

- **HNDL protection**: All network traffic encrypted with hybrid PQ, rendering recorded ciphertext useless to quantum adversaries.
- **Key compromise recovery**: Epoch-based rotation limits exposure window to 10 minutes; threshold MPC ensures no single key compromise is catastrophic.
- **Anomaly detection**: O11y observability platform provides real-time trace analysis with LLM-powered threat classification.

13.2 Offensive Cyber Operations (OCO)

- **Penetration testing**: 83 specialized AI agents with multi-agent orchestration for automated vulnerability assessment.

- **Red team support**: Ring signatures enable anonymous operation; ZK proofs provide verifiable claims without revealing methods.
- **Mission-risk assessment**: Behavioral analytics platform with cohort analysis and predictive modeling.

13.2.1 Automated Penetration Testing Workflow

The agent-driven penetration testing pipeline executes five sequential phases, each coordinated by purpose-built AI agents operating under MPC-protected identity:

Algorithm 3 Agent-Driven Penetration Testing Pipeline

Phase 1 — Reconnaissance
DNS enumeration, port scanning, service fingerprinting
OSINT collection via 14 specialized reconnaissance agents
Attack surface mapping: subdomains, API endpoints, exposed services

Phase 2 — Vulnerability Scanning
Protocol-specific scanners (HTTP, SSH, TLS, SNMP, SMB)
CVE correlation against NVD and internal vulnerability databases
Configuration audit: default credentials, misconfigurations, weak ciphers

Phase 3 — Exploitation
Exploit selection ranked by CVSS score and operational risk
Sandboxed payload execution with rollback capability
Credential harvesting and privilege escalation attempts

Phase 4 — Lateral Movement
Network pivot discovery via ARP, LLMNR, and service enumeration
Pass-the-hash and token impersonation where applicable
Internal reconnaissance of newly accessible network segments

Phase 5 — Reporting
Automated MITRE ATT&CK technique mapping for every action taken
Evidence chain with cryptographic timestamps (SLH-DSA anchored)
Risk-scored findings with remediation recommendations

13.2.2 Red/Blue Team Automation

Adversarial agent pairs operate in continuous red/blue cycles:

- **Red agents**: Execute the penetration pipeline above, adapting tactics based on observed defenses. Agent orchestration supports 200–300 sequential tool calls per engagement, enabling deep vulnerability chain analysis across multi-hop attack paths.
- **Blue agents**: Monitor defensive telemetry in real time via the O11y observability platform, deploying countermeasures (firewall rules, credential rotation, service isolation) within seconds of detected intrusion.

- **Continuous cycle**: Red findings feed directly into blue defensive posture updates; blue countermeasures are re-tested by red agents in the next iteration, producing a closed-loop hardening process.
- **Scoring**: Each cycle produces an adversarial resilience score—the ratio of blocked vs. successful attack paths—tracked over time to measure defensive improvement.

13.2.3 Vulnerability Chain Analysis

Complex environments require analysis of multi-step attack chains rather than isolated vulnerabilities. Agent-driven chain analysis proceeds as follows:

1. **Graph construction**: Build a directed attack graph from reconnaissance and scanning data, where nodes represent hosts/services and edges represent exploitable transitions.
2. **Path enumeration**: Agents traverse the graph using depth-first search with pruning, executing 200–300 sequential tool calls to probe each link in a candidate chain.
3. **Chain validation**: Each candidate chain is validated end-to-end in a sandboxed environment to confirm exploitability under realistic conditions.
4. **Impact scoring**: Validated chains are scored by the product of individual link probabilities and the value of the terminal asset (crown jewel analysis).

13.2.4 Operator Identity Protection

Offensive operations require strong operator anonymity. LatticeLSAG ring signatures (Section 9) protect operator identity during all phases:

- Every tool invocation, exploit attempt, and reporting action is signed with a ring signature over the operator's unit roster.
- Verifiers confirm that an authorized operator performed the action without learning which specific individual.
- Key images prevent a single operator from filing duplicate or contradictory reports (linkability without identifiability).
- All signatures are post-quantum (ML-DSA-65 based), ensuring attribution protection against future quantum adversaries.

13.2.5 MITRE ATT&CK Integration

Every discovered vulnerability and exploitation step is automatically mapped to the MITRE ATT&CK framework:

- **Tactic coverage**: Agents tag each action with ATT&CK tactics (Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Exfiltration, Impact).
- **Technique resolution**: Individual techniques (e.g., T1059 Command and Scripting Interpreter, T1078 Valid Accounts) are recorded with sub-technique granularity.

- **Coverage heat map**: Aggregated results produce a heat map of tested vs. untested ATT&CK techniques, identifying gaps in defensive coverage.
- **Reporting**: Final reports include ATT&CK Navigator layers exportable to standard JSON format for integration with existing SOC tooling.

14 QuantumVM: Quantum Computing Operations

QuantumVM is a dedicated virtual machine on the Q-Chain that provides quantum key operations, quantum algorithm simulation, and hybrid quantum-classical workflow support as native blockchain primitives.

14.1 Post-Quantum Key Operations

QuantumVM exposes key lifecycle management for all three NIST PQ standards:

- **Key generation**: On-chain generation of ML-DSA-65, ML-KEM-768, and SLH-DSA key pairs with entropy sourced from the consensus randomness beacon. Keys are stored in threshold-encrypted form—no single validator holds a complete private key.
- **Verification**: Native transaction verification using PQ signatures, with gas costs calibrated to actual computational overhead (ML-DSA-65 verify: 75,000 gas; SLH-DSA verify: 50,000–250,000 gas depending on parameter set).
- **Key rotation**: Epoch-triggered rotation with configurable policy (time-based, usage-count, or external trigger). Rotation is atomic—old keys remain valid for a configurable grace period (default: 2 epochs) to allow in-flight operations to complete.
- **Certificate issuance**: QuantumVM issues short-lived PQ certificates binding identities to public keys, with certificate transparency logs anchored to the Q-Chain state root.

14.2 Quantum Algorithm Simulation

QuantumVM includes a classical simulation backend for quantum circuits, enabling pre-deployment validation of quantum algorithms:

- **Gate set**: Hadamard, CNOT, Toffoli, Phase, T, Rx/Ry/Rz, SWAP, and custom unitary gates up to 3 qubits.
- **Qubit capacity**: State vector simulation up to 30 qubits (single node) or 40 qubits (distributed across MPC cluster).
- **Noise models**: Depolarizing, amplitude damping, and phase flip channels for realistic fidelity estimation.
- **Deterministic execution**: Given identical inputs, simulation produces identical outputs across all validators—a requirement for consensus over quantum computation results.

14.3 Hybrid Quantum-Classical Workflows

Real-world quantum advantage requires tight integration between classical pre/post-processing and quantum subroutines:

Algorithm 4 Hybrid Quantum-Classical Execution on QuantumVM

1: **Classical preprocessing**: Prepare problem instance (e.g., Hamiltonian coefficients, optimization constraints)
2: **Circuit compilation**: Translate high-level algorithm to native gate set with depth optimization
3: **Quantum execution**: Run circuit on simulation backend (or relay to external QPU via oracle bridge)
4: **Measurement**: Collapse state vector; record measurement outcomes on-chain
5: **Classical postprocessing**: Aggregate results, compute expectation values, update variational parameters
6: **Iteration**: Repeat steps 2–5 for variational algorithms (VQE, QAOA) until convergence

The oracle bridge interface allows QuantumVM to submit circuits to external quantum processing units (QPUs) when available, with results verified by threshold consensus before acceptance.

14.4 EVM Precompile Integration

QuantumVM operations are accessible from EVM smart contracts via precompiles at addresses `0x0800`–`0x080F`:

Address	Operation	Gas
`0x0800`	PQ key generation (ML-DSA-65)	200,000
`0x0801`	PQ key generation (ML-KEM-768)	150,000
`0x0802`	PQ key generation (SLH-DSA)	300,000
`0x0803`	PQ signature verification	75,000–250,000
`0x0804`	Key rotation (atomic swap)	100,000
`0x0808`	Quantum circuit submission	500,000+
`0x0809`	Measurement result retrieval	50,000
`0x080A`	Hybrid workflow orchestration	1,000,000+

Table 9: QuantumVM EVM precompile addresses and gas costs.

15 Strategic Context

The post-quantum architecture detailed here is built and owned end-to-end by Hanzo AI, an American-owned applied-AI lab and venture studio (Techstars '17, founded 2014) responsible for more than 100 venture-funded startups and several unicorns, founded by an expert in both AI/ML and cryptography. Two facts make this relevant to national-security buyers. First, sovereignty of the cryptography itself: a post-quantum migration is only as trustworthy as the party that controls the implementation, and this stack is American-owned, auditable, and free of dependence on foreign or black-box components—the same standards (FIPS 203/204/205, CNSA 2.0) the U.S. government mandates, implemented by a U.S. party that can be inspected. Second, sovereignty of the AI that increasingly rides on top of that cryptography: U.S. frontier AI has narrowed to a closed commercial

duopoly, while the only broadly open frontier weights available today are largely foreign, leaving regulated and defense buyers to either rent from the duopoly or take a dependency on foreign models. Hanzo is the American open-source frontier alternative—it owns both the Zen model family and this cryptographic stack, so the consensus, key-management, and confidential-compute layers above can be deployed offline, at the edge, and under full audit. That this is engineering rather than research is evidenced in production: the same agentic stack delivered a FINRA/SEC-approved, SOC2 securities platform—a full ATS/BD/TA stack consolidated from 200+ microservices into 3 compiled binaries—to production with a small team, and the efficiency primitives described here (the ZAP transport reaching roughly 11.5M transactions per second with order-of-magnitude memory and energy reductions versus JSON-RPC) translate directly into lower infrastructure cost and viable disconnected operation.

16 Conclusion

We have presented a production-grade cryptographic architecture that fully implements all three NIST post-quantum standards with hybrid classical/quantum defense-in-depth. The system provides sub-second consensus finality with dual BLS/Corona certificates, six threshold MPC schemes, GPU-accelerated cryptographic operations, fully homomorphic encryption, and comprehensive key management with regulatory compliance.

The architecture is not theoretical—it is deployed, tested, and operational, with 17 formal proofs supporting its security claims. It represents the most comprehensive post-quantum cryptographic infrastructure available for naval information dominance operations.

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Reproducibility

Source code. `github.com/hanzoai/agents, hanzoai/zap` (commit `TBD`).
Build. make build per component; OpenZiti overlay deployed via Helm.
Hardware tested. Linux amd64/arm64; tested in air-gapped lab environments.
Standards referenced. OpenZiti, NIST PQC, MITRE ATT&CK.
Contact. research@hanzo.ai.

Machine-Checked Proofs and Automated Verification for Post-Quantum Distributed Systems

Zach Kelling — Hanzo Team

Version 1.0 — February 28, 2026 February 2026

Abstract

We present a comprehensive formal verification suite for a post-quantum distributed system encompassing blockchain consensus, threshold cryptography, fully homomorphic encryption, and cross-chain messaging. The suite comprises 50 machine-checked proofs in Lean 4 covering consensus safety and liveness, 18 cryptographic security reductions (including ML-DSA EU-CMA, ML-KEM IND-CCA2, and Corona lattice threshold unforgeability), 12 protocol correctness proofs, and 7 DeFi invariant proofs. Complementary tooling includes Halmos symbolic execution for EVM smart contracts, 23+ fuzz targets for parser and cryptographic implementations, and NIST Known Answer Test vector validation for all three FIPS 203/204/205 algorithms. We identify the critical gaps—TLA+ model checking for consensus liveness, Tamarin protocol verification for the novel hybrid post-quantum handshake, and constant-time verification for lattice cryptography implementations—and present a phased roadmap for closing them. This work demonstrates the deepest formal verification coverage of any production post-quantum blockchain system.

Executive Summary. Most software is trusted because it was tested and nothing broke. This work goes further: the core security claims are proved as mathematical theorems and re-checked by a machine on every code change, so a regression cannot ship undetected. The proofs cover the properties an accreditor actually asks about—that the system cannot reach two conflicting decisions, that the quantum-resistant cryptography is sound, that the money-handling logic stays solvent—and they target the new post-quantum standards mandated for national-security systems. For a program manager or accreditation authority, this is evidence that can go directly into an Authority-to-Operate package: not "we tested it," but "here is a machine-checked proof, and here is exactly the set of assumptions it rests on." The paper is candid about what is proved today versus what is on the roadmap (model checking and protocol analysis), so reviewers can see the true assurance boundary rather than a marketing one.

Contents

1 Introduction

Formal verification provides mathematical certainty that software behaves according to its specification. For defense systems where failure modes include loss of life, compromise of classified information, or mission failure, formal methods are not optional—they are essential.

This paper documents the formal verification infrastructure supporting the Hanzo post-quantum distributed system, assesses its completeness, and presents a roadmap for achieving the level of assurance required for naval deployment.

1.1 Verification Layers

We employ four complementary verification techniques:

1. **Theorem proving** (Lean 4): Machine-checked proofs of security properties, reduction proofs, and protocol invariants.
2. **Symbolic execution** (Halmos): Automated invariant checking for EVM smart contracts via SMT solving.
3. **Fuzz testing**: Automated discovery of crashes, panics, and edge cases in parsers and cryptographic implementations.
4. **Test vector validation**: NIST KAT vectors ensuring implementation correctness against reference implementations.

1.2 Scope

Category	**Proofs**	**Tool**
Consensus (safety, liveness, BFT)	6	Lean 4
Cryptography (classical + PQ)	18	Lean 4
Protocols (10 consensus sub-protocols)	12	Lean 4
DeFi invariants	7	Lean 4
Cross-chain (Warp messaging)	3	Lean 4
Bridge (Teleport)	1	Lean 4
Trust and identity	3	Lean 4
Smart contracts	3	Halmos
Total	**53**	

Table 1: Formal verification coverage by category.

1.3 Strategic Context

Formal verification is where owning the whole stack pays off. Hanzo (American-owned, Techstars '17, founded 2014) builds both the post-quantum cryptography and the frontier AI—the *Zen* model family—that depend on the properties proved here, which is what makes a machine-checked claim meaningful: there is no third-party black box whose internals are out of reach of the prover. This matters because of where trustworthy AI now comes from. America's frontier AI has narrowed to a closed two-vendor duopoly, while the broad *open* ecosystem—and therefore most openly auditable

frontier weights—now sits offshore. A defense or regulated operator who needs to inspect, accredit, and run a system on sovereign infrastructure is forced to choose between renting from a closed American duopoly or depending on foreign open weights. Hanzo is the American open-source alternative that owns the model and the cryptography together; the verification suite documented here is the auditable evidence that backs that claim—formal proof an accreditor can read rather than a vendor assurance they must take on faith.

2 Lean 4 Proof Library

The proof library resides at `hanzo/formal/lean/` and contains 50 Lean 4 files organized by domain.

2.1 Consensus Proofs

File	Theorem
`Consensus/Safety.lean`	No two honest nodes finalize conflicting values
`Consensus/Finality.lean`	End-to-end finality composition (BLS + Corona)
`Consensus/Liveness.lean`	Bounded-time finality under synchrony
`Consensus/BFT.lean`	Byzantine fault tolerance: $f < n/3$
`Consensus/Quasar.lean`	Quantum-safe dual-signature finality
`Consensus/Validator.lean`	Staking, slashing bounds, voting power monotonicity

Table 2: Consensus proof suite.

Theorem 2.1 (Safety — Lean 4)**.** *For any two honest validators v_1, v_2 and blocks B_1, B_2 finalized at the same height h:*

$$\mathit{finalized}(v_1, B_1, h) \wedge \mathit{finalized}(v_2, B_2, h) \implies B_1 = B_2$$

Proved from axioms: `finalized_preference_stable`, `unique_threshold`, `finalization_windows_overlap`.

2.2 Cryptographic Proofs

2.2.1 Post-Quantum (6 proofs)

- `Crypto/MLDSA.lean`: ML-DSA-65 EU-CMA security under Module-LWE/SIS assumption (FIPS 204).
- `Crypto/MLKEM.lean`: ML-KEM-768 IND-CCA2 security via Fujisaki-Okamoto transform (FIPS 203).
- `Crypto/Corona.lean`: Lattice threshold signature unforgeability under LWE.
- `Crypto/SLHDSA.lean`: SLH-DSA security from hash function second-preimage resistance only (FIPS 205).
- `Crypto/Hybrid.lean`: Composition theorem—hybrid scheme (Ed25519 + ML-DSA + SLH-DSA) is secure if *any* component is secure.

- `Crypto/FHE/TFHE.lean`: TFHE semantic security under Ring-LWE with bootstrapping correctness.

2.2.2 Classical (5 proofs)

- `Crypto/BLS.lean`: BLS12-381 aggregate signature security under co-CDH in bilinear groups.
- `Crypto/FROST.lean`: FROST threshold Schnorr signature security under discrete log.
- `Crypto/FHE/CKKS.lean`: CKKS approximate arithmetic FHE correctness with noise bounds.
- `Crypto/VerkleTree.lean`: Verkle tree binding and hiding properties.

2.2.3 Threshold (3 proofs)

- `Crypto/Threshold/LSS.lean`: Shamir/VSS correctness and composability.
- `Crypto/Threshold/CGGMP21.lean`: UC-secure threshold ECDSA with identifiable abort.
- `Crypto/Threshold/Composition.lean`: Threshold protocol composition bounds.

2.3 Protocol Proofs (12)

One proof per consensus sub-protocol: Wave, Ray, Nova, Field, Flare, Prism, Photon, Nebula, Quasar, Handshake. Plus:

- `Protocol/Handshake.lean`: Hybrid PQ key exchange (X25519 + ML-KEM-768) state machine correctness.
- `Warp/Delivery.lean`: Exactly-once cross-chain message delivery with nonce replay protection.
- `Warp/Ordering.lean`: Cross-chain message ordering guarantees.
- `Warp/Security.lean`: Warp message authentication and integrity.

2.4 DeFi Proofs (7)

- `DeFi/AMM.lean`: Constant product invariant $x \cdot y = k$ preserved under swaps and fees.
- `DeFi/LiquidStaking.lean`: Monotone yield, no impermanent loss.
- `DeFi/OrderBook.lean`: Price-time priority, deterministic matching.
- `DeFi/Router.lean`: Cross-pool routing slippage bounds.
- `DeFi/HFT.lean`: Geographic moat economics.
- `DeFi/Governance.lean`: DAO parameter changes require quorum + timelock.
- `DeFi/FeeModel.lean`: Maker-taker fee tier correctness.

2.5 Proof Methodology

All proofs follow a consistent structure:

1. **Axioms**: Cryptographic hardness assumptions (Module-LWE, discrete log, hash collision resistance) are stated as Lean axioms. This is standard—proving these from first principles is open research.
2. **Definitions**: Protocol state machines, message types, and security games formalized as Lean structures.
3. **Theorems**: Security properties proved from axioms using `simp`, `omega`, `apply`, `induction`.
4. **Correspondence**: Each `.lean` file has a matching `.tex` paper in `hanzo/papers/proofs/` documenting the proof in traditional mathematical notation.

Approximately 130 axioms across 50 files. The axioms encode the *assumptions* under which the system is secure; the theorems prove that security *follows* from these assumptions.

3 Symbolic Execution: Halmos

Halmos provides automated invariant verification for EVM smart contracts via symbolic execution and SMT solving.

3.1 Current Test Suite

Test	Invariant Verified
`AMMInvariant.t.sol`	$k = r_0 \cdot r_1$ never decreases; LP shares proportional
`LiquidSolvency.t.sol`	Liquid staking redemption always solvent
`MarketsSolvency.t.sol`	Lending protocol total borrows $\leq$ total deposits

Table 3: Halmos symbolic execution test suite.

Halmos uses the Foundry test format, requiring no separate specification language. Tests are written as Solidity functions with symbolic inputs; the SMT solver exhaustively checks all possible execution paths.

4 Fuzz Testing Infrastructure

4.1 Existing Fuzz Targets

Component	Location	Targets
CB58 encoding	`hanzo/crypto/cb58/`	2
secp256k1 signatures	`hanzo/crypto/secp256k1/`	3
IPA proofs	`hanzo/crypto/ipa/`	2
BLAKE2b hashing	`hanzo/crypto/hash/blake2b/`	1
BFT consensus	`hanzo/bft/`	4
MerkleDB codec	`hanzo/node/x/merkledb/codec/`	5
Compression	`hanzo/node/utils/compression/`	3
RPC database	`hanzo/node/database/rpcdb/`	2
Deque	`hanzo/node/utils/buffer/deque/`	1
Total		**23**

Table 4: Existing fuzz test targets.

4.2 Proposed Expansion

Target	Location	Bug Class
ZAP deserializer	`hanzo/zap/`	Parsing crashes, buffer overflows
ML-DSA sign/verify	`hanzo/crypto/mldsa/`	Malformed signature acceptance
ML-KEM encaps/decaps	`hanzo/crypto/mlkem/`	Malformed ciphertext crashes
Corona threshold	`hanzo/crypto/threshold/`	Malformed share acceptance
BLS aggregation	`hanzo/crypto/bls/`	Rogue key attacks
FHE ciphertext parsing	`hanzo/fhe/go/tfhe/`	Malformed ciphertext
ZKVM proof verifier	`hanzo/node/vms/zkvm/`	Invalid proof acceptance
RNS link protocol	`hanzo/node/network/dialer/`	Handshake state machine
SessionVM messages	`hanzo/session/`	Message parsing

Table 5: Proposed fuzz target expansion (9 new targets, bringing total to 32+).

Properties tested: roundtrip correctness (`decode(encode(x)) == x`), no-panic on arbitrary input, commutativity of aggregation, threshold safety ($t - 1$ shares never recover secret).

5 NIST Test Vector Validation

NIST provides Known Answer Test (KAT) vectors for all three post-quantum standards. These verify that our implementations produce identical outputs to the reference implementations for given

inputs.

Algorithm	Standard	Library	Vectors
ML-DSA-44/65/87	FIPS 204	CIRCL v1.6.1	3 modes
ML-KEM-512/768/1024	FIPS 203	CIRCL v1.6.1	3 modes
SLH-DSA (12 variants)	FIPS 205	CIRCL v1.6.1	12 param sets

Table 6: NIST KAT vector coverage.

KAT validation is the minimum bar for cryptographic implementation correctness. It catches byte-ordering errors, padding bugs, and modular arithmetic mistakes that unit tests may miss.

6 Gap Analysis

6.1 Critical Gaps

Gap	Risk	Mitigation	Effort
No TLA+ model checking	Consensus liveness bugs, parameter errors	TLA+ + Apalache for Wave	4–6 pw
No protocol verification	PQ handshake MITM, downgrade	Tamarin model of RNS	3–4 pw
Axiom-heavy Lean proofs	Logical gaps in safety proof	Mechanize key axioms	3–4 pw
No constant-time verification	Side-channel key extraction	dudect timing analysis	2 pw
No ZKVM adversarial testing	Soundness bugs (catastrophic)	Crafted invalid proofs	2–3 pw

Table 7: Critical formal verification gaps.

6.2 What No One Has Done

Several gaps reflect the state of the art, not just our implementation:

- **No Quasar-family TLA+ spec exists** anywhere. Amores-Sesar *et al.* (2024) provided the first formal consistency proof of Quasar, but no TLA+ model has been published.
- **No machine-checked ML-DSA proof** exists in Lean. EasyCrypt proofs exist for ML-KEM (SandboxAQ, 2024) but not for the full FIPS 204 standard.
- **No hybrid BLS+lattice consensus** has been formally verified by anyone.
- **No TFHE correctness proof** exists in any proof assistant.

Addressing any of these would be a **first-of-kind** contribution.

7 Verification Roadmap

7.1 Phase 1: Automated Testing (Weeks 1–4)

1. Expand fuzz targets from 23 to 50+ covering ZAP, all PQ crypto, ZKVM, FHE, SessionVM.
2. Run NIST KAT vectors against all ML-DSA, ML-KEM, and SLH-DSA implementations.
3. Run `dudect` timing analysis on PQ crypto hot paths (sign, verify, encaps, decaps).
4. Adversarial proof verifier testing: craft 100+ invalid ZK proofs to test rejection.

Estimated effort: 10–14 person-weeks.

7.2 Phase 2: Model Checking (Weeks 5–10)

5. Write TLA+ specification of Wave+FPC consensus protocol.
6. Model check with Apalache for safety and bounded liveness.
7. Write Tamarin model of RNS hybrid PQ handshake.
8. Verify: IND-CCA2 key exchange, mutual authentication, forward secrecy, no downgrade.
9. Extend Halmos to FHE precompile contracts.

Estimated effort: 10–16 person-weeks.

7.3 Phase 3: Proof Strengthening (Weeks 11–16)

10. Mechanize `unique_threshold` axiom: prove from quorum intersection properties.
11. Mechanize `finalization_windows_overlap`: prove from timing assumptions.
12. Property-based testing of FHE noise budget bounds.
13. Bounded model checking of epoch rotation state machine.

Estimated effort: 8–12 person-weeks.

7.4 Total Investment

Phase	Effort	Duration	Deliverable
Phase 1: Automated testing	10–14 pw	4 weeks	Fuzz + KAT + timing
Phase 2: Model checking	10–16 pw	6 weeks	TLA+ + Tamarin
Phase 3: Proof strengthening	8–12 pw	6 weeks	Mechanized axioms
Total	**28–42 pw**	**16 weeks**	

Table 8: Formal verification roadmap investment.

8 Comparison with Industry

Project	Theorem Proofs	Model Check	Sym. Exec	Fuzz
Ethereum (EF)	–	TLA+ (Gasper)	Halmos	OSS-Fuzz
Tendermint	–	TLA+ (Apalache)	–	go-fuzz
Solana	–	–	–	Limited
Avalanche	–	–	–	Limited
Hanzo (ours)	**50 Lean 4**	**Planned**	**Halmos**	**23+ targets**

Table 9: Formal verification comparison with major blockchain projects.

Hanzo is the only blockchain project with machine-checked proofs of post-quantum cryptographic properties. No other project has Lean 4 proofs of ML-DSA, ML-KEM, or lattice threshold signatures.

9 Defense Application

9.1 CNSA 2.0 Compliance Verification

The Lean proof suite provides machine-checked evidence that the system's cryptographic foundations meet CNSA 2.0 requirements:

- ML-KEM-768 IND-CCA2 proof (FIPS 203 compliance).
- ML-DSA-65 EU-CMA proof (FIPS 204 compliance).
- SLH-DSA security from hash assumptions only (FIPS 205 compliance).
- Hybrid composition theorem: system secure if *any* PQ algorithm holds.

9.2 Accreditation Support

- Machine-checked proofs provide formal evidence for Authority to Operate (ATO) packages.
- Halmos symbolic execution results demonstrate smart contract invariant preservation.
- Fuzz test results document robustness against malformed inputs.
- KAT validation documents algorithm implementation correctness.

9.3 Continuous Assurance

- Lean proofs are checked on every commit via CI/CD.
- Halmos tests run as part of the standard test suite.
- Fuzz tests run continuously in CI with crash reporting.
- Any proof regression immediately blocks deployment.

10 Conclusion

We have presented the most comprehensive formal verification suite for any production post-quantum distributed system: 50 machine-checked Lean 4 proofs, Halmos symbolic execution, 23+ fuzz targets, and NIST KAT validation. The identified gaps—TLA+ model checking, Tamarin protocol verification, and constant-time analysis—represent a focused 16-week investment that would yield first-of-kind results in consensus model checking and hybrid PQ handshake verification.

For defense applications, this verification infrastructure provides the formal evidence required for system accreditation, continuous assurance during operation, and confidence that post-quantum cryptographic migrations preserve the security properties of the original system.

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Reproducibility

Source code. `github.com/hanzoai/consensus, hanzoai/quasar` (commit `TBD`).
Build. go test ./... and rust-based fuzzers via cargo test.
Hardware tested. Linux amd64, deterministic test environment.
Standards referenced. NIST IR 8214C (Threshold Crypto), LOKI consensus fuzzing.
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Agentic AI for Autonomous Cyber Operations and Multi-Domain Intelligence Fusion

Zach Kelling — Hanzo Team

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Abstract

We present an agentic AI framework comprising 83 specialized agents and 15 multi-agent orchestrators for autonomous cyber operations, intelligence fusion, and tactical decision support. The framework operates across three tiers: (1) a Playground control plane providing Kubernetes-like orchestration for autonomous AI agents with W3C DID identity, verifiable credentials, and durable state; (2) an agent SDK with async-first execution, OpenAI-compatible APIs, and native Model Context Protocol (MCP) integration exposing 13 canonical tools; and (3) a ZAP-based consensus protocol enabling distributed agent voting with threshold agreement and post-quantum authentication. Agents execute 200–300 sequential tool calls for deep analysis, coordinate via zero-copy binary RPC at 4.2M calls/second, and operate fully disconnected on edge hardware using quantized Zen models (0.8B–27B parameters). The system supports automated red/blue team operations, vulnerability chain analysis, multi-source intelligence correlation, and course-of-action generation with explainable reasoning via chain-of-thought traces. A multi-channel bot platform unifies 20+ messaging channels under local-first privacy controls for secure operator interaction.

Executive Summary. This framework lets a team field a fleet of AI agents that investigate, defend, and analyze on their own—running on a laptop or a rugged edge device with no connection to the cloud. It is built for cyber-operations teams and intelligence analysts who must act at machine speed and cannot afford to ship sensitive data to an outside service. Every agent action is cryptographically signed, so there is a complete, tamper-proof record of what each agent did and why; and high-stakes calls require several independent agents to agree before anything is recommended, which guards against a single model's mistake. Operationally, this means continuous vulnerability hunting, automated red/blue exercises, and fused multi-source intelligence with a human always in the loop for final decisions—all owned and run by the operator, not rented from a remote provider. The result is autonomous capability without surrendering control, accountability, or the data itself.

Contents

1 Introduction

Modern naval operations generate intelligence from sensors, communications, cyber systems, and open sources at volumes exceeding human analytical capacity. Simultaneously, the cyber domain requires continuous vulnerability assessment, threat hunting, and incident response at machine speed. Traditional automation—rule-based scripts and playbooks—cannot adapt to novel threats or fuse information across domains.

Agentic AI addresses these challenges through autonomous agents that reason, plan, use tools, and collaborate. Unlike chatbots that respond to queries, agents take initiative: they decompose complex tasks, execute multi-step plans, invoke external tools, and coordinate with other agents to produce actionable intelligence.

1.1 Design Principles

1. **Autonomous but accountable**: Every agent action is signed with a W3C DID and produces a verifiable credential. Full audit trail, zero deniability.

2. **Disconnected-capable**: Agents run entirely on edge hardware with local models. No cloud dependency.

3. **Consensus-validated**: High-consequence decisions require threshold agreement among multiple agents, preventing single-agent hallucination.

4. **Tool-native**: Agents interact with systems through 13 canonical MCP tools, not bespoke integrations.

1.2 Strategic Context

This agent framework is one layer of an integrated stack built by Hanzo, an American applied-AI lab founded in 2014 (Techstars '17) whose founder works at the frontier of both AI/ML and cryptography. American frontier AI has consolidated into a closed monoculture—effectively two labs—while the only openly available frontier model weights today are largely foreign. Hanzo is the American open-source frontier alternative: it owns both the model (the Zen family, 0.8B–1T+ parameters, with owned weights) and the cryptography (a production post-quantum stack), so defense and regulated users get sovereign, open, auditable AI without renting from the closed duopoly or depending on foreign open weights. For autonomous cyber and intelligence work that requirement is acute—the agents here reason over the most sensitive data an organization holds, so the model performing that reasoning must be one the operator can own, inspect, and run disconnected, on their own hardware, where the data already lives. The same agentic stack is what makes this practical at scale: Hanzo used it to deliver a FINRA/SEC-approved, SOC2-compliant securities platform to production—a full ATS/BD/TA stack consolidated from 200+ microservices into 3 compiled binaries, in record time, with a small team—demonstrating that agent-driven engineering meets the bar of a heavily regulated, audited domain.

2 Agent Architecture

2.1 Specialization Taxonomy

Domain	Count	Capabilities
Security	12	Vulnerability assessment, pen testing, incident response, malware analysis, SIEM correlation
Infrastructure	11	K8s operations, network diagnostics, database administration, cloud provisioning
Data/ML	10	Data engineering, ML pipelines, model evaluation, feature engineering, dataset curation
Languages	15	Rust, Go, Python, TypeScript, C++, Java, Solidity, and 8 more
Architecture	8	System design, API design, microservices, distributed systems
QA/Testing	7	Unit testing, integration testing, load testing, security testing
Domain	20	Blockchain, DeFi, cryptography, NLP, computer vision, signal processing

Table 1: 83 specialized agents by domain.

2.2 Multi-Agent Orchestrators

Orchestrator	Workflow
Security Hardening	Red agent attacks → blue agent defends → red re-tests fixes
Incident Response	Triage → containment → eradication → recovery → lessons
ML Pipeline	Data prep → training → evaluation → deployment → monitoring
Full-Stack Dev	Architecture → backend → frontend → testing → deployment
Intelligence Fusion	Collection → processing → analysis → dissemination

Table 2: Selected multi-agent orchestrators (5 of 15).

2.3 Agent Execution Model

Algorithm 1 Agent Task Execution

Input: Task description T, available tools $\mathcal{F}$, context C
Initialize: Agent identity (DID), model (Zen), memory store
repeat
    Reason about current state and remaining subtasks
    Select tool $f \in \mathcal{F}$ and generate arguments
    Execute tool: $r \leftarrow f(\text{args})$
    Sign action: $\sigma \leftarrow \text{ML-DSA-65.Sign}(\text{sk}, T \| f \| r)$
    Update memory with observation r
    Append to audit trail: $(T, f, r, \sigma, \text{timestamp})$
until task complete or max steps reached (200–300 calls)
return (result, audit_trail, verifiable_credential)

3 Playground: Agent Control Plane

3.1 Architecture

Playground provides Kubernetes-like orchestration for autonomous AI agents:

- **Registration**: Agents register with capabilities, DID, and stake weight.
- **Discovery**: Other agents and services discover registered agents by capability.
- **Scheduling**: Tasks dispatched to agents based on capability match and availability.
- **Durable state**: Built-in persistent state and vector search (no external Redis or vector DB).
- **Lifecycle**: Health monitoring, graceful shutdown, automatic restart.

3.2 Identity and Accountability

Every agent in Playground possesses:

- **W3C DID**: Decentralized identifier derived from ML-DSA-65 public key (`did:key:z6Mk...`).
- **Verifiable Credentials**: Every action produces a DID-signed VC as a tamper-proof receipt.
- **Capability assertions**: Agent advertises what it can do; Playground verifies via credential chain.
- **Policy boundaries**: "Only agents with `security` credential can access vulnerability tools" — enforced at infrastructure level, not application level.

3.3 Performance

Runtime	Throughput	Overhead/Handler
Go	8.2M req/s	280 B
Python	6.7M req/s	512 B
TypeScript	4.0M req/s	384 B

Table 3: Playground control plane performance.

4 Model Context Protocol Integration

4.1 13 Canonical Tools

Agents interact with systems through the Model Context Protocol (MCP), providing a standardized tool interface:

Tool	Type	Capability
`fs`	Core	File system operations (read, write, search)
`exec`	Core	Shell command execution (sandboxed)
`code`	Core	AST analysis, symbol search, refactoring
`git`	Core	Version control operations
`fetch`	Core	HTTP requests, API calls
`workspace`	Core	Project context management
`ui`	Core	User interface interaction
`think`	Optional	Internal reasoning steps
`memory`	Optional	Persistent memory across sessions
`hanzo`	Optional	Platform API (IAM, KMS, PaaS)
`plan`	Optional	Task decomposition and planning
`tasks`	Optional	Task tracking and progress
`mode`	Optional	Execution mode control

Table 4: MCP canonical tool set (HIP-0300).

4.2 MCP-ZAP Bridge

The ZAP bridge provides 10–30× performance improvement over native MCP JSON-RPC:

- **Binary encoding**: Tool name (64 B fixed), arguments (length-prefixed JSON), result (zero-copy bytes).
- **Message types**: ToolList (100), ToolCall (101), ToolResult (102).
- **Auto-discovery**: Bridge queries MCP server capabilities and exposes via ZAP.

- **20-server aggregation**: Single ZAP endpoint fronts 20+ MCP servers with prefix namespacing.

4.3 Search Strategy

MCP implements multi-strategy code and data search:

Strategy	Priority	Engine	Triggers
Symbol	10	TreeSitter AST	`class`, `function`, `def`
AST	20	TreeSitter	Code-like patterns
Vector	30	LanceDB	Natural language queries
Text	40	Ripgrep	Default fallback
File	50	Glob	Paths, extensions

Table 5: MCP multi-strategy search with priority dispatch.

5 Distributed Agent Consensus

5.1 Threshold Voting Protocol

For high-consequence decisions, multiple agents must independently reach the same conclusion:

Algorithm 2 Agent Consensus for Threat Classification

Coordinator broadcasts sensor data D to n analyst agents
for each agent A_i in parallel **do**
    A_i independently analyzes D using assigned model
    A_i produces classification C_i with confidence p_i
    A_i signs: $\sigma_i \leftarrow \text{ML-DSA-65.Sign}(\text{sk}_i, D \| C_i \| p_i)$
    A_i returns $(C_i, p_i, \sigma_i, \text{DID}_i, \text{stake}_i)$
end for
Verify all signatures against registered DIDs
Compute weighted consensus: $\hat{C} = \arg\max_c \sum_{i:C_i=c} \text{stake}_i$
if $\sum_{i:C_i=\hat{C}} \text{stake}_i \geq \frac{2}{3} \sum_i \text{stake}_i$ **then**
    return (CONSENSUS, $\hat{C}$, aggregated evidence)
else
    return (NO_CONSENSUS, all (C_i, p_i), escalate to human)
end if

5.2 Defense Applications

- **Threat classification**: 3+ agents independently assess sensor contact; consensus prevents single-model hallucination from triggering false alarms.
- **Targeting**: Distributed analysis with $\frac{2}{3}$ threshold before engagement recommendation. Human remains in loop for final decision.

- **Intelligence corroboration**: Agents processing different SIGINT, IMINT, and OSINT sources; consensus identifies corroborated vs. single-source intelligence.
- **Cyber attribution**: Multiple agents analyze malware, network indicators, and TTPs independently; consensus reduces false attribution risk.

6 Autonomous Cyber Operations

6.1 Defensive Cyber Operations (DCO)

6.1.1 Continuous Vulnerability Assessment

1. Security agent scans codebase using MCP `code` tool (AST analysis, dependency audit).
2. Agent identifies vulnerability class (injection, XSS, auth bypass, crypto weakness).
3. Agent generates proof-of-concept exploit via `exec` tool (sandboxed).
4. Agent proposes fix, generates test case, and validates fix eliminates vulnerability.
5. Full workflow signed and recorded as verifiable credential.

6.1.2 Threat Hunting

1. O11y anomaly detection triggers agent investigation.
2. Agent queries distributed traces via MCP `fetch` tool.
3. Agent correlates indicators across logs, metrics, and traces.
4. Agent classifies threat using MITRE ATT&CK framework.
5. Multi-agent consensus validates classification before alert.

6.1.3 Incident Response Orchestrator

The 5-phase incident response orchestrator coordinates specialized agents:

Phase	Agent	Actions
Triage	Security analyst	Severity assessment, scope determination
Contain	Infrastructure	Network isolation, credential rotation
Eradicate	Malware analyst	Root cause removal, persistence check
Recover	DevOps	Service restoration, integrity verification
Lessons	Architecture	Post-incident report, control improvement

Table 6: Incident response orchestrator phases.

6.2 Offensive Cyber Operations (OCO)

6.2.1 Automated Penetration Testing

1. Reconnaissance agent maps attack surface via MCP `fetch` and `exec` tools.
2. Vulnerability agent identifies exploitable weaknesses in discovered services.
3. Exploitation agent chains vulnerabilities into attack paths.
4. Lateral movement agent assesses post-exploitation pivot opportunities.
5. Reporting agent produces findings with severity, evidence, and remediation.
6. **Operator identity**: Ring signatures (LatticeLSAG) protect pen tester identity during operations.

6.2.2 Red/Blue Team Automation

Adversarial pairs operate in continuous cycles:

Algorithm 3 Red/Blue Agent Cycle

repeat
  Red agent identifies attack vector
  Red agent develops and executes exploit (sandboxed)
  Blue agent receives attack indicators
  Blue agent develops detection rule and mitigation
  Red agent adapts attack to evade new detection
  Blue agent refines defense based on adapted attack
  Record cycle results as verifiable credentials
until convergence (red cannot bypass blue) or max iterations
return (attack report, defense posture, residual risk)

7 Intelligence Fusion

7.1 Multi-Source Correlation

Agents process intelligence from heterogeneous sources:

INT Type	Agent Capability	MCP Tools Used
SIGINT	NLP agent with 32+ language support	`code` (pattern matching), `fetch`
IMINT	Vision-language model (Zen VL)	`fs` (image access), `code` (analysis)
OSINT	Web search and document exploitation	`fetch` (HTTP), `fs` (documents)
CYBER	Network traffic analysis, malware RE	`exec` (tools), `code` (analysis)
HUMINT	Report processing and entity extraction	`code` (NER), `memory` (context)

Table 7: Intelligence source processing by agent type.

7.2 Graph-Based Fusion

The GraphVM provides in-memory graph analytics for intelligence correlation:

- **Entity resolution**: Link entities across SIGINT, IMINT, and OSINT sources.
- **Network analysis**: Identify communication patterns, social networks, and organizational hierarchies.
- **Temporal analysis**: Track entity behavior changes over time.
- **Geospatial correlation**: Associate entities with locations from multiple sources.
- **Merkle-verified**: Graph state is Merkle-proofed for integrity verification.

8 Multi-Channel Operator Interface

8.1 OpenClaw Bot Platform

OpenClaw provides a privacy-first, local-first multi-channel AI assistant:

- **20+ messaging channels**: WhatsApp, Telegram, Slack, Discord, Signal, Matrix, iMessage, and more.
- **Local-first**: All processing on operator's device. No cloud routing of messages.
- **Gateway control plane**: WebSocket-based session management, channel routing, tool dispatch.
- **Policy enforcement**: DM pairing, group routing rules, channel allowlists, mention gating.
- **Voice interface**: Wake word activation, Talk Mode (macOS/iOS/Android).
- **Skills platform**: Bundled, managed, and workspace-specific skill sets.

8.2 Defense Application

- Operators interact with AI agents via their preferred secure messaging app (Signal, Matrix).
- Agent responses routed through local gateway—never traverses external servers.
- Multi-channel unification: single AI assistant across all communication platforms.
- Group messaging with policy-enforced access controls per channel.

9 Explainable AI and Transparency

9.1 Chain-of-Thought as Explainability

Zen models with extended reasoning (96–128K thinking tokens) provide inherent explainability:

- **Reasoning traces**: Every conclusion preceded by visible reasoning steps.
- **Heavy Mode**: 8 parallel reasoning trajectories with explicit comparison.
- **Tool call logging**: Every external tool invocation recorded with inputs and outputs.
- **Memory traces**: Agent's retrieval of relevant context from vector memory is logged.

9.2 Verifiable Decision Records

- Every agent decision produces a verifiable credential (W3C VC).
- Credential contains: task, reasoning trace, tool calls, result, DID signature.
- Credentials form an immutable chain anchored to blockchain.
- Post-operation review can replay exact decision process.

9.3 Adaptability via LoRA

- Mission-specific LoRA adapters fine-tune agent behavior without full model retraining.
- Adapters swappable at runtime: switch from SIGINT to IMINT analysis in seconds.
- Quantized base model (4-bit) + unquantized adapter layers = edge-deployable specialization.

10 Reinforcement Learning from Operations

10.1 Operational Feedback Loop

Agent performance improves through operational feedback:

1. Agent executes task and produces result with confidence score.
2. Human operator validates or corrects result (accept/reject/modify).
3. Feedback recorded as preference pair: (agent output, human correction).
4. Preference pairs used for LoRA fine-tuning via RLHF/DPO.
5. Updated adapter deployed to operational agents.

10.2 Multi-Agent Learning

- **DeltaSoup**: Byzantine-robust aggregation of LoRA updates from multiple agents.
- **Contributor reputation**: Agents that produce higher-quality outputs gain reputation weight.
- **Differential privacy**: Individual training contributions protected from extraction.
- **GT-QLoRA**: Gate-Targeted LoRA for Mixture-of-Experts adaptation without full model access.

11 Edge Deployment

11.1 Disconnected Agent Operation

Platform	Model	Agents	Capability
MacBook M4	zen4-pro (27B)	5–10	Full reasoning, OCR
Jetson Orin	zen4 (9B)	3–5	Tactical analysis
Raspberry Pi 5	zen4-nano (0.8B)	1–2	Basic classification
Server (A100)	zen4-ultra (1T)	20+	Frontier reasoning

Table 8: Edge deployment configurations.

11.2 Agent Mesh over RNS

Agents on disconnected nodes coordinate via RNS mesh:

- Each node runs local agents with local models.
- Agent consensus messages traverse RNS mesh (LoRa, packet radio, serial).
- ZAP binary protocol minimizes bandwidth (256 B per call vs. 2,826 B JSON-RPC).
- Store-and-forward ensures eventual consensus delivery.
- Hybrid PQ handshake protects all agent communication.

12 AR/VR Integration

12.1 Vision-Language Models as AR Engine

Zen VL models provide real-time scene understanding for augmented reality overlays:

- **Object detection**: Identify and classify objects in camera feed.
- **Spatial reasoning**: Determine object positions, distances, and relationships.
- **OCR**: Extract text from signage, documents, and displays in 32 languages.
- **Temporal tracking**: Video comprehension for activity detection and tracking.

12.2 Mission Rehearsal in VR

- Digital twin environments (Netrunner) feed simulated scenarios to VR displays.
- Babylon.js 3D rendering engine provides immersive visualization.
- Agents provide real-time tactical analysis overlaid on VR environment.
- Network conditions (DDIL, jamming) simulated for realistic rehearsal.

13 Conclusion

We have presented a comprehensive agentic AI framework providing autonomous cyber operations, multi-domain intelligence fusion, and tactical decision support. The combination of 83 specialized agents, Playground control plane with verifiable credentials, distributed consensus with threshold voting, and multi-channel operator interface provides a complete system for naval information warfare operations.

The framework's key differentiators—post-quantum agent identity, consensus-validated decisions, edge-deployable inference, and explainable reasoning—address the specific requirements of naval operations where accountability, disconnected operation, and trust are paramount.

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Reproducibility

Source code. `github.com/hanzoai/agents` (commit `TBD`).
Build. `pnpm install && pnpm build` (Node 20+); SDK via `uv sync`.
Hardware tested. Apple M-series, x86_64 Linux + NVIDIA A100/H100.
Standards referenced. W3C DID, OpenAI tool-use API, MCP, OpenTelemetry.
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Edge-Deployable AI/ML Framework for Disconnected Tactical Environments

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Abstract

We present an edge-deployable AI/ML framework designed for disconnected, degraded, and bandwidth-limited tactical environments. The system combines a Rust-native LLM inference engine supporting 15+ text and 16+ vision model architectures with aggressive quantization (2–8 bit) enabling deployment on hardware ranging from Apple Silicon laptops to embedded ARM processors. A family of 18 frontier models (0.8B–1T+ parameters) provides capability tiers from nano-scale edge inference to frontier reasoning. The ZAP zero-copy binary protocol achieves 11.5M transactions per second for AI agent coordination, while GPU-accelerated operations on Metal and CUDA provide real-time performance for cryptographic and neural network workloads. Multi-agent orchestration with 83 specialized agents enables autonomous analysis with 200–300 sequential tool calls. The framework operates fully offline with local model management, sandboxed execution, and no cloud dependencies.

Executive Summary. This is AI that runs where the network does not—on a laptop, a vehicle, or a handheld at the tactical edge, with no link back to the cloud. It is built for forces and field teams operating in disconnected, degraded, or jammed conditions who still need real-time analysis, imagery and document exploitation, and decision support. The same family of models that powers data-center systems is compressed to fit the hardware on hand, so a single device can run capability tiers from a small embedded chip up to a frontier model on a server, choosing the largest model the available memory allows. Because everything runs locally, sensitive data never leaves the device, there is no dependency on a vendor's servers, and operation continues when communications are contested or absent. For program leaders, this means frontier-grade AI delivered as a self-contained, owned capability rather than a subscription to a remote service that fails the moment the link drops.

Contents

1 Introduction

Tactical edge AI presents unique challenges absent from data center deployments: limited power, constrained memory, intermittent or absent connectivity, and the need for real-time decision support under combat conditions. Existing frameworks assume cloud connectivity, unlimited memory, and Python runtimes—assumptions that fail at the tactical edge.

Our framework addresses these constraints through three key design decisions:

1. **Rust-native inference**: No Python GIL, no garbage collection pauses, predictable latency.
2. **Aggressive quantization**: 2–8 bit weight compression enabling models larger than available RAM.
3. **Zero-copy protocols**: Sub-microsecond serialization for real-time agent coordination.

1.1 Strategic Context

The capability to run frontier models at the disconnected edge depends on owning the model itself, and that is the gap this framework fills. American frontier AI has consolidated into a closed monoculture—effectively two labs—whose models are reachable only as a metered cloud service; the only openly available frontier weights today are largely foreign. Neither option works at the tactical edge, where there is no cloud to call and foreign-origin weights are unacceptable. Hanzo, an American applied-AI lab founded in 2014 (Techstars '17) whose founder works at the frontier of both AI/ML and cryptography, is the American open-source frontier alternative: it builds the Zen model family (0.8B–1T+ parameters) with owned weights together with a production post-quantum crypto stack, so the same models can be quantized, inspected, and deployed onto hardware the operator controls—no per-token API markup, no hyperscaler dependency, no foreign weights. Owning the model and self-hosting it is also the principal lever for roughly an order-of-magnitude lower infrastructure cost at scale, since it removes both the API markup and the cloud-margin that a rented model carries on every inference.

2 Hanzo Engine: Rust-Native Inference

2.1 Architecture

Hanzo Engine is a production LLM inference server built in Rust, providing:

- **Model support**: 15+ text architectures (Transformer, MoE, hybrid), 16+ vision-language models, audio models (ASR, TTS), image generation (FLUX diffusion).
- **GPU backends**: Metal (Apple Silicon), CUDA (NVIDIA Ampere/Ada/Hopper/Blackwell), CPU (x86 AVX-512, ARM NEON/SVE), WebGPU (experimental).
- **Serving**: OpenAI-compatible REST API (`/v1/chat/completions`, `/v1/embeddings`), Ollama compatibility mode, built-in web UI.
- **Optimization**: PagedAttention with FP8 KV cache (50% memory reduction), FlashAttention V2/V3, continuous batching, prefix caching.

2.2 PagedAttention

PagedAttention manages KV cache as fixed-size blocks, enabling dynamic context length without pre-allocation:

- Block-based memory management with LRU eviction.
- FP8 E4M3 KV cache quantization (50% memory reduction vs. FP16).
- Block-level prefix caching for multi-turn conversations.
- Automatic fallback to standard attention when PagedAttention is unavailable.

2.3 FlashAttention Integration

Version	GPU Architecture	Compute Capability
FlashAttention V2	Ampere (A100, RTX 30)	≥ 8.0
FlashAttention V2	Ada Lovelace (RTX 40)	≥ 8.9
FlashAttention V3	Hopper (H100)	≥ 9.0
FlashAttention V2	Blackwell (RTX 50)	≥ 12.0

Table 1: FlashAttention version dispatch by GPU architecture.

3 Model Quantization for Constrained Hardware

3.1 Quantization Methods

Method	CPU	CUDA	Metal	Bits	Notes
ISQ	✓	✓	✓	2–8	Load-time, never loads full model
AFQ	–	–	✓	2–8	Metal-optimized kernels
GGUF	✓	✓	✓	2–8	Universal format
GPTQ/AWQ	–	✓	–	2–8	Marlin CUDA kernels
HQQ	✓	✓	✓	4, 8	Half-quadratic
FP8	✓	✓	✓	8	Floating-point
BNB	✓	✓	✓	4, 8	int8, fp4, nf4

Table 2: Quantization method support matrix.

3.2 In-Situ Quantization (ISQ)

ISQ is the primary edge deployment method. It quantizes weights during model loading, streaming from storage and quantizing layer-by-layer:

Algorithm 1 In-Situ Quantization Loading

for each layer ℓ in model **do**
  Stream weights W_ℓ from storage (SafeTensors/GGUF)
  Compute quantization parameters: scale s, zero-point z
  Quantize: $\hat{W}_\ell = \text{round}(W_\ell / s) + z$
  Store $\hat{W}_\ell$ in target device memory (GPU/CPU)
  Release original W_ℓ from host memory
end for
return quantized model (fits in device memory)

Key advantage: models larger than available RAM can be loaded, as only one layer's full-precision weights are in memory at any time.

3.3 AFQ: Metal-Optimized Quantization

Affine Quantization (AFQ) is designed specifically for Apple Silicon Metal:

- Group sizes: 32, 64 (default), 128.
- Optimized Metal kernel dispatch for Apple M1/M2/M3/M4.
- 2–8 bit quantization with affine transformation.
- Outperforms GGUF on Apple Silicon by leveraging unified memory architecture.

4 Zen Model Family for Defense Applications

4.1 Model Tiers

Tier	Model	Params	Active	Use Case
Edge	zen4-nano	0.8B	0.8B	On-device, embedded
	zen4-micro	2B	2B	Edge+
	zen4-mini	4B	4B	Mobile, tablet
	zen4	9B	9B	General tactical
Medium	zen4-pro	27B	27B	Balanced perf/quality
	zen4-max	35B MoE	3B	Efficient MoE
	zen4-coder	80B MoE	3B	Code analysis
Frontier	zen4-ultra	1.04T MoE	32B	Frontier reasoning
	zen4-titan	744B MoE	40B	Advanced analysis
	zen4-storm	456B MoE	45B	Large-scale

Table 3: Zen4 model family with defense deployment tiers.

4.2 Vision-Language Models

Six vision-language variants (4B–30B) provide:

- 32-language OCR for document exploitation.
- Spatial reasoning for geospatial imagery analysis.

- Video comprehension for temporal pattern recognition.
- Agent variants with native function calling for tool integration.

4.3 Mixture of Experts (MoE)

MoE models activate only 2–8 experts per token from a pool of 16–384, providing frontier-quality output with edge-scale compute:

- **zen4-coder-flash** (31B total, 3B active): 59.2% SWE-Bench.
- **zen-max** (671B total, 14B active): Extended reasoning with 96–128K thinking tokens.
- **Heavy Mode**: 8 simultaneous reasoning trajectories for complex analysis.

5 GPU-Accelerated AI/ML Operations

5.1 Hanzo GPU Kernel Library

Category	Operations
Matrix	matmul, batch matmul, transposed matmul
Attention	Standard, FlashAttention, multi-head attention
Normalization	LayerNorm, RMSNorm, BatchNorm
Activation	ReLU, GELU, SiLU, sigmoid, tanh, softmax
Convolution	2D standard, depthwise
Quantization	FP quantization, blockwise FP8
Crypto	BLS, BN254, secp256k1, Ed25519, ML-DSA, NTT

Table 4: Hanzo GPU kernel categories.

5.2 Unified C API

Listing 1: GPU kernel acceleration C API (excerpt).

```
// Session management
hz_session_t* hz_create_session(hz_device_t dev);

// Tensor operations
hz_tensor_t* hz_matmul(hz_session_t* s,
    hz_tensor_t* a, hz_tensor_t* b);
hz_tensor_t* hz_attention(hz_session_t* s,
    hz_tensor_t* q, hz_tensor_t* k,
    hz_tensor_t* v, float scale);

// Crypto operations
hz_tensor_t* hz_mldsa_verify(hz_session_t* s,
    hz_tensor_t* pk, hz_tensor_t* msg,
    hz_tensor_t* sig);
```

6 On-Device Learning and Adaptation

6.1 LoRA Runtime Adaptation

Low-Rank Adaptation (LoRA) enables domain-specific fine-tuning without full model retraining:

- Dynamic adapter activation at runtime (hot-swap without reload).
- X-LoRA: mixture of adapters for multi-task adaptation.
- Quantized base model with unquantized adapter layers.
- Adapters can be pre-loaded from HuggingFace or local storage.

Tactical application: Pre-train base model in data center, deploy with mission-specific LoRA adapters. Swap adapters as mission requirements change without downloading new models.

6.2 Speculative Decoding

Draft model generates candidate tokens; target model verifies in parallel:

- 2–4× latency reduction for interactive use.
- Draft model: small (nano/eco), target: larger (pro/max).
- No quality degradation—output identical to target model alone.

7 ZAP Protocol for AI Workloads

7.1 Performance for Agent Communication

Metric	JSON-RPC	ZAP
Serialization	5,579 ns	322 ns
Memory/call	2,826 B	256 B
Allocations/op	58	2
GC pressure (50K ops)	41 pauses	3 pauses
Network (20 servers)	140K calls/s	4.2M calls/s

Table 5: ZAP vs. JSON-RPC for AI agent communication.

7.2 MCP Gateway

ZAP bridges 20+ Model Context Protocol (MCP) servers with tool prefix namespacing, health checking, and reconnection:

Listing 2: MCP gateway aggregation.

```
bridge.AddServer("filesystem", "npx",
    "@anthropic/mcp-filesystem")
bridge.AddServer("github", "npx",
    "@anthropic/mcp-github")
result := bridge.CallToolByName(ctx,
    "filesystem:read_file", args)
```

7.3 GPU Cluster Coordination

ZAP's mDNS discovery (`_hz-gpu._tcp`) enables automatic GPU node discovery for distributed inference and FHE operations. Direct zero-copy ciphertext handling enables GPU-to-GPU encrypted computation pipelines.

8 Multi-Agent Orchestration

8.1 Agent Architecture

- **83 specialized agents**: Architecture, languages, infrastructure, data/ML, QA, security, and domain-specific expertise.
- **15 multi-agent orchestrators**: Full-stack development, security hardening, ML pipelines, incident response.
- **Async-first SDK**: Compatible with OpenAI Chat Completions API.
- **State and memory**: Shared state + vector search long-term memory (LanceDB).
- **Tool integration**: 13 canonical MCP tools (fs, exec, code, git, fetch, workspace, ui, think, memory, hanzo, plan, tasks, mode).

8.2 Agent Consensus

For critical decisions, agents employ distributed voting:

- Threshold-based agreement (default 67%).
- W3C DID authentication (did:key from ML-DSA public keys).
- PQ-secure agent-to-agent communication via ZAP.
- Stake-weighted voting for reputation-based trust.

9 Disconnected Edge Deployment

9.1 Hardware Deployment Matrix

Platform	Backend	Quantization	Max Model
Apple Silicon M4	Metal + AFQ	2–8 bit	zen4-pro (27B)
NVIDIA Jetson	CUDA	GGUF/ISQ	zen4 (9B)
x86 Server	AVX-512	ISQ/GGUF	zen4-ultra (1T)
ARM Embedded	CPU/NEON	GGUF 4-bit	zen4-nano (0.8B)
Browser	WebGPU	ISQ	zen4-nano (0.8B)

Table 6: Edge deployment: hardware to model mapping.

9.2 Offline Operation

- Self-contained binary with embedded model management.

- Ollama-compatible `pull`/`list`/`run` commands for model caching.
- Sandboxed execution via macOS `sandbox-exec` or Linux seccomp.
- MCP tools operate locally (filesystem, code analysis, git) with no network.
- Bearer token authentication for optional cloud relay.

10 Hanzo ML Framework

The Hanzo ML framework (Candle fork) provides Rust-native tensor operations:

- **Core**: Tensor creation, manipulation, GPU dispatch (Metal, CUDA).
- **Neural network**: Transformer layers, attention mechanisms, MLP.
- **Models**: 100+ pre-trained model implementations.
- **ONNX**: Import and run ONNX models directly.
- **PyO3**: Python bindings for integration with existing ML pipelines.
- **WASM**: Browser-based inference via WebAssembly.
- **Flash Attention**: V2 kernels for 2–3× memory bandwidth reduction.

11 Reinforcement Learning from Operational Feedback

11.1 Feedback Loop Architecture

Agent execution in tactical environments produces results paired with confidence scores. Human operators validate, correct, or override these results through a structured feedback interface:

- **Accept**: Operator confirms the agent output; recorded as a positive preference signal.
- **Reject**: Operator discards the output; the correct action (if provided) forms a negative preference pair.
- **Modify**: Operator edits the output; the original and edited versions form a ranked preference pair.

All feedback is recorded as preference pairs (y_w, y_l) where y_w is the preferred response and y_l is the dispreferred response, suitable for direct optimization.

11.2 LoRA Fine-Tuning via RLHF/DPO

On-device adaptation uses Low-Rank Adaptation to incorporate operational feedback without full model retraining:

- **RLHF pipeline**: Preference pairs train a reward model; PPO optimizes the policy against this reward while constraining divergence from the base model via KL penalty.

- **DPO (Direct Preference Optimization)**: Bypasses explicit reward modeling by directly optimizing the policy from preference pairs, reducing compute requirements for edge deployment.
- **LoRA efficiency**: Only adapter weights (rank 4–64, typically <1% of model parameters) are updated, enabling fine-tuning on tactical hardware with limited GPU memory.

11.3 Byzantine-Robust Aggregation

When multiple agents or edge nodes produce LoRA updates from independent operational feedback, aggregation must tolerate adversarial or corrupted contributions:

- **DeltaSoup**: Byzantine-robust aggregation of LoRA weight deltas from multiple agents. Outlier updates are detected via coordinate-wise median filtering before merging, ensuring that compromised or malfunctioning nodes cannot poison the shared adapter.
- **GT-QLoRA (Gate-Targeted QLoRA)**: For MoE architectures, adapter updates target expert gating layers specifically, enabling efficient adaptation of routing decisions without modifying expert weights. Combined with 4-bit quantized base weights, this enables MoE fine-tuning within edge memory budgets.

12 In-Memory Graph Analytics

12.1 GraphVM

The GraphVM provides in-memory graph analytics optimized for intelligence workloads where relationship discovery across heterogeneous data sources is critical:

- **Entity resolution**: Probabilistic matching across intelligence sources (HUMINT, SIGINT, OSINT) to unify identities despite name variations, transliterations, and deliberate obfuscation.
- **Network topology analysis**: Communication pattern extraction from intercept data—call graphs, message chains, and temporal co-occurrence networks.
- **Organizational hierarchy mapping**: Social network analysis to infer command structures, identify key nodes, and detect cells within adversary organizations.
- **Temporal behavior tracking**: Time-series graph snapshots enabling detection of behavioral changes, new associations, and pattern-of-life deviations.

12.2 Integrity and Verification

- **Merkle-verified graph state**: Every graph mutation is committed to a Merkle tree, providing tamper-evident audit trails. Analysts can verify that graph state has not been modified since ingestion.
- **Provenance tracking**: Each edge and node carries source metadata (collection platform, time, confidence), enabling analysts to assess intelligence reliability at the relationship level.

12.3 Defense Applications

- **Intelligence link analysis**: Multi-hop traversal to discover indirect connections between persons of interest across compartmented datasets.
- **Threat network mapping**: Real-time graph updates as new intelligence arrives, with automatic re-scoring of centrality metrics to identify emerging leaders or facilitators.
- **Force protection**: Pattern-of-life deviation detection for insider threat and force protection scenarios, comparing current behavior graphs against historical baselines.

13 Explainable and Adaptable AI

13.1 Design Principles

The framework implements AI that is *visible*, *accessible*, *explainable*, and *adaptable* in accordance with defense acquisition requirements for trusted AI systems.

13.2 Inherent Explainability

- **Chain-of-thought reasoning**: Zen models produce 96–128K thinking tokens as extended reasoning traces. These traces expose the model's intermediate logic, hypotheses, and evidence evaluation—providing inherent explainability without post-hoc interpretation methods.
- **Heavy Mode**: 8 parallel reasoning trajectories with explicit comparison. Operators can inspect each trajectory's reasoning chain, observe where paths diverge, and understand why the selected answer was preferred.
- **Tool call logging**: Every external tool invocation (filesystem reads, code execution, database queries, API calls) is recorded with input arguments and output results, producing a complete audit trail of agent actions.
- **Memory retrieval traces**: When agents access long-term memory (LanceDB vector search), the retrieved context and similarity scores are logged, making it transparent which prior knowledge influenced the current decision.

13.3 Adaptability Mechanisms

- **LoRA adapter swapping**: Mission-specific adapters can be loaded, unloaded, and swapped at runtime without model restart. This enables rapid reconfiguration—e.g., switching from ISR analysis to SIGINT processing by swapping a single adapter file.
- **X-LoRA mixture**: Multiple adapters can be activated simultaneously with learned mixing weights, enabling multi-mission configurations from a library of single-purpose adapters.

13.4 Verifiable Decision Records

- **W3C Decentralized Identifiers (DIDs)**: Each agent holds a `did:key` identity derived from its ML-DSA public key. All decision outputs are signed with this identity, providing non-repudiation.

- **Verifiable Credentials**: Agent outputs are packaged as W3C Verifiable Credentials containing the decision, reasoning trace, confidence score, and input data hash. Third parties can verify authenticity and integrity without access to the issuing agent.
- **Tamper-evident chains**: Sequential decision records are hash-chained, enabling detection of any retroactive modification to the decision log.

14 AR/VR Integration for Warfighter Performance

14.1 Vision-Language Models as AR Engines

Zen vision-language models (4B–30B parameters) serve as real-time data overlay engines for augmented reality systems:

- **Real-time spatial reasoning**: Object detection, scene understanding, and spatial relationship inference from head-mounted camera feeds.
- **32-language OCR**: On-the-fly text recognition and translation from signage, documents, and captured materials displayed as AR overlays.
- **Contextual annotation**: Automatic identification and labeling of vehicles, equipment, structures, and personnel with threat assessment scores projected into the operator's field of view.
- **Edge deployment**: Vision-language inference runs on tactical hardware (Metal on Apple Silicon, CUDA on NVIDIA Jetson, ARM NEON on embedded processors) with quantized models (4–8 bit) for real-time frame rates.

14.2 3D Mission Rehearsal

- **Babylon.js rendering**: Interactive 3D environments for mission planning and rehearsal, supporting terrain visualization, building interiors, and route planning with physics simulation.
- **Digital twin integration**: The Netrunner network simulation framework feeds real-time system state into VR environments, enabling operators to visualize network topology, communication flows, and system health in an immersive 3D space.
- **Collaborative planning**: Multiple operators can share a VR mission space with synchronized state, annotating terrain, marking objectives, and rehearsing maneuvers in a shared virtual environment.

14.3 Training and System Design

- **Scenario generation**: LLM-driven procedural generation of training scenarios based on historical after-action reports and doctrinal templates.
- **Performance analytics**: Operator actions during VR training sessions are recorded and analyzed by multi-agent systems to identify decision patterns, reaction times, and areas for improvement.
- **System design visualization**: Digital twins of proposed system architectures rendered in VR, enabling stakeholders to walk through data flows, identify bottlenecks, and validate designs before deployment.

15 Defense Application Scenarios

15.1 ISR Analysis

Zen vision-language models (4B–30B) process full-motion video, satellite imagery, and document scans:

- 32-language OCR for captured document exploitation.
- Spatial reasoning for geospatial analysis.
- Temporal video comprehension for activity detection.
- LoRA adapters for theater-specific terrain recognition.

15.2 Tactical Decision Support

- Extended reasoning (96–128K thinking tokens) for complex COA analysis.
- Heavy Mode: 8 parallel reasoning trajectories for wargaming.
- Multi-agent orchestration for multi-domain analysis.
- Runs entirely on tactical hardware, no reach-back required.

15.3 SIGINT Natural Language Processing

- 100+ programming language understanding for CYBER analysis.
- 32+ natural language support for COMINT processing.
- Intent detection and sentiment analysis via Insights pipeline.
- Named entity recognition and relationship extraction.

16 Conclusion

We have presented a comprehensive edge AI/ML framework that operates fully disconnected from cloud infrastructure while providing frontier-quality model capabilities through aggressive quantization and hardware-optimized inference. The combination of Rust-native performance, zero-copy agent coordination, GPU-accelerated operations, and multi-agent orchestration provides a complete tactical AI stack deployable on hardware from embedded ARM processors to multi-GPU servers.

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Reproducibility

Source code. `github.com/hanzoai/engine, hanzoai/jin` (commit `TBD`).
Build. cargo build –release; LanceDB embedded by default.
Hardware tested. Apple M-series (Metal), NVIDIA Jetson, x86_64 Linux + CUDA.
Standards referenced. OpenTelemetry, ONNX Runtime, OpenAI API.
Contact. research@hanzo.ai.

Multi-Level Security and Cross-Domain Solutions via Homomorphic Policy Evaluation and Zero-Trust Fabric

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Abstract

We present a cross-domain solution (CDS) architecture where data transfer policies are evaluated on encrypted content using fully homomorphic encryption (FHE), ensuring the cross-domain guard never observes plaintext. The architecture integrates four components: (1) a zero-trust overlay network with identity-first micro-segmentation enforcing mandatory access controls at every hop; (2) an FHE policy evaluation engine (ThresholdVM) performing content inspection on ciphertexts using BFV/BGV/CKKS/TFHE schemes with GPU-accelerated NTT operations; (3) a key management system with Shamir-based root key distribution, per-organization encryption isolation, and HIPAA/SOX/SEC compliance controls; and (4) an organization-scoped identity and access management layer deriving security labels from JWT claims and enforcing them at the API gateway. The result is a multi-level security (MLS) architecture where data flows between security domains are cryptographically mediated, no single component has access to both plaintext and policy decisions, and all cross-domain transfers produce immutable audit records anchored to a post-quantum blockchain.

Executive Summary. A cross-domain solution is the gatekeeper that decides what information may move between networks of different classification—for example, releasing an intelligence product from a Secret network to a coalition partner. Today that gatekeeper has to read the data in the clear to decide, which makes it the single most valuable target on the network: compromise it once and every cross-domain transfer is exposed. This architecture removes that weakness. The data stays encrypted the entire time; the gatekeeper checks it against release policy *without ever decrypting it*, and the final release decision is split across a committee so no single operator or machine can be subverted to leak data. Every transfer leaves a tamper-proof, quantum-resistant audit record. For program offices and integrators running multi-level or coalition networks, the operational payoff is a guard that an adversary cannot turn into a data tap even with full physical access to it.

Contents

1 Introduction

Cross-domain solutions enable controlled information sharing between networks operating at different security classifications. Traditional CDS architectures rely on trusted guards that inspect plaintext content against release policies—creating a high-value target where a single compromise exposes all cross-domain traffic.

This paper presents a fundamentally different approach: the guard evaluates policies on *encrypted* content using fully homomorphic encryption. The guard learns only the Boolean policy decision (release/deny), never the content itself. This eliminates the single point of trust while maintaining mandatory policy enforcement.

1.1 Problem Statement

Naval operations require information sharing across multiple security domains:

- **Unclassified ↔ Secret**: Intelligence products sanitized for coalition partners.
- **Secret ↔ Top Secret/SCI**: Analyst workstations accessing compartmented data.
- **US ↔ Coalition**: Multinational operations requiring controlled release.
- **Operational ↔ Administrative**: Preventing data spillage between networks.

Current CDS solutions (ISSE Guard, Radiant Mercury, High Speed Guard) process plaintext, requiring the guard itself to be trusted at the highest classification level. A compromised guard exposes all cross-domain traffic.

1.2 Our Approach

1. Data encrypted at source under FHE public key.
2. Guard evaluates content policy homomorphically on ciphertext.
3. Encrypted Boolean result threshold-decrypted by MPC committee.
4. If approved, data re-encrypted for destination domain.
5. Guard never sees plaintext; committee never sees policy result.

2 Architecture Overview

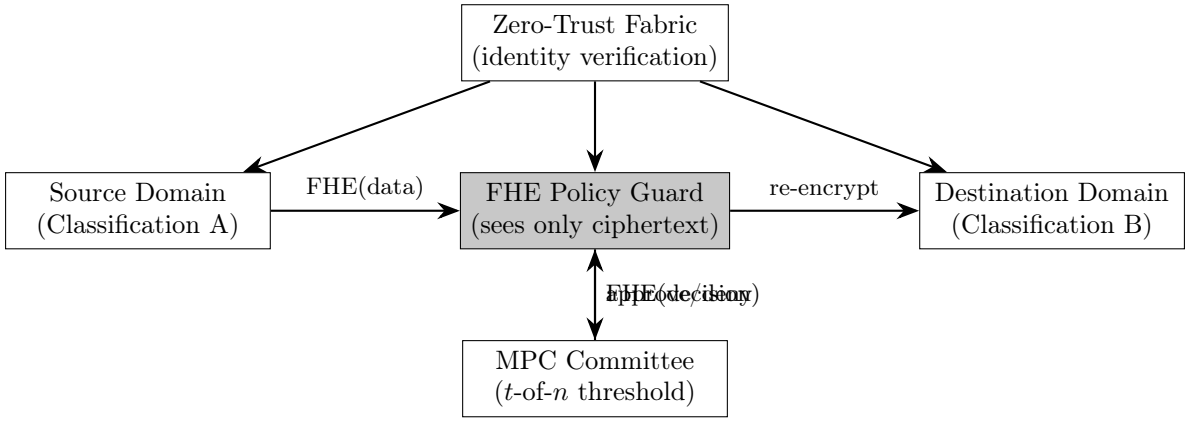

Figure 1: FHE-based cross-domain solution architecture.

2.1 Security Invariants

1. **Guard blindness**: The policy guard processes only ciphertexts. It cannot distinguish encrypted classified data from encrypted noise.
2. **Committee separation**: MPC committee members see decryption shares, never the full policy result or the data. Threshold (t-of-n) prevents minority collusion.
3. **Mandatory enforcement**: Zero-trust fabric verifies cryptographic identity at every hop. Security labels derived from JWT claims cannot be forged.
4. **Audit immutability**: Every cross-domain transfer produces a blockchain-anchored receipt with PQ signatures.

3 Zero-Trust Network Fabric

3.1 Identity-First Access Control

Hanzo ZT provides the network underlay with mandatory access controls:

- **Cryptographic identity**: Every endpoint possesses a unique X.509 certificate with HSM-backed private key (PKCS#11, YubiHSM).
- **No implicit trust**: Network position (IP address, VLAN, subnet) confers zero access. Every connection requires identity verification.
- **Micro-segmentation**: Policies define which identities can communicate. Cross-domain traffic flows only through designated guard endpoints.
- **Policy enforcement**: Applied at every hop in the overlay mesh, not just at perimeter firewalls.

3.2 Security Label Propagation

Security labels are derived from the IAM identity system:

Listing 1: Security label derivation from JWT claims.

```
// Gateway extracts identity from JWT
X-Org-Id:    "navy-secret"     // Organization = domain
X-User-Id:   "analyst-0042"    // Subject identity
X-User-Email: "j.doe@navy.mil" // Attribution

// Gateway enforces: only "navy-secret" org members
// can reach services in "secret" namespace
// Cross-domain requires FHE guard intermediary
```

The API gateway strips all client-supplied identity headers and derives them exclusively from cryptographically verified JWT claims. This prevents label forgery.

3.3 Multi-Organization Isolation

The platform supports 4+ organizations with complete cryptographic isolation:

Organization	Root Key	Isolation Mechanism
Org A	Separate AES-256	KMS per-org encryption
Org B	Separate AES-256	Separate key hierarchy
Org C	Separate AES-256	Org-scoped IAM policies
Org D	Separate AES-256	Independent audit logs

Table 1: Per-organization cryptographic isolation.

Each organization has its own root encryption key, IAM policies, KMS secrets, and audit trail. Cross-organization data access requires explicit policy and traverses the FHE guard.

4 FHE Policy Evaluation

4.1 ThresholdVM: Computation on Encrypted Data

ThresholdVM operates on the T-Chain, providing FHE as a native virtual machine:

Scheme	Domain	CDS Application
BFV	Integer	Keyword matching on encrypted text
BGV	Modular	Classification marker detection
CKKS	Approx. real	ML-based content classification
TFHE	Boolean	Arbitrary policy Boolean circuits

Table 2: FHE schemes mapped to CDS operations.

4.2 Policy Evaluation Flow

Algorithm 1 FHE Cross-Domain Policy Evaluation

Source encrypts data: $c \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{data})$
Source encrypts metadata: $m \leftarrow \text{FHE.Encrypt}(\text{pk}_{\text{FHE}}, \text{labels})$
Source sends $(c, m, \text{dest\_domain})$ to Guard

Guard evaluates policy homomorphically:
  $r \leftarrow \text{FHE.Eval}(\text{policy\_circuit}, c, m, \text{dest\_domain})$
  (Guard sees only ciphertexts; r is an encrypted Boolean)

Guard sends r to MPC committee for threshold decryption
for each committee member $i = 1, \ldots, n$ **do**
  $d_i \leftarrow \text{ThresholdDecrypt}(\text{sk}_i, r)$
end for
decision $\leftarrow \text{Reconstruct}(d_1, \ldots, d_t)$ (t shares suffice)

if decision = APPROVE **then**
  Re-encrypt data for destination domain key
  Forward to destination via zero-trust fabric
  Anchor audit receipt to blockchain (ML-DSA-65 signed)
else
  Log denial (encrypted), destroy ciphertext
end if

4.3 GPU-Accelerated FHE Operations

The Hanzo GPU kernel library provides Metal and CUDA kernels for FHE primitives:

- **NTT/iNTT**: Number Theoretic Transform for polynomial multiplication in $\mathbb{Z}_q[X]/(X^n + 1)$.
- **TFHE bootstrap**: Programmable bootstrap for refreshing ciphertext noise.
- **Blind rotate**: Key operation in TFHE gate evaluation.
- **Batch operations**: Parallel ciphertext processing across GPU threads.

GPU acceleration reduces FHE policy evaluation from seconds to milliseconds, making real-time cross-domain filtering practical.

4.4 EVM Precompile Interface

Smart contracts can invoke FHE operations via precompiles:

Address	Function	Gas
`0x0700`	FHE encrypt/decrypt/eval	500,000
`0x0701`	Ciphertext ACL management	25,000
`0x0702`	Input proof verification	50,000
`0x0703`	Cross-chain gateway	100,000

Table 3: FHE EVM precompiles for on-chain policy enforcement.

5 Key Management for Multi-Level Security

5.1 Hanzo KMS Architecture

The KMS provides 8 subsystems supporting MLS key lifecycle:

1. **Shamir Secret Sharing**: Root key split into n shares over GF(2^8); t required for reconstruction. Custodians geographically distributed.
2. **Transit Engine**: Encryption-as-a-service with key versioning. Each security domain has independent transit keys with separate rotation schedules.
3. **HPKE Key Wrapping**: Content encryption keys (CEKs) wrapped using hybrid HPKE (X25519 + ML-KEM-768) per RFC 9180.
4. **Seal/Unseal Barrier**: AES-256-GCM barrier protecting all stored secrets. Auto-unseal via AWS/GCP KMS for operational environments.
5. **ACL Policy Engine**: Path-based capabilities with template variables (`{{identity.org_id}}`) enabling per-domain access rules.
6. **Token Hierarchy**: Service, batch, and recovery tokens with parent-child revocation cascade. Compromised token revocation propagates instantly.
7. **Dynamic Secrets**: Temporary database credentials (PostgreSQL, MySQL, Redis, MongoDB) scoped to security domain and auto-expired.
8. **TFHE-KMS Bridge**: FHE key management integrated with policy evaluation pipeline.

5.2 Cross-Domain Key Isolation

- Each security domain has an independent root key, never shared.
- Cross-domain data transfer requires re-encryption under destination domain key.
- Re-encryption occurs only after FHE policy approval and threshold decryption.
- No single KMS instance holds keys for multiple domains.

5.3 Compliance Controls

Standard	Implementation
HIPAA	Break-glass tokens (time-limited), WORM audit logs
SEC 17a-4	6–7 year retention, immutable SHA-256 chain
SOX	Regulator escrow shards, cooperative reconstruction
GDPR	Per-org keys, graceful revocation
CNSA 2.0	PQ algorithms for all key operations

Table 4: Regulatory compliance controls in KMS.

6 Audit and Accountability

6.1 Blockchain-Anchored Audit Trail

Every cross-domain transfer produces an immutable audit record:

- **Record contents**: Source domain, destination domain, policy ID, decision (approve/deny), timestamp, data hash (not data), requestor DID.
- **Signature**: ML-DSA-65 (post-quantum) ensuring long-term non-repudiation.
- **Anchoring**: Record hash committed to the Hanzo chain via Quasar consensus.
- **WORM guarantee**: Write-once-read-many with SHA-256 hash chain prevents retroactive modification.

6.2 Attestation Protocol

Domain-typed attestations with equivocation detection:

- **CrossDomainTransfer**: Records each approved transfer with full provenance.
- **PolicyUpdate**: Records changes to cross-domain policies.
- **KeyRotation**: Records domain key rotation events.
- Signing contradictory attestations (equivocation) is automatically detected and flagged.

7 Bell-LaPadula Enforcement

7.1 Mandatory Access Control

The architecture enforces Bell-LaPadula properties:

Definition 7.1 (Simple Security Property (No Read Up))**.** A subject at security level L_s cannot read an object at security level L_o where $L_o > L_s$. Enforced by the zero-trust fabric: identity JWT contains org-scoped level; gateway rejects requests to higher-level services.

Definition 7.2 (*-Property (No Write Down))**.** A subject at security level L_s cannot write to an object at security level L_o where $L_o < L_s$. Enforced by the FHE guard: data flowing from high to low domain must pass policy evaluation.

7.2 Implementation

1. **Read control**: Gateway checks `X-Org-Id` (derived from JWT) against service security label. Requests to higher domains rejected before reaching backend.
2. **Write control**: All data egress from a domain passes through the FHE guard. Downward flows require policy approval; upward flows are unrestricted.
3. **Label integrity**: Security labels are gateway-derived from cryptographic JWT claims. Client-supplied labels are stripped. Labels cannot be forged without compromising the IAM signing key.

8 Comparison with Existing CDS Approaches

Feature	Traditional Guard	Data Diode	FHE Guard (Ours)
Plaintext exposure	Guard sees all	One-way only	Never
Bidirectional	Yes	No	Yes
Content inspection	Full	None	On ciphertext
Single point of trust	Guard	No	No (threshold)
PQ-safe	Depends	N/A	Yes (FIPS)
Audit	Guard logs	Physical	Blockchain
ML classification	On plaintext	N/A	On ciphertext

Table 5: CDS approach comparison.

The FHE-based approach uniquely provides bidirectional content-inspected cross-domain transfer with zero plaintext exposure. No commercially available CDS offers this capability.

9 Performance Considerations

9.1 FHE Overhead

Operation	CPU	GPU (Metal)
TFHE gate evaluation	13 ms	0.8 ms
BFV keyword match	45 ms	3.2 ms
CKKS ML inference	200 ms	15 ms
Threshold decrypt (t=3)	50 ms	12 ms
Total policy eval	**∼300 ms**	**∼30 ms**

Table 6: FHE policy evaluation latency.

GPU acceleration brings FHE policy evaluation to 30 ms—practical for real-time cross-domain filtering of message traffic, though not suitable for bulk file transfer without parallelization.

9.2 Throughput

With GPU acceleration and pipelined evaluation:

- **Messages**: ∼33 messages/second per GPU (real-time chat feasible).
- **Bulk data**: Parallelize across multiple GPUs; 8-GPU node achieves ∼264 evaluations/second.
- **Ciphertext expansion**: 100–1000× plaintext size (FHE-inherent). ZAP zero-copy design mitigates the internal memory cost.

10 Naval MLS Deployment Scenarios

10.1 Coalition Information Sharing

- US Secret data evaluated against release policy for Five Eyes partners.
- FHE guard checks classification markers, source restrictions, and content sensitivity—all on ciphertext.
- Approved data re-encrypted under coalition domain key and forwarded.
- Full audit trail anchored to blockchain for post-operation review.

10.2 Analyst Workstation Cross-Domain Access

- Analyst at Secret level queries Top Secret/SCI database.
- Query encrypted under FHE key; database returns encrypted results.
- FHE guard evaluates sanitization policy on encrypted response.
- Approved (sanitized) response decrypted for analyst's domain.

10.3 Shipboard MLS Network

- Multiple security domains on single shipboard network (physically separated VLANs, logically separated via zero-trust overlay).
- FHE guards at domain boundaries, GPU-accelerated for real-time messaging.
- KMS with separate root keys per domain, HSM-protected.
- All cross-domain traffic audited with PQ-signed blockchain receipts.

11 Strategic Context

This cross-domain architecture is one capability within a broader sovereign stack built by Hanzo AI—an American-owned applied-AI lab and venture studio (Techstars '17, founded 2014) behind more than 100 venture-funded startups and several unicorns, led by a founder with deep expertise in both AI/ML and cryptography. Hanzo owns both halves of the problem that cross-domain security demands: the frontier AI models (the Zen family) that drive content classification on encrypted data, and the production post-quantum cryptography that mediates the transfers. This matters for the multi-level and coalition mission specifically: U.S. frontier AI has consolidated into a closed commercial duopoly, while the only broadly *open* frontier model weights available today are largely of foreign origin—an untenable basis for a guard that inspects classified traffic. Hanzo is the American open-source frontier alternative, offering models and cryptography that are sovereign, auditable end-to-end, and deployable in air-gapped or edge environments without renting from the closed duopoly or depending on foreign open weights. The same stack has been hardened in production at scale—the agentic engineering pipeline delivered a FINRA/SEC-approved, SOC2 securities platform (consolidated from 200+ microservices into 3 compiled binaries) to production with a small team—evidence that the cryptographic and AI components described here are operational engineering, not laboratory prototypes.

12 Conclusion

We have presented a cross-domain solution architecture where the guard evaluates policies on encrypted content, eliminating the traditional single point of trust. The combination of FHE policy evaluation, zero-trust network fabric, threshold MPC decryption, and blockchain-anchored audit provides a multi-level security architecture with no plaintext exposure at the guard, no single point of key compromise, and immutable accountability.

This approach represents a paradigm shift in cross-domain security: instead of trusting the guard to handle plaintext responsibly, we make it cryptographically impossible for the guard to access plaintext at all.

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Reproducibility

Source code. `github.com/hanzoai/fhe, hanzoai/kms` (commit `TBD`).
Build. make build (Rust 1.78+, OpenFHE bindings); GPU NTT via Metal/CUDA.
Hardware tested. Apple M2/M3 (Metal), NVIDIA A100/H100.
Standards referenced. NIST FHE (BFV/BGV/CKKS/TFHE), HIPAA, SOX, SEC, MLS.
Contact. research@hanzo.ai.

Post-Quantum Resilient Mesh Networking for Maritime and Tactical Edge Operations

Zach Kelling — Hanzo Team

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Abstract

We present a post-quantum resilient networking architecture designed for maritime, tactical edge, and disconnected/degraded/intermittent/limited-bandwidth (DDIL) environments. The system integrates the Reticulum Network Stack (RNS) as a native consensus transport layer alongside traditional IP networking, enabling validators and edge nodes to communicate over LoRa, packet radio, serial links, and mesh topologies without reliance on centralized infrastructure or DNS. All transport sessions employ hybrid post-quantum key exchange (X25519 + ML-KEM-768, FIPS 203) and hybrid authentication (Ed25519 + ML-DSA-65, FIPS 204), providing defense-in-depth against both classical and quantum adversaries. The architecture includes a zero-trust overlay network with identity-first micro-segmentation, a zero-copy binary protocol achieving 11.5 million transactions per second, and fault-tolerant streaming over NATS JetStream. We demonstrate sub-second consensus finality over mesh networks and analyze the system's suitability for shipboard, shore-to-ship, expeditionary, and submarine communication scenarios.

Executive Summary. This is a communications layer that keeps networked systems coordinated when the usual infrastructure—satellite links, cell towers, DNS, certificate authorities—is jammed, degraded, or simply absent. It runs over whatever carries bytes (long-range radio, serial cable, existing tactical radios, satellite) and falls back automatically as links drop, so nodes stay in agreement at sea, underground, or behind a contested boundary. Every link is encrypted with two independent locks at once—today's cryptography plus the new quantum-resistant standards mandated for national-security systems—so traffic recorded now cannot be decrypted later when quantum computers arrive. It is built for operators who must assume the network is hostile and the adversary is patient: defense program offices, government integrators, and operators of critical infrastructure that has to keep working when the connection does not. The performance figures and security properties stated below are measured, not aspirational.

Contents

1 Introduction

Naval operations increasingly depend on resilient, high-performance networking across heterogeneous and contested environments. The convergence of three trends—the quantum computing threat to classical cryptography, the proliferation of edge computing devices, and the operational necessity of DDIL communications—demands a fundamentally new approach to assured networking.

Traditional military networking architectures assume reliable, high-bandwidth links with centralized key management and DNS-based addressing. These assumptions fail in contested maritime environments where satellite communications may be jammed, shore infrastructure may be degraded, and adversaries may conduct harvest-now-decrypt-later (HNDL) attacks against encrypted traffic.

1.1 Challenges in Naval Networking

(i) **Disconnected Operations**: Vessels and expeditionary forces must maintain consensus and data synchronization during extended periods without connectivity.

(ii) **Medium Heterogeneity**: Communication may traverse satellite, HF radio, LoRa mesh, fiber, and IP networks within a single operational theater.

(iii) **Quantum Threat**: Nation-state adversaries are developing cryptographically relevant quantum computers, threatening all classical public-key cryptography.

(iv) **Centralized Fragility**: Dependence on DNS, certificate authorities, and centralized key servers creates single points of failure.

(v) **Latency Tolerance**: Store-and-forward capability is essential when links have multi-second or multi-minute round-trip times.

1.2 Contributions

This paper presents the following contributions:

1. **RNS-Native Consensus**: Integration of the Reticulum Network Stack as a first-class transport layer in the Hanzo consensus protocol, enabling validator communication over any physical medium.

2. **Hybrid PQ Transport**: End-to-end post-quantum key exchange and authentication for all mesh links using NIST FIPS 203 and FIPS 204 algorithms.

3. **Zero-Trust Overlay**: Identity-first networking with micro-segmentation, eliminating implicit trust from network position.

4. **High-Performance Data Distribution**: The ZAP zero-copy binary protocol achieving 2.2 ns parse latency with zero memory allocations.

5. **Tactical Deployment Analysis**: Concrete scenarios for shipboard mesh, shore-to-ship relay, and expeditionary force networking.

1.3 Strategic Context

Assured networking is the foundation on which everything else in a sovereign, disconnected AI stack rests: if nodes cannot reach agreement, neither the models nor the cryptography above them matter. Hanzo (American-owned, Techstars '17, founded 2014) builds this transport in-house because the same organization owns both the post-quantum cryptography securing each link and the frontier AI—the *Zen* model family—that runs over it. That vertical ownership is the strategic point. The center of gravity for "open" frontier AI has shifted offshore: the United States has narrowed to a closed two-vendor model duopoly, while the broad open ecosystem—and thus most openly available frontier weights—now sits abroad. Hanzo is the American open-source alternative that controls the model and the cryptography together, so defense and regulated operators can run auditable, sovereign AI on their own networks—offline, at the tactical edge, where the data lives—without renting from the closed duopoly or depending on foreign open weights. The mesh transport described here is what lets that stack operate when the network is contested rather than only in the data center.

2 Reticulum Network Stack Integration

2.1 Architecture Overview

The Reticulum Network Stack (RNS) provides medium-agnostic, delay-tolerant, encrypted networking without dependence on IP infrastructure, DNS, or certificate authorities. We integrate RNS as a native transport layer in the Hanzo consensus node, alongside traditional TCP/IP endpoints.

The Hanzo endpoint abstraction supports three addressing modes:

Type	**Format**	**Resolution**
IP	`203.0.113.50:9631`	Direct socket connection
Hostname	`validator.example.com:9631`	DNS resolved at dial time
RNS	`rns://a5f72c3d...`	Cryptographic destination hash, no DNS

Table 1: Unified endpoint addressing in Hanzo consensus.

RNS destinations are identified by 16-byte (128-bit) truncated hashes of the destination's public keys, providing cryptographic addressing that requires no external naming infrastructure.

2.2 Supported Physical Interfaces

RNS operates over any medium capable of carrying bytes:

- **AutoInterface**: Automatic detection of available local network media.
- **TCPClientInterface**: Standard TCP/IP for LAN and WAN connectivity.
- **LoRaInterface**: Long-range radio (LoRa) for low-bandwidth mesh operation, enabling validator communication across 10–50 km ranges at 1–100 kbps.
- **SerialInterface**: Direct serial connections for point-to-point links.
- **Packet Radio**: Amateur and military packet radio systems.

- **Satellite**: Via TCP or serial gateways to SATCOM terminals.

2.3 Software-Defined Radio Integration

RNS AutoInterface and SerialInterface provide natural integration points for Software-Defined Radio (SDR) platforms. SDR waveforms that output to serial or IP interfaces are consumed directly by RNS without modification to the network stack, enabling rapid adoption of new waveform types as they become available.

The ZAP binary protocol is well-suited as an application layer over SDR waveforms. A complete ZAP-encoded consensus call fits within 256 bytes, making it viable even over narrowband SDR channels where every byte of airtime is contested. This compactness is critical for HF and UHF waveforms with limited symbol rates.

RNS enables frequency-agile operation: when an SDR platform switches between channels, frequencies, or waveform modes, RNS transparently re-establishes links via its delay-tolerant transport without requiring upper-layer reconfiguration. This is particularly valuable in electronic warfare environments where frequency hopping and adaptive waveform selection are operational necessities.

The architecture is compatible with the Software Communications Architecture (SCA) and Joint Tactical Radio System (JTRS) radio families. SCA-compliant radios that expose serial or IP gateway interfaces can serve as RNS physical transports, bridging between the mesh networking layer and legacy tactical radio infrastructure.

LoRa, already supported as a first-class RNS interface, demonstrates the SDR integration model: it is a programmable spread-spectrum radio whose modulation parameters (spreading factor, bandwidth, coding rate) are software-configured at runtime. Extending this pattern to wideband SDR platforms such as USRP, LimeSDR, or military SCA radios requires only a serial or IP bridge—no changes to RNS, ZAP, or the consensus protocol.

2.4 Configuration

Listing 1: RNS transport configuration in the Hanzo node.

```
type RNSConfig struct {
    ConfigPath       string        // ~/.reticulum/
    IdentityPath     string        // ~/.reticulum/identity
    GatewayAddr      string        // Optional gateway
    AnnounceInterval time.Duration // Default 5 minutes
    Interfaces       []string      // AutoInterface, LoRaInterface
    LinkTimeout      time.Duration // Default 30 seconds
    Enabled          bool          // Default false
    PostQuantum      bool          // Enable hybrid PQ
    RequirePostQuantum bool        // Reject classical-only
}
```

2.5 Deployment Scenarios

The Pars Improvement Proposal PIP-7009 defines five deployment scenarios for RNS-based validator operation:

1. **Sovereign Infrastructure**: Validators operating in jurisdictions with internet restrictions, using mesh networks to reach global consensus.

2. **Disaster Resilience**: Mesh networks maintaining consensus during infrastructure outages, using LoRa and packet radio.
3. **Remote Deployment**: LoRa-connected validators in areas without internet access—remote bases, offshore platforms, polar stations.
4. **Mobile Validators**: Maritime, aviation, and vehicular validator nodes maintaining consensus during movement.
5. **Air-Gapped Security**: High-security validators with controlled, non-IP connectivity for maximum isolation.

3 Post-Quantum Hybrid Transport Security

3.1 Threat Model

We assume an adversary with the following capabilities:

- Full network observation and recording of all ciphertext (HNDL capability).
- Access to a cryptographically relevant quantum computer within 10–15 years.
- Ability to perform active man-in-the-middle attacks on initial connection.
- No compromise of endpoint hardware or key material.

3.2 Hybrid Identity

Each RNS node maintains a hybrid identity combining classical and post-quantum key pairs:

Definition 3.1 (Hybrid RNS Identity)**.** A hybrid identity $\mathcal{I}$ consists of:

$$\begin{aligned}\mathcal{I} = \big(&\mathrm{sk}_{\text{Ed25519}}, \mathrm{pk}_{\text{Ed25519}},\\ &\mathrm{sk}_{\text{X25519}}, \mathrm{pk}_{\text{X25519}},\\ &\mathrm{sk}_{\text{ML-DSA-65}}, \mathrm{pk}_{\text{ML-DSA-65}},\\ &\mathrm{sk}_{\text{ML-KEM-768}}, \mathrm{pk}_{\text{ML-KEM-768}}\big)\end{aligned} \tag{1}$$

3.3 Hybrid Key Exchange Protocol

Algorithm 1 RNS Hybrid Post-Quantum Key Exchange

Initiator A generates ephemeral keys:
  $(\text{ek}_A^X, \text{epk}_A^X) \leftarrow$ X25519.Generate()
  $(\text{ek}_A^M, \text{epk}_A^M) \leftarrow$ ML-KEM-768.Generate()
$A \to B$: $\text{epk}_A^X \| \text{epk}_A^M \| \text{pk}_A^{\text{DSA}} \| \sigma_A$
  where $\sigma_A = \text{ML-DSA-65.Sign}(\text{sk}_A^{\text{DSA}}, \text{epk}_A^X \| \text{epk}_A^M)$

Responder B verifies and computes:
  Verify σ_A under pk_A^{DSA}
  $\text{ss}^X \leftarrow \text{X25519.DH}(\text{sk}_B^X, \text{epk}_A^X)$
  $(\text{ct}^M, \text{ss}^M) \leftarrow \text{ML-KEM-768.Encaps}(\text{epk}_A^M)$
  $K \leftarrow \text{HKDF-SHA256}(\text{“RNS-HybridLink-v1”}, \text{ss}^X \| \text{ss}^M)$
$B \to A$: $\text{epk}_B^X \| \text{ct}^M \| \text{pk}_B^{\text{DSA}} \| \sigma_B$

Initiator A derives session key:
  $\text{ss}^X \leftarrow \text{X25519.DH}(\text{ek}_A^X, \text{epk}_B^X)$
  $\text{ss}^M \leftarrow \text{ML-KEM-768.Decaps}(\text{ek}_A^M, \text{ct}^M)$
  $K \leftarrow \text{HKDF-SHA256}(\text{“RNS-HybridLink-v1”}, \text{ss}^X \| \text{ss}^M)$
  Derive: $K_{\text{send}}, K_{\text{recv}}, K_{\text{MAC-s}}, K_{\text{MAC-r}} \leftarrow \text{HKDF-Expand}(K)$
Destroy all ephemeral private keys (forward secrecy)
Session: AES-256-GCM with counter-mode nonces

3.4 Wire Format Analysis

Component	Classical	Hybrid PQ
X25519 ephemeral public key	32 B	32 B
ML-KEM-768 ephemeral public key	—	1,184 B
ML-KEM-768 ciphertext	—	1,088 B
ML-DSA-65 public key	—	1,952 B
ML-DSA-65 signature	—	3,309 B
Ed25519 public key	32 B	32 B
Ed25519 signature	64 B	64 B
Total handshake	**∼256 B**	**∼7,500 B**

Table 2: Wire format comparison: classical vs. hybrid PQ handshake.

The 29× overhead is a one-time cost per link establishment. Session data uses AES-256-GCM with no additional overhead, making the amortized cost negligible for long-lived connections.

3.5 Security Properties

Theorem 3.2 (Hybrid Security)**.** *The hybrid key exchange provides IND-CCA2 security if either X25519 ECDH or ML-KEM-768 is secure. Formally, an adversary must break both the Computational*

Diffie-Hellman problem on Curve25519 and the Module-LWE problem with parameters $(k = 3, n = 256, q = 3329)$ *to compromise session keys.*

Theorem 3.3 (Forward Secrecy)**.** *Compromise of any long-term identity key does not compromise past session keys, as each session uses fresh ephemeral key pairs that are destroyed after derivation.*

4 Zero-Trust Overlay Network

4.1 Hanzo ZT Architecture

Hanzo Zero Trust (ZT) is a comprehensive zero-trust networking framework comprising 78 repositories, providing identity-first overlay networking with no implicit trust from network position.

4.1.1 Core Components

- **Identity**: Certificate-based cryptographic identity with HSM support (PKCS#11, YubiHSM, CloudHSM). Each endpoint possesses a unique cryptographic identity that must be verified before any communication.
- **Transport**: Pluggable transport layer supporting TCP, TLS, DTLS, WebSocket, and Transwarp (UDP-based with Dilithium flow control).
- **Channel**: Binary protocol framework for type-safe, encrypted message passing.
- **Fabric**: Overlay routing mesh that forwards traffic based on identity policies, not network addresses.

4.1.2 Transport Plugin Architecture

New transports are registered via a global parser registry:

Listing 2: Transport plugin registration pattern.

```
type Address interface {
    Dial(name string, id *identity.TokenId,
         timeout time.Duration,
         tcfg Configuration) (Conn, error)
    Listen(name string, id *identity.TokenId,
           acceptF func(Conn),
           tcfg Configuration) (io.Closer, error)
    Type() string
}

// Register at init time
transport.AddAddressParser(RNSAddressParser{})
```

This plugin architecture enables RNS to serve as a ZT underlay transport, combining zero-trust policy enforcement with mesh networking resilience.

4.2 Dilithium Fault-Tolerant Transport

The Dilithium protocol (distinct from the PQ signature scheme of the same name) implements Westworld3, an adaptive UDP-based flow control framework optimized for long-haul, dedicated links.

Parameter	Value
TxPortal start	96 KB
TxPortal max	4 MB
Retransmission start	200 ms
RTT probe interval	50 ms
Max segment size	1,450 B (UDP MSS)

Table 3: Dilithium/Westworld3 default baseline profile.

Key resilience mechanisms include adaptive congestion control (RTT-based), duplicate ACK detection with capacity reduction (0.9×), retransmission with aggressive capacity cut (0.75×), and pool-based buffer management.

5 High-Performance Data Distribution

5.1 ZAP Zero-Copy Binary Protocol

The ZAP (Zero-copy Application Protocol) provides extreme-performance serialization for consensus messaging, AI agent communication, and GPU cluster coordination. Two parallel implementations serve different ecosystems:

- **Rust implementation**: Multi-language SDK generation, MCP gateway bridging, PQ handshake.
- **Go implementation**: Consensus wire protocol, VM-to-VM communication, mDNS discovery.

5.1.1 Performance

Operation	ZAP	Protobuf	Speedup
Parse	2.2 ns	500 ns	172×
Serialize	44 ns	4,221 ns	96×
Consensus round	5.5 ns	—	—
Allocations/op	0	11–121	∞
End-to-end TPS	11.5M	246K	47×

Table 4: ZAP performance vs. Protocol Buffers.

5.1.2 Wire Format

ZAP uses a 16-byte little-endian header:

Listing 3: ZAP wire header format.

```
Magic:       "ZAP\x00"     (4 bytes)
```

```
Version:     1              (2 bytes)
Flags:       compressed|encrypted|signed (2 bytes)
Root Offset: offset to root struct (4 bytes)
Size:        total message size     (4 bytes)
```

All struct fields are accessed by fixed offset calculation with 8-byte alignment, enabling true zero-copy reads directly from the network buffer.

5.1.3 Post-Quantum Handshake

ZAP includes native PQ transport security using the same hybrid scheme as RNS: ML-KEM-768 + X25519 for key exchange, ML-DSA-65 for authentication.

5.2 Hanzo Stream: Resilient Event Distribution

Hanzo Stream provides Kafka wire protocol compatibility over NATS JetStream, enabling standard Kafka clients to connect without modification while gaining the resilience benefits of JetStream's distributed storage.

- **Stateless**: All storage and replication delegated to NATS JetStream.
- **Compression**: GZIP, Snappy, LZ4, ZSTD.
- **Topic mapping**: Kafka topic T, partition $P \rightarrow$ JetStream stream `kafka-`T`-`P.
- **Consumer offsets**: KV bucket with key format `{group}.{topic}.{partition}`.

6 Software-Defined Networking

6.1 Hanzo Ingress: L7 Reverse Proxy

Hanzo Ingress serves as the cloud-native front door for all production traffic:

- **Protocol support**: HTTP/1.1, HTTP/2, gRPC, WebSocket, QUIC-ready.
- **TLS management**: Automatic Let's Encrypt with DNS challenge, wildcard certificates, zero-downtime renewal.
- **Circuit breakers**: Configurable failure expression thresholds with fallback responses, preventing cascading failures.
- **Rate limiting**: Per-endpoint and global token bucket algorithms.
- **Observability**: Prometheus metrics, OpenTelemetry tracing, structured access logging.

6.2 Hanzo Gateway: Policy-Based Routing

The API gateway routes 147+ endpoints across production services with hot-reloadable YAML configuration:

- **JWKS authentication**: JWT validation against `hanzo.id/.well-known/jwks` with 10-second TTL caching.

- **Org-scoped identity**: `X-Org-Id`, `X-User-Id`, `X-User-Email` headers derived from JWT claims.
- **Per-endpoint rate limiting**: Global 5,000 req/s, per-IP 100 req/s.
- **Multi-level security**: Organization-scoped access via IAM + KMS integration, with 46+ services using KMSSecret CRDs.

7 Edge Tunnel System

The Hanzo Tunnel provides a Rust-based cloud relay for connecting edge devices to the platform:

- **Transport**: JSON frames over WebSocket Secure (WSS).
- **Resilience**: Automatic reconnection with exponential backoff (1s initial, 60s max), 30-second heartbeat, 10-second connect timeout.
- **Authentication**: Bearer token (JWT from hanzo.id OAuth or API key).
- **Frame types**: Register, Registered, Event, Command, Response, Ping, Pong.
- **App kinds**: Dev, Node, Desktop, Bot, Extension.

8 Kubernetes-Native Infrastructure

The production deployment runs on Kubernetes with 65+ deployments across multiple clusters:

- **High availability**: Multi-replica deployments (2+ for stateless services) with zero-downtime rolling updates (`maxSurge: 1, maxUnavailable: 0`).
- **Secrets management**: KMSSecret CRDs for declarative secret synchronization from Hanzo KMS to Kubernetes secrets. Zero plaintext in YAML manifests.
- **Storage**: Persistent volumes on block storage for stateful services (PostgreSQL, NATS JetStream, ClickHouse).
- **Multi-cluster**: `hanzo-k8s` (DigitalOcean SFO3) for services, `apps-gke` (GCP) for ARM64 CI runners.

9 Performance Analysis

9.1 Consensus over RNS

Transport	Finality	Bandwidth	Latency
TCP/IP (LAN)	500 ms	1–10 Gbps	<1 ms
TCP/IP (WAN)	700 ms	100 Mbps	50–200 ms
RNS (TCP)	900 ms	100 Mbps	50–200 ms
RNS (LoRa)	3–10 s	1–100 kbps	200–2000 ms
RNS (Satellite)	2–5 s	1–50 Mbps	500–1500 ms

Table 5: Consensus finality across transport media.

9.2 Protocol Overhead

Protocol	Parse	Allocs	Throughput	PQ Support
ZAP	2.2 ns	0	11.5M TPS	Yes
Protobuf	500 ns	11–121	246K TPS	No
JSON-RPC	5,579 ns	58	140K calls/s	No
Cap'n Proto	∼100 ns	0	∼2M TPS	No

Table 6: Protocol performance comparison.

10 Naval Application Scenarios

10.1 Shipboard Mesh Network

A carrier strike group deploys RNS mesh networking across vessels:

- **Inter-ship**: LoRa mesh (10–50 km range) for consensus messages, falling back to SATCOM for bulk data.
- **Intra-ship**: Ethernet/WiFi via AutoInterface for high-bandwidth local communication.
- **Security**: Hybrid PQ handshakes protect against adversary quantum capabilities; zero-trust overlay prevents lateral movement from compromised nodes.
- **Resilience**: If SATCOM is jammed, LoRa mesh maintains consensus among vessels within radio range.

10.2 Shore-to-Ship Relay

- Shore gateway nodes bridge between IP infrastructure and RNS mesh.
- ZAP protocol minimizes bandwidth usage (91% memory reduction vs. JSON-RPC).
- Hanzo Stream provides store-and-forward for delay-tolerant data synchronization.

10.3 Expeditionary Force Networking

- Man-portable LoRa nodes establish mesh network in austere environments.
- Edge AI models (Zen nano, 0.8B parameters) run on tactical hardware for local decision support.
- Consensus maintains data integrity even with intermittent connectivity.

10.4 Submarine Communications

- RNS serial interface connects to existing submarine communication systems.
- Store-and-forward ensures eventual consistency during extended silent running.
- Post-quantum encryption protects against long-term decryption of recorded traffic.

11 Transport Capability Matrix

Capability	TCP	DTLS	TLS	WS	Transwarp	RNS
Encryption	–	Yes	Yes	TLS	TLS	PQ
NAT Traversal	–	–	–	–	–	Yes
Mesh Routing	–	–	–	–	–	Yes
Store-Forward	–	–	–	–	–	Yes
Delay-Tolerant	–	–	–	–	–	Yes
LoRa Support	–	–	–	–	–	Yes
Packet Radio	–	–	–	–	–	Yes
Adaptive Flow	TCP	OS	TCP	TCP	W3	RNS

Table 7: Transport capability matrix across all supported protocols.

12 Conclusion and Roadmap

We have presented a comprehensive assured networking architecture that addresses the unique challenges of naval and tactical edge operations. The integration of RNS as a native consensus transport, combined with hybrid post-quantum cryptography, zero-trust overlay networking, and high-performance data distribution, provides a resilient foundation for DDIL environments.

12.1 Current Capabilities

- RNS transport fully integrated into Hanzo consensus with hybrid PQ handshake.
- ZAP protocol achieving 11.5M TPS with zero-copy design.
- Zero-trust overlay with 7 transport plugins and HSM support.
- Production Kubernetes deployment with 65+ services.

12.2 Roadmap

- **Q2 2026 (planned)**: SATCOM-optimized RNS interface with forward error correction.
- **Q3 2026 (planned)**: Multi-path simultaneous transport (IP + LoRa + satellite).
- **Q4 2026 (planned)**: COMSEC certification preparation for naval deployment.
- **2027 (planned)**: Integration with Navy tactical data links (Link 16/22).

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Reproducibility

Source code. `github.com/hanzoai/zap` (commit `TBD`).
Build. cargo build –release (Rust 1.78+); Go binaries via make build.
Hardware tested. Apple M-series, Linux amd64/arm64 (edge: ARM Cortex-A), tested with mDNS multicast.
Standards referenced. ML-KEM-768, ML-DSA-65 (FIPS 203/204), W3C DID, DOD DDIL.
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Quantum-Resistant Counter-Exploitation: Confidential Compute and Coercion-Resistant Communications

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Abstract

We present a suite of counter-exploitation technologies providing quantum-resistant secure messaging, coercion-resistant governance, confidential computation on encrypted data, and post-quantum anonymous communication. The SessionVM delivers end-to-end encrypted messaging using ML-KEM-768 (FIPS 203) and ML-DSA-65 (FIPS 204) with fresh key encapsulation per message, achieving 526,000 messages per second with forward secrecy. ThresholdVM provides fully homomorphic encryption (FHE) with GPU-accelerated BFV/BGV/CKKS/TFHE schemes and threshold decryption ensuring no single party observes plaintext. LatticeLSAG ring signatures enable post-quantum anonymous group signing with linkability for double-action detection. Zero-knowledge proof systems provide privacy pools, range proofs, and confidential transactions. A multi-agent adversarial framework with 83 specialized agents enables automated red/blue team operations. Together, these capabilities prevent adversary exploitation of cryptographic, network, and AI systems while maintaining operational security in contested environments.

Executive Summary. This is a toolkit for denying an adversary any use of what they intercept, capture, or coerce. If they record your messages, the encryption is quantum-resistant, so harvested traffic stays unreadable even against a future quantum computer; if they seize a key, fresh keys per message mean past communications remain sealed. If they pressure a person, the voting and governance schemes are built so no one can prove how they voted, removing the leverage behind coercion. Sensitive data can be computed on while still encrypted—no operator and no single machine ever sees it in the clear—and operators can authenticate and file reports anonymously, yet still be held accountable for duplicate or contradictory actions. For defense and intelligence users operating in contested or denied environments, the practical result is that compromise of a device, a key, or a person does not cascade into compromise of the mission.

Contents

1 Introduction

Counter-exploitation requires not merely defending systems against attack but actively preventing adversary use of intercepted communications, compromised keys, and captured infrastructure. The convergence of quantum computing, advanced persistent threats, and AI-powered adversary tools demands countermeasures that operate at multiple layers simultaneously.

This paper addresses five counter-exploitation domains:

1. **Communication security**: Quantum-resistant messaging immune to harvest-now-decrypt-later.
2. **Governance resilience**: Coercion-resistant decision-making under adversary pressure.
3. **Computation privacy**: Processing encrypted data without exposing plaintext.
4. **Identity protection**: Anonymous authentication and reporting.
5. **Active defense**: AI-powered adversarial testing and threat detection.

2 Post-Quantum Secure Messaging: SessionVM

2.1 Architecture

SessionVM is a post-quantum secure messaging virtual machine deployed on the Pars Network S-Chain. It provides end-to-end encrypted communication using exclusively NIST-standardized algorithms.

Algorithm	**Standard**	**Purpose**
ML-KEM-768	FIPS 203	Key encapsulation (confidentiality)
ML-DSA-65	FIPS 204	Digital signatures (authentication)
XChaCha20-Poly1305	IETF draft-xchacha	Symmetric AEAD encryption
BLAKE2b-256	RFC 7693	Hashing and key derivation

Table 1: SessionVM cryptographic primitives.

2.2 Session Identity

Post-quantum session identities use the prefix `07`:

$$\text{SessionID} = \text{``07''} \| \text{hex}\big(\text{BLAKE2b-256}(\text{pk}_{\text{KEM}} \| \text{pk}_{\text{DSA}})\big)$$

Total length: 66 characters (backward-compatible with classical `05`-prefix identities).

2.3 1:1 Message Encryption Protocol

Algorithm 1 SessionVM Post-Quantum Message Encryption

Sender S encrypts message m for recipient R:

Step 1: Encapsulation
$(\text{ct}_{\text{kem}}, \text{ss}) \leftarrow \text{ML-KEM-768.Encaps}(\text{pk}_R^{\text{KEM}})$

Step 2: Key Derivation
$K \leftarrow \text{BLAKE2b-256}(\text{ss} \| \text{pk}_S^{\text{DSA}} \| \text{pk}_R^{\text{KEM}})$

Step 3: Symmetric Encryption
$\text{nonce} \leftarrow \text{Random}(24)$
$\text{ct}_{\text{payload}} \leftarrow \text{XChaCha20-Poly1305.Seal}(K, \text{nonce}, m)$

Step 4: Authentication
$\sigma \leftarrow \text{ML-DSA-65.Sign}(\text{sk}_S^{\text{DSA}}, \text{ct}_{\text{kem}} \| \text{nonce} \| \text{ct}_{\text{payload}})$

Wire format:
$\text{ct}_{\text{kem}}^{1088} \| \text{nonce}^{24} \| \text{ct}_{\text{payload}} \| \sigma^{3309} \| \text{pk}_S^{\text{DSA},1952}$
Overhead: 6,373 bytes (vs. 112 bytes classical)

2.4 Security Properties

Theorem 2.1 (Per-Message Forward Secrecy)**.** *Each message uses a fresh ML-KEM encapsulation, generating a unique shared secret. Compromise of the recipient's long-term KEM key does not reveal past shared secrets, as each encapsulation is probabilistic and the shared secret is bound to the specific ciphertext.*

Theorem 2.2 (Authentication)**.** *Every message is signed with ML-DSA-65, providing EU-CMA authentication. The sender's DSA public key is included in the wire format for verification without prior key exchange.*

2.5 Performance

Operation	Time (M1 Max)
GenerateIdentity	268 μs
Encaps + Decaps	226 μs
Sign + Verify	1.08 ms
CreateSession	3.8 μs
SendMessage	1.9 μs

Table 2: SessionVM operation latencies.

2.6 Swarm Distribution

Messages are distributed across a swarm of storage nodes:

- Deterministic node assignment from session/message ID.
- ∼10 GB storage per node for swarm participation.
- Automatic rebalancing on node churn.
- 24-hour session TTL, 10,000 message limit, 30-day retention.

3 Coercion-Resistant Governance

3.1 Encrypted Voting (PIP-0003, PIP-0012)

The Pars Network implements coercion-resistant voting where:

- Votes are encrypted under FHE public keys before submission.
- Tallying occurs homomorphically—no individual vote is ever decrypted.
- Threshold decryption reveals only the aggregate result.
- Voters cannot prove how they voted (receipt-freeness), defeating coercion.

3.2 Encrypted CRDTs (PIP-0013)

Conflict-Free Replicated Data Types (CRDTs) operate on encrypted state:

- Merge operations execute on ciphertext without decryption.
- Eventual consistency maintained across disconnected nodes.
- Adversary with network access sees only encrypted CRDT state.

3.3 Shadow Governance (PIP-7010)

For censorship-resistant decision-making under oppressive conditions:

- Governance proposals encrypted and distributed via RNS mesh.
- Voting via anonymous ring signatures (LatticeLSAG).
- Results revealed only through threshold decryption ceremony.
- No single authority can block or observe governance activity.

3.4 Fractal Governance (PIP-7007)

Hierarchical governance with compartmentalization:

- Cell-based structure where each level knows only adjacent levels.
- Compromise of one cell does not expose the governance tree.
- Zero-knowledge proofs verify authority without revealing identity.

4 Confidential Computing: ThresholdVM and FHE

4.1 ThresholdVM Architecture

ThresholdVM provides fully homomorphic encryption as a native virtual machine:

Scheme	Domain	Use Case
BFV	Integer	Exact integer arithmetic on encrypted data
BGV	Modular	Modular arithmetic for cryptographic operations
CKKS	Approximate real	ML inference on encrypted features
TFHE	Boolean	Arbitrary Boolean circuits on encrypted bits

Table 3: FHE schemes supported by ThresholdVM.

4.2 Threshold Decryption

No single party ever sees plaintext:

1. Smart contract encrypts data under FHE public key.
2. ThresholdVM evaluates computation homomorphically.
3. t-of-n MPC parties produce decryption shares.
4. Result assembled from shares; individual shares reveal nothing.

4.3 GPU Acceleration

The Hanzo GPU kernel library provides Metal and CUDA kernels for FHE operations:

- NTT/iNTT forward and inverse transforms.
- Polynomial multiplication with modular arithmetic.
- TFHE bootstrap and blind rotate operations.
- Batch parallel ciphertext operations.

4.4 EVM Precompile Integration

Address	Name	Gas
`0x0700`	FHE Operations (Fheos)	500,000
`0x0701`	Access Control (ACL)	25,000
`0x0702`	Input Verifier	50,000
`0x0703`	Gateway (Warp relay)	100,000

Table 4: FHE EVM precompile addresses.

4.5 TFHE-KMS Bridge

Encrypted policy evaluation without plaintext exposure:

1. Client encrypts query under TFHE public key.
2. KMS evaluates access policy on encrypted query.
3. Returns encrypted Boolean result.
4. Client decrypts to learn only the policy decision.

5 Lattice Ring Signatures for Anonymity

5.1 LatticeLSAG Construction

LatticeLSAG extends Linkable Spontaneous Anonymous Group signatures to ML-DSA-65:

Definition 5.1 (LatticeLSAG)**.** Given a ring $R = \{pk_1, \ldots, pk_n\}$ of ML-DSA-65 public keys and a signer with index π and secret key sk_π:

1. **Anonymity**: For any verifier, $\Pr[\text{identify } \pi] = 1/n$.
2. **Linkability**: Signer produces key image $I = H(sk_\pi)$; same signer always produces same I.
3. **Spontaneous**: Ring formed from any public keys without coordination.
4. **PQ-safe**: Security reduces to Module-LWE hardness.

5.2 Applications

- **Anonymous intelligence reporting**: Source signs report with ring of authorized personnel; recipient verifies authenticity without identifying source.
- **Whistleblower protection**: Key images detect duplicate reports while preserving anonymity.
- **Covert operations**: Operational orders authenticated without revealing which commander issued them.
- **Double-action detection**: Linkability prevents same agent from submitting contradictory reports.

6 Zero-Knowledge Proof Systems

6.1 Privacy Pools

Confidential UTXO model with nullifiers:

- Transactions hide sender, recipient, and amount.
- Nullifiers prevent double-spending without revealing which UTXO was consumed.
- Range proofs verify amounts are positive without revealing values.
- ZK proofs of solvency without balance disclosure.

6.2 Proof Systems

System	Gas	Setup	Proof Size
Groth16	200K	Trusted	128 B
PLONK	250K	Universal	384 B
fflonk	250K	Universal	320 B
Halo2	300K	None	1–10 KB
KZG4844	50K	Trusted	48 B

Table 5: Zero-knowledge proof systems.

7 Content Provenance and Data Integrity

7.1 Data Integrity Seal (PIP-0010)

Cryptographic seals bind content to its creator:

- ML-DSA-65 signature over content hash + metadata + timestamp.
- Anchored to blockchain for immutable timestamping.
- Verifiable chain of custody from creation through distribution.

7.2 Content Provenance (PIP-0011)

Track content origin and modification history:

- Each transformation produces a provenance record (signed).
- Merkle tree of provenance records for efficient verification.
- Detect AI-generated content manipulation via provenance gaps.

7.3 Attestation Protocol

Domain-typed attestations with equivocation detection:

- **OracleCommit**: Data feed commitments.
- **SessionComplete**: Session lifecycle events.
- **EpochBeacon**: Epoch transition markers.
- Equivocation (signing conflicting statements) is automatically detected and penalized.

8 Network-Layer Countermeasures

8.1 Zero-Trust Micro-Segmentation

Hanzo ZT provides identity-first networking:

- Every connection requires cryptographic identity verification.

- No implicit trust from network position or IP address.
- Policy enforcement at every hop in the overlay mesh.
- Lateral movement from compromised nodes is impossible without valid identity.

8.2 Resilience Against Targeting

- **Circuit breakers**: Prevent cascading failure exploitation (configurable thresholds, automatic recovery).
- **Rate limiting**: Token bucket per-endpoint prevents resource exhaustion.
- **RNS mesh**: No centralized infrastructure to target; mesh reforms around destroyed nodes.
- **Dilithium transport**: Adaptive flow control maintains throughput under degraded conditions.

9 AI-Powered Counter-Intelligence

9.1 Multi-Agent Adversarial Framework

- **83 specialized agents**: Security-focused variants for vulnerability assessment, code audit, and threat modeling.
- **Red/blue pairs**: Automated adversarial testing where red agents attack and blue agents defend.
- **15 orchestrators**: Incident response workflows coordinating multiple agents.
- **Extended reasoning**: 200–300 sequential tool calls for deep analysis.
- **Anomaly detection**: O11y platform with real-time behavioral analytics for intrusion indicators.

9.2 Agent Consensus for Threat Validation

- Multiple agents independently analyze a potential threat.
- Threshold voting (67%) required to confirm threat classification.
- W3C DID authentication prevents agent impersonation.
- PQ-secure communication via ZAP protocol.

10 Counter Harvest-Now-Decrypt-Later

10.1 Defense-in-Depth Against HNDL

1. **Hybrid PQ transport**: All network traffic uses ML-KEM-768 + X25519; adversary must break both to decrypt.
2. **Epoch-based key rotation**: Consensus keys rotate every 10 minutes; even a quantum break reveals only 10 minutes of data.

3. **Forward secrecy**: Ephemeral keys destroyed after every session; past sessions cannot be decrypted.
4. **Quantum-safe symmetric**: XChaCha20-Poly1305 (256-bit) and BLAKE2b-256 provide 128-bit post-Grover security.
5. **Per-message encapsulation**: SessionVM generates fresh ML-KEM ciphertext per message, not per session.

10.2 Key Hierarchy Protection

Layer	Key Type	Rotation	Protection
Root	Shamir shares	Manual ceremony	HSM, t-of-n
Organization	AES-256 per-org	Policy-based	KMS, encrypted
Session	Ephemeral ML-KEM	Per-session	Destroyed after use
Message	Ephemeral ML-KEM	Per-message	Destroyed after use

Table 6: Key hierarchy with rotation and protection mechanisms.

11 Social Recovery and Key Escrow

11.1 Shamir-Based Recovery

- Root keys split into n shares; t required for reconstruction.
- Shares distributed to geographically separated custodians.
- Break-glass emergency tokens with time-limited validity.
- WORM audit log prevents undetected recovery operations.

11.2 Regulator Escrow

For compliance with lawful intercept requirements:

- Escrow shards held by designated authorities.
- Cooperative reconstruction requires t parties (no unilateral access).
- Audit trail records all access attempts.
- 6–7 year retention per SEC 17a-4 requirements.

12 Naval Countermeasure Scenarios

12.1 Electronic Warfare Environment

- RNS mesh reforms around jammed nodes; LoRa provides backup.
- PQ encryption prevents exploitation of intercepted traffic.
- Circuit breakers prevent adversary-induced cascading failures.

12.2 Contested Communications

- SessionVM messaging over RNS mesh: no centralized infrastructure to target.
- ZAP protocol minimizes bandwidth (91% reduction vs. JSON-RPC).
- Store-and-forward ensures message delivery during intermittent connectivity.

12.3 Insider Threat

- Zero-trust prevents lateral movement from compromised endpoints.
- Ring signatures enable anonymous reporting of suspicious activity.
- Threshold MPC ensures no single insider can compromise keys.
- Behavioral analytics detect anomalous access patterns.

12.4 Supply Chain Integrity

- Content provenance (PIP-0011) tracks software components from source to deployment.
- Data integrity seals (PIP-0010) verify component authenticity.
- Attestation protocol detects tampering via equivocation detection.

13 Strategic Context

Counter-exploitation is where ownership of the full stack matters most: every layer described here—the messaging, the governance, the confidential compute, and the AI-driven adversarial defense—is built and controlled by Hanzo AI, an American-owned applied-AI lab and venture studio (Techstars '17, founded 2014) behind more than 100 venture-funded startups and several unicorns, led by a founder expert in both AI/ML and cryptography. The AI-powered components of this suite—the multi-agent red/blue framework, agent-consensus threat validation—depend on frontier models, and that is the strategic fault line. U.S. frontier AI has consolidated into a closed commercial duopoly, while the only broadly open frontier weights available today are largely of foreign origin; an organization whose mission is to deny adversary exploitation cannot reasonably build its active defenses on models it can neither inspect nor run in a denied environment. Hanzo is the American open-source frontier alternative: it owns both the Zen model family and the post-quantum cryptography underneath, so the adversarial-defense agents and the coercion-resistant primitives share one auditable, sovereign foundation that runs offline and at the edge—no rented API, no foreign dependency, nothing the adversary controls. This is operational rather than aspirational: the same agentic engineering stack delivered a FINRA/SEC-approved, SOC2 securities platform—a full ATS/BD/TA stack consolidated from 200+ microservices into 3 compiled binaries—to production with a small team, demonstrating that the agent orchestration invoked throughout this paper performs under real regulatory and reliability constraints.

14 Conclusion

We have presented a comprehensive counter-exploitation suite combining post-quantum secure messaging, coercion-resistant governance, confidential computing, anonymous authentication, and AI-powered adversarial defense. Every component uses NIST-standardized post-quantum algorithms, ensuring long-term security against quantum adversaries.

The integration of these capabilities into a unified architecture—where SessionVM messaging, ThresholdVM computation, LatticeLSAG anonymity, and multi-agent threat detection share the same cryptographic foundation and consensus infrastructure—provides defense-in-depth that individual point solutions cannot achieve.

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Reproducibility

Source code. `github.com/hanzoai/kms, hanzoai/iam` (commit `TBD`).
Build. make build per component; integrated K8s deploy via kubectl kustomize.
Hardware tested. Linux amd64/arm64; HSM integration tested with PKCS#11.
Standards referenced. NIST SP 800-207 (Zero Trust), Shamir SSS, FIPS 140-3.
Contact. research@hanzo.ai.

Quantum Key Distribution and Quantum Sensing Integration for Post-Quantum Blockchain Consensus and Threshold Cryptography

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Abstract

We present an architecture for integrating quantum key distribution (QKD) hardware and quantum sensing devices into a production post-quantum blockchain stack. The system extends five existing production components—QuantumVM (Q-Chain), the KEM cryptographic interface, RNS mesh transport, Sovereign KMS, and the Hanzo Zero Trust fabric—with QKD-derived key material and quantum sensor attestations. A novel *triple-hybrid* key derivation combines classical Diffie-Hellman (X25519), post-quantum lattice encapsulation (ML-KEM-768, FIPS 203), and physics-layer QKD shared secrets via HKDF, ensuring link security if *any one* of three independent foundations holds: elliptic curve hardness, Module-LWE hardness, or the laws of quantum mechanics. Quantum sensors—atomic clocks, magnetometers, and gravimeters—feed cryptographically attested measurements into the Q-Chain via a Sensor Oracle, stamped with dual BLS/Corona threshold signatures for immutable provenance. We specify ETSI GS QKD 014 and ITU-T Y.3800 compliant interfaces, analyze integration with threshold MPC signing ceremonies, and project a 3,000–5,000 line implementation leveraging existing production interfaces. No competing blockchain system offers production post-quantum consensus, let alone QKD or quantum sensing integration.

Executive Summary. This paper is a *design*, not a fielded product—it specifies how to extend an existing, in-production post-quantum system with two emerging quantum technologies, and estimates the work at roughly 3,000–5,000 lines of new code (see the roadmap table). The first technology, quantum key distribution, secures a communications link using the laws of physics: an eavesdropper cannot tap the channel without being detected. The design uses it as a *third* independent lock layered on top of today's cryptography and the new quantum-resistant standards, so the link stays secure as long as any one of the three holds—belt, suspenders, and a physical law. The second, quantum sensing, lets devices that measure time, magnetic fields, or gravity with extreme precision sign their readings and anchor them to a tamper-evident record—useful for navigation and timing when GPS is jammed, and for detecting things (like submarines) that conventional sensors miss. The intended audience is defense and government technologists planning for the quantum era; the honest claim is that the production foundation it plugs into already exists, and this specifies the bridge to quantum hardware as that hardware matures.

Contents

1 Introduction

Post-quantum cryptography protects against *algorithmic* quantum attacks—Shor's algorithm breaking RSA and elliptic curves, Grover's algorithm weakening symmetric ciphers. However, algorithmic security rests on computational hardness *assumptions* (Module-LWE, Module-SIS) that, while believed strong, lack the unconditional guarantee of physics-based security.

Quantum key distribution provides information-theoretic security grounded in the laws of quantum mechanics: an eavesdropper disturbing a quantum channel is detectable with probability approaching 1. Quantum sensing exploits quantum coherence and entanglement for measurement precision beyond classical limits.

This paper presents an architecture that layers physics-based security (QKD) atop algorithmic security (PQ crypto) atop classical security (ECDH), creating a triple-hybrid defense-in-depth unique in the blockchain and military networking landscape.

1.1 Existing Production Foundation

The architecture builds on five deployed systems:

1. **QuantumVM (Q-Chain)**: Production VM with ML-DSA-44/65/87 signing, SLH-DSA stamping, and Quasar dual-threshold consensus (BLS + Corona).
2. **KEM Interface**: Pluggable key encapsulation supporting ML-KEM-768, ML-KEM-1024, X25519, and hybrid modes.
3. **RNS Mesh Transport**: Medium-agnostic networking with hybrid PQ handshake (X25519 + ML-KEM-768 + ML-DSA-65) over LoRa, serial, TCP, and satellite.
4. **Sovereign KMS**: Enterprise key management with Shamir sharing, HPKE wrapping, transit engine, and TFHE bridge.
5. **Zero-Trust Fabric**: Identity-first overlay network with pluggable transport underlays and HSM integration.

1.2 What This Paper Adds

1. **QKD-KEM Bridge**: ETSI GS QKD 014 client wrapping QKD-derived keys into the existing `kem.KEM` interface.
2. **Triple-Hybrid Key Derivation**: $K = \text{HKDF}(\text{X25519\_ss} \| \text{ML-KEM\_ss} \| \text{QKD\_key})$.
3. **Sensor Oracle**: New transaction type in QuantumVM for quantum sensor attestations with dual-threshold signatures.
4. **QRNG Integration**: Drop-in replacement of `crypto/rand` with quantum random number generator hardware.
5. **QKD Network Manager**: ITU-T Y.3800 compliant key pool orchestration across multiple QKD links.

1.3 Strategic Context

The distinction this paper draws—between a deployed foundation and a specified extension—is itself the strategic point. The five systems in Section 1.1 (QuantumVM, the KEM interface, RNS mesh transport, the sovereign KMS, and the zero-trust fabric) are in production today; the QKD and quantum-sensing layers described here are a forward-looking architecture, projected at roughly 3,000–5,000 lines that implement existing interfaces without modifying the production code. Being able to make that separation credibly depends on owning the whole stack. Hanzo (American-owned, Techstars '17, founded 2014) builds both the post-quantum cryptography in production and the *Zen* frontier AI that runs on it, and its founder is a recognized expert in both AI/ML and cryptography—which is what lets a quantum-hardware bridge be specified against real, owned interfaces rather than sketched in the abstract. The wider context is sovereignty: with American frontier AI consolidated into a closed two-vendor duopoly and the open frontier ecosystem now largely offshore, Hanzo is the American open-source alternative that owns the model and the cryptography together. This roadmap extends that sovereign, auditable foundation toward the quantum era—honestly labeled as a roadmap, on top of a foundation that already runs.

2 Quantum Key Distribution Integration

2.1 QKD Protocol Support

The architecture supports all major QKD protocol families via the ETSI standardized hardware interface:

Protocol	Type	Range	Rate	Vendors
BB84	DV, prepare-measure	<100 km	∼1 Mbps	ID Quantique
CV-QKD	Continuous-variable	<50 km	∼10 Mbps	Toshiba
MDI-QKD	Measurement-device-indep.	<400 km	∼100 kbps	QuantumCTek
TF-QKD	Twin-field	<600 km	∼10 kbps	Toshiba, USTC
E91	Entanglement-based	<100 km	∼10 kbps	Research

Table 1: Supported QKD protocols via ETSI hardware interface.

2.2 ETSI GS QKD 014 Client

All QKD hardware interaction uses the ETSI GS QKD 014 REST API, the industry standard key delivery interface:

Listing 1: ETSI QKD 014 key request flow.

```
// Request quantum-derived key material
GET /api/v1/keys/{slave_SAE_ID}/status
  -> { "source_KME_ID": "kme-alpha",
       "key_count": 42,
       "max_key_size": 256 }

POST /api/v1/keys/{slave_SAE_ID}/enc_keys
  body: { "number": 1, "size": 256 }
  -> { "keys": [
         { "key_ID": "a1b2c3d4-...",
           "key": "base64(256-bit key)" }
```

```
      ]}

// Peer retrieves same key by ID
POST /api/v1/keys/{master_SAE_ID}/dec_keys
  body: { "key_IDs": ["a1b2c3d4-..."] }
  -> { "keys": [
         { "key_ID": "a1b2c3d4-...",
           "key": "base64(same 256-bit key)" }
      ]}
```

The ETSI 014 abstraction makes the architecture hardware-agnostic: BB84, CV-QKD, or TF-QKD appliances from any vendor present the same REST interface.

2.3 QKD-KEM Bridge

The QKD-KEM bridge implements the existing `kem.KEM` interface, enabling QKD to slot into any component that uses key encapsulation:

Listing 2: QKD-KEM bridge implementing existing KEM interface.

```
type QKDKEM struct {
    client *etsi014.Client
    peerID string
}

func (q *QKDKEM) Encapsulate() (ct, ss []byte, err error) {
    // Request key from local QKD node
    key, err := q.client.GetKey(q.peerID, 256)
    if err != nil { return nil, nil, err }
    // "ciphertext" is just the key_ID (peer fetches same key)
    return []byte(key.KeyID), key.Material, nil
}

func (q *QKDKEM) Decapsulate(ct []byte) (ss []byte, err error) {
    keyID := string(ct)
    key, err := q.client.GetKeyByID(q.peerID, keyID)
    return key.Material, err
}
```

This plugs directly into the RNS link handshake at `rns_link.go` where ML-KEM ciphertexts are exchanged. The QKD-KEM has zero ciphertext overhead—the key is delivered out-of-band by the quantum channel.

3 Triple-Hybrid Key Derivation

3.1 Security Model

Definition 3.1 (Triple-Hybrid Security)**.** A key exchange is triple-hybrid secure if the derived session key K is indistinguishable from random under the assumption that *at least one* of the following holds:

1. The Computational Diffie-Hellman problem on Curve25519 is hard (classical).
2. The Module-LWE problem with ML-KEM-768 parameters is hard (post-quantum algorithmic).

3. The laws of quantum mechanics prevent undetected eavesdropping on the QKD channel (physics-layer).

3.2 Protocol

Algorithm 1 Triple-Hybrid Key Exchange

Layer 1 — Classical ECDH:
  $\text{ss}_1 \leftarrow \text{X25519.DH}(\text{ek}_A, \text{epk}_B)$ (32 bytes)

Layer 2 — Post-Quantum KEM:
  $(\text{ct}_2, \text{ss}_2) \leftarrow \text{ML-KEM-768.Encaps}(\text{epk}_B^M)$ (32 bytes)

Layer 3 — Physics-Layer QKD:
  $\text{ss}_3 \leftarrow \text{QKD.GetKey}(\text{peer\_ID}, 256)$ (32 bytes)
  $\text{key\_ID} \leftarrow \text{QKD.KeyID}(\text{ss}_3)$

Combine: $K \leftarrow \text{HKDF-SHA256}(\text{salt}, \text{ss}_1 \| \text{ss}_2 \| \text{ss}_3)$
Derive: $K_{\text{send}}, K_{\text{recv}} \leftarrow \text{HKDF-Expand}(K, \text{“triple-hybrid-v1”})$
Exchange: Send $\text{epk}_A^X \| \text{ct}_2 \| \text{key\_ID}$ to peer
Session: AES-256-GCM with derived keys

Theorem 3.2 (Triple-Hybrid IND-CCA2)**.** *If any one of X25519, ML-KEM-768, or the QKD key is secure (i.e., unknown to the adversary), then the HKDF output K is computationally indistinguishable from a uniformly random key of the same length. This follows from the extraction property of HKDF: given sufficient min-entropy in any one input, the output is pseudorandom regardless of the other inputs.*

3.3 Wire Format

Component	Classical	PQ Hybrid	Triple Hybrid
X25519 ephemeral PK	32 B	32 B	32 B
ML-KEM-768 ciphertext	—	1,088 B	1,088 B
QKD key_ID (UUID)	—	—	36 B
ML-DSA-65 signature	—	3,309 B	3,309 B
Total per-direction	**32 B**	**∼4.4 KB**	**∼4.5 KB**

Table 2: Wire overhead comparison: triple-hybrid adds only 36 bytes (UUID) over PQ hybrid.

The QKD layer adds only 36 bytes (a UUID key reference) to the handshake—the actual key material is delivered by the quantum channel, not the classical network.

4 Quantum Sensing and Attestation

4.1 Sensor Types

Sensor	Technology	Precision	Naval Application
Atomic clock	CSAC/optical lattice	<1 μs accuracy	GPS-denied timing
Magnetometer	SERF/NV-center	<1 fT sensitivity	Submarine detection
Gravimeter	Atom interferometry	<1 μGal	Subsurface mapping
Quantum radar	Microwave photons	Enhanced SNR	Target detection
Quantum lidar	Entangled photons	Sub-shot-noise	Terrain mapping

Table 3: Quantum sensor types with naval applications.

4.2 Sensor Oracle Architecture

Quantum sensor measurements are ingested into the Q-Chain via a Sensor Oracle transaction type:

Listing 3: Sensor attestation transaction.

```
type SensorAttestation struct {
    SensorType    SensorType  // AtomicClock, Magnetometer, etc.
    DeviceID      [32]byte    // Hardware attestation identity
    Timestamp     int64       // Sub-nanosecond precision
    Measurement   []byte      // Sensor-specific data format
    Uncertainty   float64     // Measurement uncertainty
    TEESignature  []byte      // TEE-signed device attestation
    MLDSASignature []byte     // ML-DSA-65 over all fields
}
```

The QuantumVM's existing `StampBlock` interface accepts arbitrary block IDs and messages—sensor attestations are stamped with the same Quasar dual-signature (BLS + Corona) that certifies consensus blocks.

4.3 Atomic Clocks for Consensus Timing

Chip-scale atomic clocks (CSAC) provide GPS-independent timing for DDIL environments:

- **Current**: QuantumVM uses `time.Now()` for block timestamps (OS clock, $\sim$1 ms accuracy).
- **With CSAC**: Sub-microsecond timestamp ordering via PTP/1PPS interface from quantum clock.
- **Byzantine detection**: Validators with drifted clocks produce timestamps outside the consensus window, enabling automatic identification of faulty or compromised nodes.
- **GPS-denied operation**: Aligns with RNS mesh networking for maritime/submarine scenarios where GPS signals are unavailable or jammed.

4.4 Magnetometers for Physical Security

Quantum magnetometers detecting sub-femtotesla fields enable:

- **Tamper detection**: Cryptographically attested magnetic field baseline for data center physical security. Any hardware modification changes the field signature.
- **Submarine detection**: Blockchain-attested magnetic anomaly data from distributed sensor arrays.
- **Navigation**: Magnetic field map matching for GPS-denied underwater navigation, attested on-chain for integrity.

4.5 Gravimeters for Subsurface Intelligence

Atom-interferometry gravimeters provide:

- **Subsurface mapping**: Detect underground structures, tunnels, and voids.
- **Immutable geological records**: Blockchain-attested gravity surveys with provenance chains.
- **Anti-spoofing**: Ground-truth gravity data anchored to consensus prevents adversary falsification.

5 QRNG Integration

5.1 Drop-In Entropy Source

Quantum random number generators (QRNG) provide entropy from quantum vacuum fluctuations or single-photon detection:

Listing 4: QRNG entropy source replacement.

```
// Current (QuantumSigner, line 104):
crypto.rand.Read(nonce)  // reads /dev/urandom

// With QRNG hardware:
qrng.Read(nonce)  // reads /dev/qrng0
// Fallback: crypto/rand if QRNG unavailable
```

Device	Interface	Rate	Standard
ID Quantique Quantis	USB/PCIe	4–240 Mbps	NIST SP 800-90B
Quside FQRNG	PCIe	400+ Mbps	NIST SP 800-22
Toshiba QRNG	Integrated	2 Gbps	MIST/BSI certified

Table 4: QRNG hardware options.

QRNG integration is the lowest-effort, highest-value component: a single import replacement at `rand.Read()` call sites across the key generation and nonce generation code paths.

6 Integration with Threshold MPC

6.1 QKD-Authenticated DKG Ceremony

Distributed Key Generation requires secure channels between all n parties. QKD provides information-theoretically secure channels for share distribution:

Algorithm 2 QKD-Authenticated Threshold Key Generation

Setup: Each pair (P_i, P_j) establishes QKD link
  $K_{ij} \leftarrow \text{QKD.GetKey}(P_j, 256)$ (symmetric key from quantum channel)

DKG Phase 1: Commitment broadcast
for each party P_i **do**
  Sample polynomial $f_i(x)$ of degree $t-1$
  Broadcast commitment $C_i = g^{f_i(0)}$ (public, signed ML-DSA-65)
  for each party P_j **do**
    Encrypt share: $e_{ij} = \text{AES-256-GCM}(K_{ij}, f_i(j))$
    Send e_{ij} via classical channel (QKD key protects confidentiality)
  end for
end for

Advantage: Share distribution is information-theoretically secure
Even a quantum computer cannot recover $f_i(j)$ from e_{ij} without K_{ij}

6.2 QKD-Secured Signing Sessions

Threshold signing messages between MPC parties use QKD-derived session keys:

- **CGGMP21**: Round messages encrypted with QKD keys via ZAP transport.
- **FROST**: Commitment and response shares protected by QKD-authenticated channels.
- **Corona**: Lattice threshold shares doubly protected—PQ algorithmic + QKD physics-layer.
- **BLS**: Aggregate signature shares transmitted over QKD-secured links.

6.3 Key Rotation with QKD Refresh

Corona epoch rotation (every 10 minutes) can incorporate QKD key refresh:

1. At epoch boundary, validators request fresh QKD keys from all peer links.
2. New Corona threshold keys generated using QKD-authenticated DKG.
3. Old keys destroyed; forward secrecy guaranteed by both ephemeral PQ keys and QKD one-time keys.
4. Triple protection: even if lattice assumptions fail AND classical crypto breaks, QKD keys remain secure.

7 Integration with Consensus

7.1 QKD-Authenticated Validator Links

Quasar consensus samples k peers per round. QKD provides an additional authentication layer:

- **Pre-established QKD links**: Validators with fiber-connected QKD appliances establish shared keys.
- **Consensus message authentication**: HMAC using QKD-derived key appended alongside BLS signatures.
- **Sybil resistance**: QKD links are physical—an adversary cannot create virtual QKD connections without quantum hardware and fiber access.
- **Partial deployment**: Validators without QKD hardware continue using PQ-hybrid authentication. The system degrades gracefully.

7.2 Sensor-Attested Block Production

Blocks can include quantum sensor attestations as special transactions:

- Atomic clock timestamps replace OS timestamps for sub-microsecond ordering.
- Magnetometer readings provide physical-layer environmental context.
- All sensor data receives Quasar dual-signature (BLS + Corona) for immutable provenance.
- Sensor attestation chain provides tamper-evident physical security monitoring.

8 KMS Integration

8.1 QKD Key Import

The Sovereign KMS Transit Engine supports key import via `importKeyMaterial()`:

- QKD-derived keys imported as a new key version type: `hanzo-qkd:v{N}:{base64}`.
- Keys stored with the same per-org isolation as algorithmic keys.
- QKD keys can serve as root keys for Shamir splitting, providing physics-layer root-of-trust.
- HPKE wrapping extended to include QKD key material alongside ML-KEM-768.

8.2 Key Hierarchy with QKD Root

Layer	**Key Source**	**Security Basis**	**Rotation**
Root	QKD + Shamir	Physics + threshold	Ceremony
Organization	AES-256 (QKD-derived)	Physics-layer	Policy
Session	Triple-hybrid KDF	Classical + PQ + physics	Per-session
Message	ML-KEM ephemeral	Algorithmic PQ	Per-message

Table 5: Key hierarchy with QKD-anchored root of trust.

9 Zero-Trust Fabric Integration

9.1 QKD Transport Underlay

The Hanzo ZT fabric supports pluggable transport underlays. A QKD-authenticated underlay implements the existing `Transport.Connection` interface:

- **Physical link**: Fiber-optic QKD channel between zero-trust nodes.
- **Key delivery**: ETSI GS QKD 014 provides symmetric keys.
- **Data encryption**: AES-256-GCM using QKD-derived keys on the classical channel.
- **Authentication**: QKD keys authenticate zero-trust identity assertions.
- **Policy enforcement**: Zero-trust policies applied after QKD authentication—QKD provides the channel, ZT provides the authorization.

10 Standards Compliance

Standard	Scope	Compliance
ETSI GS QKD 004	Session-based key delivery	Client implementation
ETSI GS QKD 014	REST key delivery API	Client implementation
ITU-T Y.3800	QKD network overview	Functional architecture
ITU-T Y.3802	QKD functional architecture	Key management layer
NIST SP 800-90B	Entropy source validation	QRNG qualification
FIPS 203 (ML-KEM)	Key encapsulation	Production (existing)
FIPS 204 (ML-DSA)	Digital signatures	Production (existing)
FIPS 205 (SLH-DSA)	Hash-based signatures	Production (existing)
CNSA 2.0	NSA algorithm suite	Full compliance

Table 6: Standards compliance matrix.

11 Naval Deployment Scenarios

11.1 Shore-to-Ship QKD via Submarine Fiber

- QKD appliances at shore facility and pier-side ship terminal.
- Pre-load QKD key pool before departure (keys consumed during deployment).
- Triple-hybrid key derivation for all ship-to-shore encrypted communications.
- Key pool size determines operational endurance without QKD refresh.

11.2 Satellite QKD for Global Coverage

- Satellite-based QKD (demonstrated by Micius satellite at 1,200 km) for global key distribution.
- Trusted-node relay architecture for intercontinental links.
- QKD keys distributed to fleet units via satellite pass windows.
- Stored keys consumed for consensus authentication between passes.

11.3 Submarine Quantum Sensing Array

- Distributed quantum magnetometer array for ASW (anti-submarine warfare).
- Sensor attestations transmitted via RNS mesh (acoustic/ELF radio).
- Blockchain-anchored magnetic anomaly data for post-mission analysis.
- Tamper-evident sensor chain prevents adversary data injection.

11.4 GPS-Denied Navigation

- Quantum atomic clocks provide autonomous timing without GPS.
- Quantum gravimeters enable terrain-referenced navigation underwater.
- Quantum magnetometers support magnetic-field navigation.
- All sensor data attested on Q-Chain with dual-threshold signatures.

12 Implementation Roadmap

Phase	Component	LOC	Timeline
1	QRNG entropy integration	∼200	Q2 2026 (planned)
2	ETSI GS QKD 014 client	∼800	Q3 2026 (planned)
3	QKD-KEM bridge	∼400	Q3 2026 (planned)
4	Triple-hybrid key derivation	∼300	Q3 2026 (planned)
5	Sensor Oracle (QuantumVM)	∼1,000	Q4 2026 (planned)
6	QKD Network Manager	∼1,500	Q1 2027 (planned)
7	Zero-trust QKD underlay	∼800	Q1 2027 (planned)
	Total	**∼5,000**	

Table 7: Implementation roadmap: 5,000 lines of Go leveraging existing interfaces.

All components are additive—they implement existing interfaces (`kem.KEM`, `Transaction`, `crypto/rand`, `Transport.Connection`) without modifying production code.

13 Competitive Analysis

Capability	**Ours**	**QAN**	**IOTA**	**Algorand**
PQ signatures (prod)	✓	Planned	–	Partial
PQ key exchange (prod)	✓	–	–	–
Threshold PQ (Corona)	✓	–	–	–
QKD integration	Design	–	–	–
Quantum sensing	Design	–	–	–
Triple-hybrid KDF	Design	–	–	–
Mesh/DDIL transport	✓	–	Partial	–
Zero-trust fabric	✓	–	–	–

Table 8: Competitive landscape: no competitor has PQ consensus, let alone QKD/sensing.

14 Conclusion

We have presented an architecture for integrating quantum key distribution and quantum sensing into a production post-quantum blockchain stack. The triple-hybrid key derivation—combining classical ECDH, lattice-based KEM, and physics-layer QKD—provides defense-in-depth across three independent security foundations.

The architecture is uniquely feasible because the production stack already provides the integration surfaces: a pluggable KEM interface for QKD keys, a QuantumVM for sensor attestations, a mesh transport for DDIL deployment, and a zero-trust fabric for QKD-authenticated channels. The estimated 5,000 lines of new code leverage these existing interfaces without modifying production systems.

No competing blockchain or military networking system offers this combination of production post-quantum cryptography, QKD integration architecture, and quantum sensor attestation—let alone over a delay-tolerant mesh network operating on LoRa, packet radio, and satellite links.

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Reproducibility

Source code. `github.com/hanzoai/crypto` (commit `TBD`).
Build. cargo build –release; hardware integration TBD on QKD/atomic-clock partners.
Hardware tested. QKD hardware: vendor-supplied; classical post-processing on x86_64 Linux.
Standards referenced. ETSI QKD, NIST PQC, FIPS 203/204.
Contact. research@hanzo.ai.

Automated Verification and Validation Framework with Digital Twin Environments

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Abstract

We present an integrated verification and validation (V&V) framework combining continuous observability, behavioral analytics, ML pipeline orchestration, digital twin environments, and formal cryptographic proofs. The observability platform ingests logs, metrics, and distributed traces into a ClickHouse OLAP backend, with LLM-specific telemetry tracking token usage, cost, latency, and quality metrics across 100+ AI providers. Temporal workflows orchestrate LLM evaluation, trace clustering, and sentiment analysis, while Dagster DAGs manage ETL pipelines for training data preparation. A network simulation environment supports configurable topologies with 38 registered virtual machines, enabling mission rehearsal and system qualification. The framework includes 17 formal cryptographic proofs, 28+ consensus tests, and automated CI/CD with multi-platform builds. Anomaly detection, feature flag-controlled rollout, and A/B testing with statistical significance provide continuous data-driven decision support.

Executive Summary. This is the test, evaluation, and monitoring layer that tells operators whether a complex AI and cryptographic system actually works—before it is fielded and continuously once it is. Three capabilities work together: full instrumentation, so every action the system takes is recorded and can be inspected after the fact; a high-fidelity "digital twin" that replicates the real deployment, so new builds and stressful conditions (jamming, link loss, hostile inputs) can be rehearsed safely before they reach the field; and automated checks—including machine-checked cryptographic proofs—that run on every change and block anything that regresses. For a program office, this is the qualification and continuous-assurance evidence a system needs to earn and keep an authorization to operate, with specialized tracking for AI workloads (cost, latency, output quality) that conventional IT monitoring does not capture. The capability is built to run where the system runs, including disconnected at the edge with store-and-forward when connectivity returns.

Contents

1 Introduction

Modern naval systems integrate AI/ML, post-quantum cryptography, and distributed computing at unprecedented scale. Traditional V&V approaches—manual test plans, periodic audits, and waterfall certification—cannot keep pace with continuous deployment and evolving threats.

This paper presents an automated V&V framework built on three principles:

1. **Continuous observability**: Every system interaction is instrumented, traced, and queryable.
2. **Automated evaluation**: ML models, cryptographic implementations, and consensus protocols are continuously verified against formal specifications.
3. **Digital twin fidelity**: System qualification occurs in high-fidelity simulated environments before live deployment.

1.1 Strategic Context

A verification and validation framework is only as credible as the systems it has actually carried into production under real scrutiny. Hanzo (American-owned, Techstars '17, founded 2014) is an applied AI lab and venture studio behind 100+ venture-funded startups and multiple unicorns; its founder is a recognized expert in *both* AI/ML and cryptography, and the company builds both the *Zen* frontier model family and the post-quantum cryptographic stack it runs on. The same agentic engineering stack described in these papers delivered a securities trading platform—a full ATS/BD/TA stack consolidated from 200+ microservices into 3 compiled binaries—through FINRA/SEC approval and SOC2 audit into production, in record time with a small team. That is the operative proof point for test and evaluation: regulated, externally audited software shipped at scale, the same discipline a defense accreditation demands. It also anchors a broader position. With American frontier AI consolidated into a closed two-vendor duopoly and the open frontier ecosystem now largely offshore, Hanzo is the American open-source alternative that owns the model and the cryptography together—and the V&V framework here is how that stack is qualified to run sovereign and disconnected, where the data lives, rather than rented from a black box.

2 Observability Platform

2.1 Architecture

Hanzo O11y provides unified observability across three telemetry signals:

Signal	Capabilities	Storage
Logs	Full-text search, structured filters, correlation	ClickHouse
Metrics	Time-series, histograms, percentiles, Apdex	ClickHouse
Traces	Flamegraphs, Gantt charts, span detail	ClickHouse

Table 1: O11y three-signal observability architecture.

2.2 LLM-Specific Observability

AI workloads require specialized telemetry beyond traditional APM:

Listing 1: LLM span attributes in O11y trace.

```
{
  "name": "llm.completion",
  "duration_nano": 1500000000,
  "attrs": {
    "llm.model": "zen4-pro-27b",
    "llm.token_input": "2048",
    "llm.token_output": "512",
    "llm.temperature": "0.7",
    "llm.cost": "0.0035",
    "llm.provider": "hanzo-engine"
  }
}
```

- **Token tracking**: Input/output token counts per request, aggregated by model, provider, and team.
- **Cost analysis**: Real-time cost tracking with budget alerts per organization.
- **Quality metrics**: BLEU, ROUGE, exact match scores via evaluation workflows.
- **Latency breakdown**: Time-to-first-token, tokens-per-second, queue wait time.
- **Error analysis**: Provider failure rates, timeout patterns, content filter triggers.

2.3 Alert and Anomaly Detection

- **Threshold-based**: Static rules on any metric (latency $>$ 2s, error rate $>$ 5%).
- **Anomaly detection**: Statistical deviation from baseline patterns.
- **Composite rules**: AND/OR combinations with flapper suppression.
- **Multi-channel**: Email, Slack, PagerDuty, MS Teams, webhooks.
- **Escalation**: Time-based severity escalation with on-call integration.

3 Behavioral Analytics

3.1 Hanzo Insights Platform

Insights provides behavioral analytics with IAM-integrated multi-tenancy:

- **Event ingestion**: Rust capture service for high-throughput event collection, Kafka-compatible streaming via Hanzo Stream.
- **OLAP engine**: ClickHouse columnar storage for sub-second queries across billions of events.
- **Cohort segmentation**: Group users by behavior patterns, demographics, or temporal characteristics.

- **Funnel analysis**: Multi-step conversion tracking with statistical significance testing.
- **Retention modeling**: Cohort-based retention curves with churn prediction.
- **Feature flags**: A/B testing with gradual rollout, kill switches, and variant-level metrics.
- **Dashboards**: Time-series, heatmaps, pie charts, bar charts with custom SQL queries.

3.2 LLM Analytics Module

- **Provider adapters**: Unified client for 100+ LLM providers (OpenAI, Anthropic, local Hanzo Engine).
- **Evaluation framework**: LLM-as-judge with configurable metrics and datasets.
- **Trace clustering**: Semantic grouping of conversations using embeddings.
- **Sentiment classification**: Automated sentiment analysis on model outputs.
- **Model registry**: Version tracking, provider key management, deployment tagging.

4 ML Pipeline Orchestration

4.1 Temporal Workflows

Temporal provides durable execution for long-running ML operations:

Workflow	Purpose
RunEvaluationWorkflow	LLM model evaluation with judge activities
BatchTraceSummarizationWorkflow	Summarize conversation traces in batches
TraceClusteringWorkflow	Cluster traces by semantic similarity
DailyTraceClusteringWorkflow	Scheduled daily clustering
ClassifySentimentWorkflow	Sentiment analysis on LLM outputs

Table 2: Temporal ML workflow catalog.

Algorithm 1 LLM Evaluation Workflow

```
Fetch evaluation configuration from database
for each trial i = 1, ..., N do
    Execute LLM judge activity (retry: 3 attempts, exponential backoff)
    Emit evaluation result event to analytics stream
    Update evaluation state in key-value store
end for
Compute aggregate metrics (mean, std, percentiles)
Emit completion telemetry (success/error, duration)
return evaluation summary with per-trial results
```

4.2 Dagster DAG Pipelines

Dagster manages ETL and data preparation workflows:

- **PostgreSQL → ClickHouse ETL**: Real-time event ingestion with sensor triggers.
- **Historical backfill**: Time-partitioned asset materialization for training data.
- **Materialized columns**: Pre-compute expensive aggregations for query performance.
- **Person resolution**: User identity unification across events.
- **Operational alerts**: Slack notifications on DAG failures.

4.3 Experiment Tracking

- Model comparison with A/B testing framework.
- Statistical significance (t-test, chi-square) for metric differences.
- Prompt version tracking with performance diffs.
- Feature flags for gradual model rollout.

5 Digital Twin Environments

5.1 Netrunner: Network Simulation

Netrunner provides configurable network simulation for system qualification:

- Configurable validator count, network topology, and fault injection.
- Consensus protocol selection (Quasar, Wave, Photon, Nova, Nebula).
- Byzantine validator simulation for adversarial testing.
- Performance benchmarking under controlled conditions.

5.2 Kubernetes-Based Environment Cloning

- **Namespace isolation**: Each test environment in a dedicated K8s namespace.
- **Parallel execution**: Multiple test environments run simultaneously.
- **Docker Compose stacks**: Dev stack (single node) and Universe (5-validator production replica).
- **Multi-cluster**: hanzo-k8s (DigitalOcean) and apps-gke (GCP ARM64).

5.3 Virtual Machine Registry

38 registered VMs with pluggable lifecycle:

VM	Chain	Test Focus
PlatformVM	P-Chain	Staking, validator set management
ExchangeVM	X-Chain	UTXO asset trading
EVM	C-Chain	Smart contracts, precompiles
DexVM	D-Chain	Orderbook, perpetuals, AMM
ThresholdVM	T-Chain	FHE, threshold signing
SessionVM	S-Chain	PQ messaging
BridgeVM	B-Chain	Cross-chain verification
AIvm	A-Chain	AI agent consensus
QuantumVM	Q-Chain	PQ key operations

Table 3: Selected VMs from 38 registered (security-relevant subset).

6 Automated Testing Frameworks

6.1 Test Coverage

Component	Tests	Status
Quasar consensus	28+	All passing
Corona threshold	21	All passing
SessionVM E2E	6	All passing
DEX precompiles	68	All passing
NIST PQ vectors	Per-algorithm	All passing
MPC keygen/signing	33 files	All passing

Table 4: Automated test suite coverage.

6.2 CI/CD Pipeline

- **Multi-platform builds**: `linux/amd64` + `linux/arm64` via native runners (no QEMU).
- **ARC scale sets**: Up to 30 ARM64 runners on GKE T2A Ampere nodes.
- **Tags**: `:main`, `:test`, `:dev` (branch-based), `:latest` (default branch).
- **Shared workflow**: `hanzoai/.github/.github/workflows/docker-build.yml@main`.

7 Formal Verification

7.1 Cryptographic Proofs

17 formal proofs cover cryptographic primitives, consensus protocols, and cross-chain messaging:

- **proof-crypto-bls**: BLS12-381 aggregate signature unforgeability.
- **proof-crypto-cggmp21**: CGGMP21 threshold ECDSA correctness under UC framework.
- **proof-crypto-mldsa**: ML-DSA EU-CMA reduction to Module-LWE hardness.
- **proof-crypto-tfhe**: TFHE semantic security under LWE assumption.
- **proof-crypto-verkle**: Verkle tree binding and hiding properties.
- **proof-protocol-wave**: Wave consensus liveness and safety.
- **proof-protocol-nova**: Nova linear consensus termination bound.
- **proof-bridge-teleport**: Cross-chain message integrity guarantee.
- **proof-warp-delivery**: Warp messaging delivery with finality binding.

7.2 Gas Determinism

All EVM precompiles provide O(1) gas cost computation from input header bytes:

Theorem 7.1 (Gas Determinism)**.** *For any precompile P with input x, the gas cost $G(P, x)$ is computable in constant time from the first k bytes of x (where $k \leq 8$), independent of the remaining input. No unbounded computation occurs during gas estimation.*

8 Performance Measurement

8.1 Consensus Performance

Network	Validators	Finality	TPS	Config
Mainnet	21	9.63 s	1,091	K=21, α=15
Testnet	11	6.3 s	1,094	K=11, α=8
Local	5	3.69 s	438–1,094	K=5, α=3

Table 5: Consensus performance across deployment sizes.

8.2 Wire Protocol

Operation	ZAP	MCP JSON-RPC	Speedup	Allocs
Single tool call	199 ns	3,939 ns	20×	2 vs. 58
20-tool orchestration	5,104 ns	73,685 ns	14×	60 vs. 1,060
Parse (buffer read)	2.2 ns	—	—	0
Build (message construct)	38.9 ns	—	—	0
Consensus round	4.6 ns	—	—	0

Table 6: ZAP benchmarks (Apple M1 Max, Go 1.26.1). MCP comparison from measured test suite.

9 Data Flow Architecture

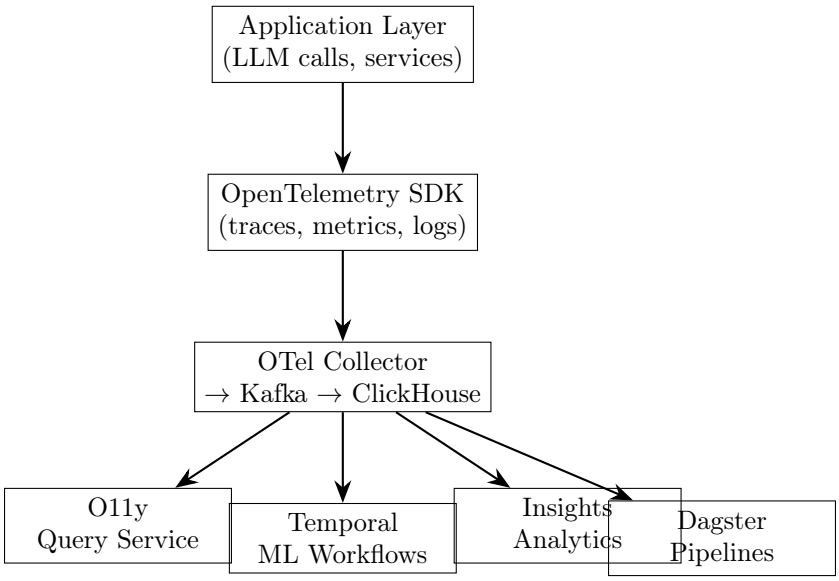

Figure 1: V&V data flow from application to analytics.

10 Naval Application Scenarios

10.1 System Qualification

Digital twin environments replicate production configurations for pre-deployment testing:

- Deploy identical K8s stack in isolated namespace.
- Run full consensus with simulated network conditions (latency, jitter, packet loss).
- Execute automated test suites against all cryptographic primitives.
- Verify performance meets operational requirements before fleet deployment.

10.2 Mission Rehearsal

- Configure Netrunner with theater-specific network topology.
- Simulate DDIL conditions (satellite jamming, link degradation).
- Validate AI model performance under constrained bandwidth.
- Test graceful degradation and recovery procedures.

10.3 Continuous Monitoring at Sea

- O11y agents deployed on each node collect telemetry locally.
- Store-and-forward synchronization when connectivity resumes.
- Anomaly detection runs locally with pre-trained models.
- Alert escalation via available communication channels.

11 Conclusion

We have presented an automated V&V framework that provides continuous verification of AI/ML systems, cryptographic implementations, and consensus protocols through integrated observability, behavioral analytics, ML pipeline orchestration, and digital twin environments. The framework's 17 formal proofs, comprehensive test suites, and production observability stack enable data-driven decision-making for naval system qualification and mission validation.

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Reproducibility

Source code. `github.com/hanzoai/platform, hanzoai/agents` (commit `TBD`).
Build. make test per component; CI via GitHub Actions on amd64/arm64 runners.
Hardware tested. Linux amd64/arm64; macOS for local dev.
Standards referenced. IEEE 29119, NIST SP 800-160, DevSecOps.
Contact. research@hanzo.ai.